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14. PROJECT DIRECTOR/PRINCIPAL INVESTIGATOR CONTACT INFORMATION				
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Phone Number: 530-754-8258		Fax Number:	Email: ofiehn@ucdavis.edu	
15. ESTIMATED PROJECT FUNDING		16. IS APPLICATION SUBJECT TO REVIEW BY STATE EXECUTIVE ORDER 12372 PROCESS?		
a. Total Federal Funds Requested \$3,186,284.00		a. YES ☐ THIS PREAPPLICATION/APPLICATION WAS MADE AVAILABLE TO THE STATE EXECUTIVE ORDER 12372 PROCESS FOR REVIEW ON:		
b. Total Non-Federal Funds \$0.00		DATE:		
c. Total Federal & Non-Federal Funds \$3,186,284.00		b. NO ☐ PROGRAM IS NOT COVERED BY E.O. 12372; OR		
d. Estimated Program Income \$0.00		☒ PROGRAM HAS NOT BEEN SELECTED BY STATE FOR REVIEW		
17. By signing this application, I certify (1) to the statements contained in the list of certifications* and (2) that the statements herein are true, complete and accurate to the best of my knowledge. I also provide the required assurances * and agree to comply with any resulting terms if I accept an award. I am aware that any false, fictitious, or fraudulent statements or claims may subject me to criminal, civil, or administrative penalties. (U.S. Code, Title 18, Section 1001)				
☒ I agree				
The list of certifications and assurances, or an Internet site where you may obtain this list, is contained in the announcement or agency specific instructions.				
18. SFLLL or other Explanatory Documentation. File Name: Mime Type:				
19. Authorized Representative				
Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
	Alyssa		Bunn	
Position/Title: Contracts and Grants Officer		Organization Name: The Regents of the University of California (Davis)		
Department: OVCR - Sponsored Programs				
Division: VC Research				
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Signature of Authorized Representative			Date Signed	
20. Pre-application File Name: Mime Type:				
21. Cover Letter Attachment File Name: Mime Type:				

Project/Performance Site Location(s)

Project/Performance Site Primary Location

Organization Name: The Regents of the University of California, Davis

* Street1: One Shields Ave.

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* City: Davis

County: Yolo

* State: CA: California

Province:

* Country: USA: UNITED STATES

* Zip / Postal Code: 95616-5270

UEI: TX2DAGQPENZ5

* Project/Performance Site Congressional District: CA-004

Project/Performance Site Location 1

Organization Name: Battell-PNWD, Pacific Northwest National Laboratory

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* Zip / Postal Code: 99354-1793

UEI: CWKJEXDG79A7

* Project/Performance Site Congressional District: WA-004

Project/Performance Site Location 2

Organization Name: The Regents of the University of Michigan

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Province:

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* Zip / Postal Code: 48109-1274

UEI: GNJ7BBP73WE9

* Project/Performance Site Congressional District: MI-006

File Name

Mime Type

Additional Location(s)

RESEARCH & RELATED Other Project Information

1. * Are Human Subjects Involved? ☐ Yes ☒ No	
1.a. If YES to Human Subjects Is the Project Exempt from Federal regulations? ☐ Yes ☐ No If yes, check appropriate exemption number Exemption Number: <u> 1 </u> <u> 2 </u> <u> 3 </u> <u> 4 </u> <u> 5 </u> <u> 6 </u> <u> 7 </u> <u> 8 </u> If no, is the IRB review Pending? ☐ Yes ☐ No IRB Approval Date: Human Subject Assurance Number	
2. * Are Vertebrate Animals Used? ☐ Yes ☒ No	
2.a. If YES to Vertebrate Animals Is the IACUC review Pending? ☐ Yes ☐ No IACUC Approval Date: Animal Welfare Assurance Number	
3. * Is proprietary/privileged information ☐ Yes ☒ No included in the application?	
4.a. * Does the Project have an Actual or Perceived Impact – positive or negative – on the environment? ☐ Yes ☒ No 4.b. If yes, please explain: 4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ☐ Yes ☐ No 4.d. If yes, please explain:	
5.a. * Is the research performance site designated, or eligible to be designated, as a historic place? ☐ Yes ☒ No 5.b. If yes, please explain:	
6.a. * Does this project involve activities outside the U.S. or partnership with International Collaborators? ☐ Yes ☒ No 6.b. If yes, identify countries: 6.c. Optional Explanation:	
7. Project Summary/Abstract	PS_Abs_Fiehn_R01_PNNL_UM_09_20231041306879.pdf mime type: application/pdf
8. Project Narrative	PN__Fiehn_R01_LC_Bi_UM_PNNL_9_231041306880.pdf mime type: application/pdf
9. Bibliography & References Cited	Bibliography_R01_DB_09_26_20231041564444.pdf mime type: application/pdf
10. Facilities & Other Resources	Fiehn_R01__Facil_Res_01_20231041306897.pdf mime type: application/pdf
11. Equipment	Equip_Fiehn_R01_UMich_PNNL1041306887.pdf mime type: application/pdf

Project Summary

Scientific teams led by Prof. Oliver Fiehn (UC Davis), Prof. Charles Evans (University of Michigan) and Dr. Tom Metz (Pacific Northwest National Laboratory) will develop and validate a unified, cloud-based data processing workflow and database for high resolution LC-MS/MS metabolomics and lipidomics data. This new database, LC-BinBase, will serve as cornerstone to harmonize and standardize metabolomics data reports. Through both experimental and computational robustness tests, we will show that different metabolomics laboratories can generate consensus data sets if the same set of data acquisition methods are used, even if instruments from different vendors are used.

This work will lead the way for next-generation metabolomics, towards faster, more reproducible and robust data processing that leads to standardized data reports and improved interpretability for biomedical researchers. To this end, we will develop algorithms to apply LC-BinBase with calculated metabolite annotation confidence scores for both hydrophilic interaction- and reversed phase chromatography, and for both QTOF and orbital ion trap high resolution mass spectrometers. These confidence scores will be calculated from deviations of probability distributions of experimental and predicted retention times, accurate masses, MS/MS entropy similarity scores and ion mobility data. We will validate the usability of LC-BinBase reports for biomedical researchers by interlaboratory comparison studies on diverse sets of mouse organs, plus different cell types and biofluids. Data usability will be improved by a unified compound metadata list for all annotated metabolites.

We will organize events and workshops to invite the scientific community to beta-test LC-BinBase.

Project Narrative

The Fiehn laboratory at UC Davis will collaborate with the Metz laboratory at PNNL and the Evans laboratory at University of Michigan to develop and validate the LC-BinBase database as generic workflow for liquid chromatography-mass spectrometry data in metabolomics and lipidomics. We aim to standardize metabolomics data processing across different mass spectrometry instruments, for different data acquisition methods, and across laboratories. We will validate result data for biofluids (plasma, urine, saliva), for different animal tissues and for cell cultures to yield reproducible and harmonized data sets that are easier to interpret for biomedical scientists.

BIBLIOGRAPHY and LITERATURE CITED

1. Bonini, P.; Kind, T.; Tsugawa, H.; Barupal, D. K.; Fiehn, O., Retip: retention time prediction for compound annotation in untargeted metabolomics. *Analytical Chemistry* 2020.
2. Tsugawa, H.; Kind, T.; Nakabayashi, R.; Yukihiro, D.; Tanaka, W.; Cajka, T.; Saito, K.; Fiehn, O.; Arita, M., Hydrogen rearrangement rules: computational MS/MS fragmentation and structure elucidation using MS-FINDER software. *Analytical chemistry* 2016, *88* (16), 7946-7958.
3. Kind, T.; Fiehn, O., Seven Golden Rules for heuristic filtering of molecular formulas obtained by accurate mass spectrometry. *BMC bioinformatics* 2007, *8* (1), 105.
4. Kind, T.; Tsugawa, H.; Cajka, T.; Ma, Y.; Lai, Z.; Mehta, S. S.; Wohlgemuth, G.; Barupal, D. K.; Showalter, M. R.; Arita, M., Identification of small molecules using accurate mass MS/MS search. *Mass Spectrometry Reviews* 2017.
5. Ma, Y.; Kind, T.; Yang, D.; Leon, C.; Fiehn, O., MS2Analyzer: A software for small molecule substructure annotations from accurate tandem mass spectra. *Analytical chemistry* 2014, *86* (21), 10724-10731.
6. Li, Y.; Kind, T.; Folz, J.; Vaniya, A.; Mehta, S. S.; Fiehn, O., Spectral entropy outperforms MS/MS dot product similarity for small-molecule compound identification. *Nature methods* 2021, *18* (12), 1524-1531.
7. Colby, S. M.; Thomas, D. G.; Nuñez, J. R.; Baxter, D. J.; Glaesemann, K. R.; Brown, J. M.; Pirrung, M. A.; Govind, N.; Teeguarden, J. G.; Metz, T. O., ISICLE: a quantum chemistry pipeline for establishing in silico collision cross section libraries. *Analytical chemistry* 2019, *91* (7), 4346-4356.
8. Lee, J.-Y.; Bilbao, A.; Conant, C. R.; Bloodsworth, K. J.; Orton, D. J.; Zhou, M.; Wilson, J. W.; Zheng, X.; Webb, I. K.; Li, A., AutoCCS: automated collision cross-section calculation software for ion mobility spectrometry-mass spectrometry. *Bioinformatics* 2021, *37* (22), 4193-4201.
9. Blaženović, I.; Shen, T.; Mehta, S. S.; Kind, T.; Ji, J.; Piparo, M.; Cacciola, F.; Mondello, L.; Fiehn, O., Increasing Compound Identification Rates in Untargeted Lipidomics Research with Liquid Chromatography Drift Time-Ion Mobility Mass Spectrometry. *Analytical chemistry* 2018, *90* (18), 10758-10764.
10. Sud, M.; Fahy, E.; Cotter, D.; Azam, K.; Vadivelu, I.; Burant, C.; Edison, A.; Fiehn, O.; Higashi, R.; Nair, K. S., Metabolomics Workbench: An international repository for metabolomics data and metadata, metabolite standards, protocols, tutorials and training, and analysis tools. *Nucleic acids research* 2016, *44* (D1), D463-D470.
11. O'Shea, K.; Misra, B. B., Software tools, databases and resources in metabolomics: updates from 2018 to 2019. *Metabolomics* 2020, *16* (3), 36.
12. Fahy, E.; Subramaniam, S., RefMet: a reference nomenclature for metabolomics. *Nature methods* 2020, *17* (12), 1173-1174.
13. Tsugawa, H.; Cajka, T.; Kind, T.; Ma, Y.; Higgins, B.; Ikeda, K.; Kanazawa, M.; VanderGheynst, J.; Fiehn, O.; Arita, M., MS-DIAL: data-independent MS/MS deconvolution for comprehensive metabolome analysis. *Nature methods* 2015, *12* (6), 523-526.
14. Tsugawa, H.; Ikeda, K.; Takahashi, M.; Satoh, A.; Mori, Y.; Uchino, H.; Okahashi, N.; Yamada, Y.; Tada, I.; Bonini, P., A lipidome atlas in MS-DIAL 4. *Nature Biotechnology* 2020, *38* (10), 1159-1163.
15. Kind, T.; Liu, K.-H.; Lee, D. Y.; DeFelice, B.; Meissen, J. K.; Fiehn, O., LipidBlast in silico tandem mass spectrometry database for lipid identification. *Nature methods* 2013, *10* (8), 755-758.
16. Kind, T.; Okazaki, Y.; Saito, K.; Fiehn, O., LipidBlast templates as flexible tools for creating new in-silico tandem mass spectral libraries. *Analytical chemistry* 2014, *86* (22), 11024-11027.
17. DeFelice, B. C.; Mehta, S. S.; Samra, S.; Cajka, T. S.; Wancewicz, B.; Fahrman, J. F.; Fiehn, O., Mass Spectral Feature List Optimizer (MS-FLO): A Tool To Minimize False Positive Peak Reports in Untargeted Liquid Chromatography-Mass Spectroscopy (LC-MS) Data Processing. *Analytical chemistry* 2017, *89* (6), 3250-3255.
18. Fiehn, O.; Wohlgemuth, G.; Scholz, M. In *Setup and annotation of metabolomic experiments by integrating biological and mass spectrometric metadata*, International workshop on data integration in the life sciences, Springer: 2005; pp 224-239.
19. Scholz, M.; Fiehn, O., Setup X-A public study design database for metabolomic projects. In *Biocomputing 2007*, World Scientific: 2007; pp 169-180.
20. Barupal, D. K.; Haldiya, P. K.; Wohlgemuth, G.; Kind, T.; Kothari, S. L.; Pinkerton, K. E.; Fiehn, O., MetaMapp: mapping and visualizing metabolomic data by integrating information from biochemical pathways and chemical and mass spectral similarity. *BMC bioinformatics* 2012, *13* (1), 99.

21. Blaženović, I.; Kind, T.; Sa, M. R.; Ji, J.; Vaniya, A.; Wancewicz, B.; Roberts, B. S.; Torbašinović, H.; Lee, T.; Mehta, S. S.; Showalter, M. R.; Song, H.; Kwok, J.; Jahn, D.; Kim, J.; Fiehn, O., Structure annotation of all mass spectra in untargeted metabolomics. *Analytical chemistry* 2019, 91 (3), 2155-2162.
22. Fan, S.; Kind, T.; Cajka, T.; Hazen, S. L.; Tang, W. W.; Kaddurah-Daouk, R.; Irvin, M. R.; Arnett, D. K.; Barupal, D. K.; Fiehn, O., Systematic error removal using random forest for normalizing large-scale untargeted lipidomics data. *Analytical chemistry* 2019, 91 (5), 3590-3596.
23. Jeffryes, J. G.; Colastani, R. L.; Elbadawi-Sidhu, M.; Kind, T.; Niehaus, T. D.; Broadbelt, L. J.; Hanson, A. D.; Fiehn, O.; Tyo, K. E.; Henry, C. S., MINEs: open access databases of computationally predicted enzyme promiscuity products for untargeted metabolomics. *Journal of cheminformatics* 2015, 7 (1), 44.
24. Kind, T.; Wohlgemuth, G.; Lee, D. Y.; Lu, Y.; Palazoglu, M.; Shahbaz, S.; Fiehn, O., FiehnLib: mass spectral and retention index libraries for metabolomics based on quadrupole and time-of-flight gas chromatography/mass spectrometry. *Analytical chemistry* 2009, 81 (24), 10038-10048.
25. Lai, Z.; Tsugawa, H.; Wohlgemuth, G.; Mehta, S.; Mueller, M.; Zheng, Y.; Ogiwara, A.; Meissen, J.; Showalter, M.; Takeuchi, K., Identifying metabolites by integrating metabolome databases with mass spectrometry cheminformatics. *Nature methods* 2018, 15 (1), 53.
26. Ma, Y.; Kind, T.; Vaniya, A.; Gennity, I.; Fahrman, J. F.; Fiehn, O., An in silico MS/MS library for automatic annotation of novel FAHFA lipids. *Journal of cheminformatics* 2015, 7 (1), 53.
27. Wang, S.; Kind, T.; Bremer, P. L.; Tantillo, D. J.; Fiehn, O., Beyond the Ground State: Predicting Electron Ionization Mass Spectra Using Excited-State Molecular Dynamics. *Journal of Chemical Information and Modeling* 2022.
28. Wohlgemuth, G.; Haldiya, P. K.; Willighagen, E.; Kind, T.; Fiehn, O., The Chemical Translation Service—a web-based tool to improve standardization of metabolomic reports. *Bioinformatics* 2010, 26 (20), 2647-2648.
29. Wohlgemuth, G.; Mehta, S. S.; Mejia, R. F.; Neumann, S.; Pedrosa, D.; Pluskal, T.; Schymanski, E. L.; Willighagen, E. L.; Wilson, M.; Wishart, D. S., SPLASH, a hashed identifier for mass spectra. *Nature biotechnology* 2016, 34 (11), 1099-1101.
30. Keshet, U.; Kind, T.; Lu, X.; Devi, S.; Fiehn, O., Acyl-CoA Identification in Mouse Liver Samples Using the In Silico CoA-Blast Tandem Mass Spectral Library. *Analytical Chemistry* 2022, 94 (6), 2732-2739.
31. Fiehn, O.; Wohlgemuth, G.; Scholz, M.; Kind, T.; Lee, D. Y.; Lu, Y.; Moon, S.; Nikolau, B., Quality control for plant metabolomics: reporting MSI-compliant studies. *The Plant Journal* 2008, 53 (4), 691-704.
32. Bremer, P. L.; Wohlgemuth, G.; Fiehn, O., The BinDiscover database: a biology-focused meta-analysis tool for 156,000 GC-TOF MS metabolome samples. *Journal of Cheminformatics* 2023, 15 (1), 1-14.
33. Nothias, L.-F.; Petras, D.; Schmid, R.; Dührkop, K.; Rainer, J.; Sarvepalli, A.; Protzyuk, I.; Ernst, M.; Tsugawa, H.; Fleischauer, M., Feature-based molecular networking in the GNPS analysis environment. *Nature methods* 2020, 17 (9), 905-908.
34. Wang, M.; Jarmusch, A. K.; Vargas, F.; Aksenov, A. A.; Gauglitz, J. M.; Weldon, K.; Petras, D.; da Silva, R.; Quinn, R.; Melnik, A. V., Mass spectrometry searches using MASST. *Nature biotechnology* 2020, 38 (1), 23-26.
35. Gilbert, J. A.; Quinn, R. A.; Debelius, J.; Xu, Z. Z.; Morton, J.; Garg, N.; Jansson, J. K.; Dorrestein, P. C.; Knight, R., Microbiome-wide association studies link dynamic microbial consortia to disease. *Nature* 2016, 535 (7610), 94-103.
36. Gauglitz, J. M.; West, K. A.; Bittremieux, W.; Williams, C. L.; Weldon, K. C.; Panitchpakdi, M.; Di Ottavio, F.; Aceves, C. M.; Brown, E.; Sikora, N. C., Enhancing untargeted metabolomics using metadata-based source annotation. *Nature biotechnology* 2022, 40 (12), 1774-1779.
37. Haug, K.; Salek, R. M.; Conesa, P.; Hastings, J.; De Matos, P.; Rijnbeek, M.; Mahendraker, T.; Williams, M.; Neumann, S.; Rocca-Serra, P., MetaboLights—an open-access general-purpose repository for metabolomics studies and associated meta-data. *Nucleic acids research* 2013, 41 (D1), D781-D786.
38. Hastings, J.; Owen, G.; Dekker, A.; Ennis, M.; Kale, N.; Muthukrishnan, V.; Turner, S.; Swainston, N.; Mendes, P.; Steinbeck, C., ChEBI in 2016: Improved services and an expanding collection of metabolites. *Nucleic acids research* 2016, 44 (D1), D1214-D1219.
39. Davidson, R. L.; Weber, R. J.; Liu, H.; Sharma-Oates, A.; Viant, M. R., Galaxy-M: A Galaxy workflow for processing and analyzing direct infusion and liquid chromatography mass spectrometry-based metabolomics data. *Gigascience* 2016, 5 (1), s13742-016-0115-8.
40. da Silva, R. R.; Dorrestein, P. C.; Quinn, R. A., Illuminating the dark matter in metabolomics. *Proceedings of the National Academy of Sciences* 2015, 112 (41), 12549-12550.

41. Ding, J.; Ji, J.; Rabow, Z.; Shen, T.; Folz, J.; Brydges, C. R.; Fan, S.; Lu, X.; Mehta, S.; Showalter, M. R., A metabolome atlas of the aging mouse brain. *Nature communications* 2021, 12 (1), 1-12.
42. Folz, J.; Culver, R. N.; Morales, J. M.; Grembi, J.; Triadafilopoulos, G.; Relman, D. A.; Huang, K. C.; Shalon, D.; Fiehn, O., Human metabolome variation along the upper intestinal tract. *Nature Metabolism* 2023, 1-12.
43. Cumeras, R.; Shen, T.; Valdiviez, L.; Tippins, Z.; Haffner, B. D.; Fiehn, O., Differences in the Stool Metabolome between Vegans and Omnivores: Analyzing the NIST Stool Reference Material. *Metabolites* 2023, 13 (8), 921.
44. Sumner, L. W.; Amberg, A.; Barrett, D.; Beale, M. H.; Beger, R.; Daykin, C. A.; Fan, T. W.-M.; Fiehn, O.; Goodacre, R.; Griffin, J. L., Proposed minimum reporting standards for chemical analysis. *Metabolomics* 2007, 3 (3), 211-221.
45. Savitski, M. M.; Wilhelm, M.; Hahne, H.; Kuster, B.; Bantscheff, M., A scalable approach for protein false discovery rate estimation in large proteomic data sets [s]. *Molecular & Cellular Proteomics* 2015, 14 (9), 2394-2404.
46. Elias, J. E.; Gygi, S. P., Target-decoy search strategy for mass spectrometry-based proteomics. *Proteome bioinformatics* 2010, 55-71.
47. Rodríguez, J. P.; Guijas, C.; Astudillo, A. M.; Rubio, J. M.; Balboa, M. A.; Balsinde, J., Sequestration of 9-hydroxystearic acid in FAHFA (fatty acid esters of hydroxy fatty acids) as a protective mechanism for colon carcinoma cells to avoid apoptotic cell death. *Cancers* 2019, 11 (4), 524.
48. Parker, B. J.; Wearsch, P. A.; Veloo, A. C.; Rodriguez-Palacios, A., The genus Alistipes: gut bacteria with emerging implications to inflammation, cancer, and mental health. *Frontiers in immunology* 2020, 11, 906.
49. Walker, A.; Pfitzner, B.; Harir, M.; Schaubeck, M.; Calasan, J.; Heinzmann, S. S.; Turaev, D.; Rattei, T.; Endesfelder, D.; Castell, W. z., Sulfonolipids as novel metabolite markers of Alistipes and Odoribacter affected by high-fat diets. *Scientific reports* 2017, 7 (1), 1-10.
50. Makishima, M.; Okamoto, A. Y.; Repa, J. J.; Tu, H.; Learned, R. M.; Luk, A.; Hull, M. V.; Lustig, K. D.; Mangelsdorf, D. J.; Shan, B., Identification of a nuclear receptor for bile acids. *Science* 1999, 284 (5418), 1362-1365.
51. Stofan, M.; Guo, G. L., Bile acids and FXR: novel targets for liver diseases. *Frontiers in medicine* 2020, 7, 544.
52. Shen, T.; Conway, C.; Rempfert, K. R.; Kyle, J. E.; Colby, S. M.; Gaul, D. A.; Habra, H.; Kong, F.; Bloodsworth, K. J.; Allen, D.; Evans, B. S.; Du, X.; Fernandez, F. M.; Metz, T. O.; Fiehn, O.; Evans, C. R., The unknown lipids project: harmonized methods improve compound identification and data reproducibility in an inter-laboratory untargeted lipidomics study. *bioRxiv* 2023, *BIORXIV/2023/526566*, 1-16.
53. Blaženović, I.; Kind, T.; Torbašinović, H.; Obrenović, S.; Mehta, S. S.; Tsugawa, H.; Wermuth, T.; Schauer, N.; Jahn, M.; Biedendieck, R.; Jahn, D.; Fiehn, O., Comprehensive comparison of in silico MS/MS fragmentation tools of the CASMI contest: database boosting is needed to achieve 93% accuracy. *Journal of cheminformatics* 2017, 9 (1), 1-12.
54. Cooper, B. T.; Yan, X.; Simón-Manso, Y.; Tchekhovskoi, D. V.; Mirokhin, Y. A.; Stein, S. E., Hybrid search: a method for identifying metabolites absent from tandem mass spectrometry libraries. *Analytical chemistry* 2019, 91 (21), 13924-13932.
55. Li, Y.; Fiehn, O., Flash entropy search to query all mass spectral libraries in real time. *Nature methods* 2023, 20 (in press).
56. Kachman, M.; Habra, H.; Duren, W.; Wigginton, J.; Sajjakulnukit, P.; Michailidis, G.; Burant, C.; Karnovsky, A., Deep annotation of untargeted LC-MS metabolomics data with Binner. *Bioinformatics* 2020, 36 (6), 1801-1806.
57. Habra, H.; Kachman, M.; Bullock, K.; Clish, C.; Evans, C. R.; Karnovsky, A., metabCombiner: Paired Untargeted LC-HRMS Metabolomics Feature Matching and Concatenation of Disparately Acquired Data Sets. *Analytical Chemistry* 2021, 93 (12), 5028-5036.
58. Xing, S.; Shen, S.; Xu, B.; Huan, T., Molecular formula discovery via bottom-up MS/MS interrogation. *bioRxiv* 2022, 2022.08. 03.502704.
59. Dührkop, K.; Fleischauer, M.; Ludwig, M.; Aksenov, A. A.; Melnik, A. V.; Meusel, M.; Dorrestein, P. C.; Rousu, J.; Böcker, S., SIRIUS 4: a rapid tool for turning tandem mass spectra into metabolite structure information. *Nature methods* 2019, 16 (4), 299-302.
60. Maas, P.; Wegrzyn, A.; Harms, A.; Hankemeier, T.; Kindt, A. *mzQuality: A tool for quality monitoring and reporting of mass spectrometry measurements.* , GitHub, 2022.

61. Colby, S. M.; Chang, C. H.; Bade, J. L.; Nunez, J. R.; Blumer, M. R.; Orton, D. J.; Bloodsworth, K. J.; Nakayasu, E. S.; Smith, R. D.; Ibrahim, Y. M., DEIMoS: an open-source tool for processing high-dimensional mass spectrometry data. *Analytical Chemistry* 2022, 94 (16), 6130-6138.
62. Barupal, D. K.; Fiehn, O., Generating the blood exposome database using a comprehensive text mining and database fusion approach. *Environmental health perspectives* 2019, 127 (9), 097008.
63. Wishart, D. S.; Bartok, B.; Oler, E.; Liang, K. Y.; Budinski, Z.; Berjanskii, M.; Guo, A.; Cao, X.; Wilson, M., MarkerDB: an online database of molecular biomarkers. *Nucleic Acids Research* 2021, 49 (D1), D1259-D1267.
64. Wishart, D. S.; Feunang, Y. D.; Guo, A. C.; Lo, E. J.; Marcu, A.; Grant, J. R.; Sajed, T.; Johnson, D.; Li, C.; Sayeeda, Z., DrugBank 5.0: a major update to the DrugBank database for 2018. *Nucleic acids research* 2018, 46 (D1), D1074-D1082.
65. Scalbert, A.; Brennan, L.; Manach, C.; Andres-Lacueva, C.; Dragsted, L. O.; Draper, J.; Rappaport, S. M.; van der Hooft, J. J.; Wishart, D. S., The food metabolome: a window over dietary exposure. *The American journal of clinical nutrition* 2014, 99 (6), 1286-1308.
66. Errico, F.; Napolitano, F.; Nisticò, R.; Usiello, A., New insights on the role of free D-aspartate in the mammalian brain. *Amino acids* 2012, 43 (5), 1861-1871.
67. Liebisch, G.; Fahy, E.; Aoki, J.; Dennis, E. A.; Durand, T.; Ejsing, C. S.; Fedorova, M.; Feussner, I.; Griffiths, W. J.; Köfeler, H., Update on LIPID MAPS classification, nomenclature, and shorthand notation for MS-derived lipid structures. *Journal of lipid research* 2020, 61 (12), 1539-1555.
68. Djoumbou Feunang, Y.; Eisner, R.; Knox, C.; Chepelev, L.; Hastings, J.; Owen, G.; Fahy, E.; Steinbeck, C.; Subramanian, S.; Bolton, E., ClassyFire: automated chemical classification with a comprehensive, computable taxonomy. *Journal of cheminformatics* 2016, 8 (1), 1-20.
69. Xing, S.; Shen, S.; Xu, B.; Li, X.; Huan, T., BUDDY: molecular formula discovery via bottom-up MS/MS interrogation. *Nature Methods* 2023, 1-10.
70. Blaženović, I.; Kind, T.; Torbašinović, H.; Obrenović, S.; Mehta, S. S.; Tsugawa, H.; Wermuth, T.; Schauer, N.; Jahn, M.; Biedendieck, R.; Jahn, D.; Fiehn, O., Comprehensive comparison of in silico MS/MS fragmentation tools of the CASMI contest: database boosting is needed to achieve 93% accuracy. *Journal of Cheminformatics* 2017, 9 (1), 32.
71. Dührkop, K.; Nothias, L.-F.; Fleischauer, M.; Reher, R.; Ludwig, M.; Hoffmann, M. A.; Petras, D.; Gerwick, W. H.; Rousu, J.; Dorrestein, P. C., Systematic classification of unknown metabolites using high-resolution fragmentation mass spectra. *Nature Biotechnology* 2021, 39 (4), 462-471.
72. Barupal, D. K.; Fiehn, O., Chemical Similarity Enrichment Analysis (ChemRICH) as alternative to biochemical pathway mapping for metabolomic datasets. *Scientific reports* 2017, 7 (1), 1-11.
73. Lipscomb, C. E., Medical subject headings (MeSH). *Bulletin of the Medical Library Association* 2000, 88 (3), 265.
74. Bremer, P. L.; Fiehn, O., SMetaS: A Sample Metadata Standardizer for Metabolomics. *Metabolites* 2023, 13 (8), 941.
75. Aron, A. T.; Gentry, E. C.; McPhail, K. L.; Nothias, L.-F.; Nothias-Esposito, M.; Bouslimani, A.; Petras, D.; Gauglitz, J. M.; Sikora, N.; Vargas, F., Reproducible molecular networking of untargeted mass spectrometry data using GNPS. *Nature protocols* 2020, 15 (6), 1954-1991.
76. Goodman, J. M.; Pletnev, I.; Thiessen, P.; Bolton, E.; Heller, S. R., InChI version 1.06: now more than 99.99% reliable. *Journal of Cheminformatics* 2021, 13 (1), 1-8.
77. Fabregat, A.; Jupe, S.; Matthews, L.; Sidiropoulos, K.; Gillespie, M.; Garapati, P.; Haw, R.; Jassal, B.; Korninger, F.; May, B., The reactome pathway knowledgebase. *Nucleic acids research* 2018, 46 (D1), D649-D655.
78. Caspi, R.; Billington, R.; Keseler, I. M.; Kothari, A.; Krummenacker, M.; Midford, P. E.; Ong, W. K.; Paley, S.; Subhraveti, P.; Karp, P. D., The MetaCyc database of metabolic pathways and enzymes-a 2019 update. *Nucleic acids research* 2020, 48 (D1), D445-D453.
79. Wishart, D. S.; Tian, S.; Allen, D.; Oler, E.; Peters, H.; Lui, V. W.; Gautam, V.; Djoumbou-Feunang, Y.; Greiner, R.; Metz, T. O., BioTransformer 3.0—a web server for accurately predicting metabolic transformation products. *Nucleic Acids Research* 2022.
80. Kim, S.; Thiessen, P. A.; Bolton, E. E.; Chen, J.; Fu, G.; Gindulyte, A.; Han, L.; He, J.; He, S.; Shoemaker, B. A., PubChem substance and compound databases. *Nucleic acids research* 2016, 44 (D1), D1202-D1213.

81. Gaud, C.; Sousa, B. C.; Nguyen, A.; Fedorova, M.; Ni, Z.; O'Donnell, V. B.; Wakelam, M. J.; Andrews, S.; Lopez-Clavijo, A. F., BioPAN: a web-based tool to explore mammalian lipidome metabolic pathways on LIPID MAPS. *F1000Research* 2021, 10.
82. Aimo, L.; Liechti, R.; Hyka-Nouspikel, N.; Niknejad, A.; Gleizes, A.; Götz, L.; Kuznetsov, D.; David, F. P.; van der Goot, F. G.; Riezman, H., The SwissLipids knowledgebase for lipid biology. *Bioinformatics* 2015, 31 (17), 2860-2866.
83. Wishart, D. S.; Guo, A.; Oler, E.; Wang, F.; Anjum, A.; Peters, H.; Dizon, R.; Sayeeda, Z.; Tian, S.; Lee, B. L., HMDB 5.0: the human metabolome database for 2022. *Nucleic Acids Research* 2022, 50 (D1), D622-D631.
84. Bansal, P.; Morgat, A.; Axelsen, K. B.; Muthukrishnan, V.; Coudert, E.; Aimo, L.; Hyka-Nouspikel, N.; Gasteiger, E.; Kerhornou, A.; Neto, T. B., Rhea, the reaction knowledgebase in 2022. *Nucleic acids research* 2022, 50 (D1), D693-D700.
85. Kim, S.; Thiessen, P. A.; Cheng, T.; Yu, B.; Bolton, E. E., An update on PUG-REST: RESTful interface for programmatic access to PubChem. *Nucleic Acids Research* 2018, 46 (W1), W563-W570.
86. Caspi, R.; Billington, R.; Ferrer, L.; Foerster, H.; Fulcher, C. A.; Keseler, I. M.; Kothari, A.; Krummenacker, M.; Latendresse, M.; Mueller, L. A., The MetaCyc database of metabolic pathways and enzymes and the BioCyc collection of pathway/genome databases. *Nucleic acids research* 2016, 44 (D1), D471-D480.
87. Oliveira, A.; Cunha, E.; Cruz, F.; Capela, J.; Sequeira, J. C.; Sampaio, M.; Sampaio, C.; Dias, O., Systematic assessment of template-based genome-scale metabolic models created with the BiGG Integration Tool. *Journal of Integrative Bioinformatics* 2022, 19 (3).
88. Yang, Q.; Ji, H.; Fan, X.; Zhang, Z.; Lu, H., Retention time prediction in hydrophilic interaction liquid chromatography with graph neural network and transfer learning. *Journal of Chromatography A* 2021, 1656, 462536.
89. Osipenko, S.; Botashev, K.; Nikolaev, E.; Kostyukovich, Y., Transfer learning for small molecule retention predictions. *Journal of Chromatography A* 2021, 1644, 462119.
90. Fedorova, E. S.; Matyushin, D. D.; Plyushchenko, I. V.; Stavrianidi, A. N.; Buryak, A. K., Deep learning for retention time prediction in reversed-phase liquid chromatography. *Journal of Chromatography A* 2022, 1664, 462792.

Facilities and Other Resources

(1) UC Davis

UC Davis, founded in 1905 as a land grant university, provides an exceptionally rich and dynamic environment for multidisciplinary investigations. UC Davis is geographically the largest of the ten campuses of the UC system (approximately 5300 acres), second largest in overall budget, and third largest in student enrollment. Located near the California state capital of Sacramento, UC Davis has more than 35,000 students and 231,000 living alumni with degrees. UC Davis is widely known for its collaborative culture, multidisciplinary research infrastructure and partnerships that stretch across 104 undergraduate majors, 96 graduate programs, 4 colleges (the Colleges of Engineering, Letters and Science, Biological Sciences, and Agricultural & Environmental Sciences) and 6 professional schools (Schools of Medicine, Education, Law, Management, Veterinary Medicine, and the Betty Irene Moore School of Nursing). With over \$941 million in extramural research awards in fiscal year 2019-2020, UC Davis is one of the nation's leading research institutions.

Fiehn laboratory at UC Davis

Space: *Davis campus facility:* The Fiehn laboratory is part of the UC Davis West Coast Metabolomics Center (WCMC). The laboratory is located on the Davis campus of the UC Davis Genome Center building. The Fiehn metabolomics laboratory extends to 3,500 ft² for laboratory, five separate offices for 30 staff and students, plus his own office and a dedicated meeting room for 12 persons on the same floor. The lab is provided with all the necessary equipment such as freezers, balances, homogenizers, lyophilizers, shakers, centrifuges, speedvacs and sonicators. *Sacramento campus metabolomics core facility:* Since 2020, the Fiehn lab uses additional space on the UC Davis' clinical campus in Sacramento, as dedicated ThermoFisher Center of Excellence in Clinical Metabolomics. ThermoFisher provides continuous updates of instrumentation, instrument repair and maintenance, direct access to ThermoFisher software developers and access to the ThermoFisher clinical and biological network. The Sacramento lab gives us access to the needs and expertise of the UC Davis Health clinics in Sacramento and the UC Davis School of Medicine researchers. Here, the Fiehn laboratory has direct access to researchers and patients of the UC Davis Comprehensive Cancer Center that is situated next to the main UC Davis hospitals. The Sacramento lab extends to 1,500 square feet at the Oak Park building, with adjacent offices and a 35-seat seminar room for training and outreach activities. The WCMC Sacramento lab is provided with all necessary metabolomic equipment, from freezers, fridges, shakers, balances and a separate supply of pressured air for feeding nitrogen generators for five high resolution LC-MS instruments.

Chemical and Informatics Resources: The Fiehn lab has purchased over 5,000 chemical reference standards for identification and quantifications that are stored and accessed through the laboratory's LIMS system. Each chemical is maintained at proper temperature regimes (room temperature, +4°C or -20°C) and catalogued routinely by undergraduate staff members. The Fiehn laboratory hosts 18 metabolomics databases and software resources: *MassBank of North America* for more than 2 million spectra (including all experimental MS/MS entries from MassBank, ReSpect, GNPS, HMDB, GolmDB, BinBase), *LipidBlast*, *FiehnLib* with mass spectra and retention indices of more than 1,200 authentic primary compounds, *BinBase* storing data for over 200,000 samples and more than 18,000 metabolites (including unknowns), *miniX* as LIMS system, the *Chemical Translation Service* DB to translate between more than 200 public databases of biochemical and chemical species, *MetaMapp* and *ChemRICH* as biochemical mapping software, and *MetaBox (metDA)* as genomics integration platform.

Intellectual and training environment: Students and postdocs attend the Fiehnlab weekly "RapidFire" project meetings, the weekly lab seminars (for which they present their work twice a year), and the monthly invited speaker webinar (as part of the West Coast Metabolomics Center). For this series, Dr. Fiehn organizes opportunities for students and postdocs to meet with the invited speakers after the seminar in online settings, to discuss ideas and concepts in a more in-depth setting. Fiehnlab scientist can also participate in the yearly Genome Center's Storer Lecture and the Genome Center's Halloween symposium, plus in multiple other seminar series offered on campus.

Administrative support: The Fiehn lab has shares five administrative staff with the other Genome Center laboratories, to assists with ordering requests, room reservations, purchasing, staff management and other general support as needed. The Genome Center provides administrative support as well through building management and facilities maintenance.

The Fiehn laboratory is part of the UC Davis Genome Center that integrates experimental and computational approaches to address key problems at the forefront of genomics. The UC Davis Genome Center integrates

experimental and computational approaches to address key problems at the forefront of genomics. The metabolomics facility is housed in a 6-story research building with state-of-the-art computational and laboratory facilities. A 150-seat auditorium, three 30-seat seminar rooms and five 12-seat conference rooms are available for courses, seminars and workshops. All four Genome Center core facilities, including the metabolomics core, are in adjacent laboratory and office spaces in the first floor of the UC Davis GBSF building on the main campus in Davis. In addition, the West Coast Metabolomics Center integrates three further mass spectrometry-based laboratories to boost expertise and instrument redundancies, with the USDA laboratory of Prof. John Newman (WHNRC), Prof. Bruce Hammock, and Prof. Ameer Taha.

The Fiehn laboratory is housed in the GBSF research building with state-of-the-art computational and laboratory facilities. A 150-seat auditorium, three 30-seat seminar rooms and five 12-seat conference rooms are available for courses, seminars and workshops. The bioinformatics and genomics facilities of the Genome Center core facilities are in adjacent laboratory and office spaces in the first floor of the UC Davis GBSF building, right next to the Fiehn metabolomics laboratory:

(a) The UC Davis Genome Center's bioinformatics core managed by Dr. Jessie Li to provide expertise and infrastructure for the acquisition, curation, analysis, and distribution of large complex datasets as well as develops and performs computations, analyses and simulations addressing a wide variety of biological questions from genomics to network biology. The Bioinformatics Core has staff members with overlapping expertise in computing infrastructure, web/database, scientific programming, biological annotation and statistics. The core provides bioinformatics support for the wetlab service cores as well as for researchers with independent bioinformatics needs. The bioinformatics core space uses 8 cubicles for software developers, statisticians and bioinformaticists on the first floor of the Genome Center as part of a 4,000 ft² office space which is shared by the Genome Center's system administrators and the Metabolomics core researchers. The Genome Center provides a server room with cooling power for 20 industry standard racks which are at 90% capacity. The Bioinformatics Core data center is a 444 square foot facility in room 4340 in the Genomics and Biomedical Sciences Facility (GBSF) that was specifically designed as a machine room. Optical fiber and copper interconnections provide for either 1Gb/s or 10 Gb/s data transfer rates within the Core, which is connected to the 10 Gb/s Corporation for Education Network Initiatives in California (CENIC) network for fast transfer off campus. This room sufficient air conditioning for cooling up to 12Kw per rack. *Data Storage* uses two distinct classifications of data storage: *i*) a high performance storage system (> 3.6PB total spread across several filesystems) that uses fast disk drives and the ZFS file system that allows us to group storage components into logical units and create lab shares for users. This makes it possible to access portions of data from multiple storage devices and aggregates data reading and writing across multiple disk drives and network connections, thereby increasing overall performance. Each individual disk storage system (array) combines multiple disks into a single logical unit (triple parity RAID), which provides redundancy on a disk level. The file servers are connected to the compute nodes with 10Gb/sec Ethernet. *ii*) a backup storage systems (approximately 540TB total) located in the Academic Surge building to back up select data nightly to commodity disks. We maintain incremental changes for 60 days on disk and monthly snapshots for at least 1 year. *High Performance Computing*: We maintain multiple large computer clusters for life science research and omics data analysis. We maintain six high performance compute clusters which provide over four thousand processor cores and over 20 terabytes of system memory. The servers that comprise the clusters are connected with 1Gb/sec Ethernet. The modular design of these clusters, enclosures (blade chassis) and blade servers, makes it possible to service or upgrade components without interrupting our users, further with the use of Lights Out Management we can reboot individual nodes remotely. Removable and redundant fans and power supplies located at the back of the enclosure provide easy access and replacement without powering down individual systems, and each enclosure contains its own network components to maximize inter-enclosure server communication. In addition to the high performance compute cluster we also maintain a GPU cluster of 4 nodes (48 cores and 48Gb of RAM total) and 16 GPUs each with 448 cores and 6 Gb of memory (7,168 GPU cores). *Large Memory Servers*: We maintain over 11 servers that vary in size with 512 GB to 1 TB of resident shared memory.

(b) The UC Davis Genome Center's genomics core directed by Dr Lutz Froenicke offers library construction and high-throughput sequencing for Illumina MiSeq & HiSeq and PacBio RSII platforms. The core currently operates one Illumina HiSeq2500, three MiSeq, PacBio RSII, and liquid handling robots (Caliper/PerkinElmer Sciclone NGS G3 and Intenex Apollo324), and Covaris E220 ultra-focused 96-well plate sonicator for high-throughput sample preparation. Use of the latest analytic software and reagents ensures that the core is getting as much quality output from these instruments as is currently possible. The next generation sequencing platforms have been in operation in the cores since August 2007 and have provided exceptional data for a variety of

projects including de novo sequencing, re-sequencing, exome-seq, RNA-Seq, ChIP-Seq, MDBseq, genotyping by sequencing (GBS), and mutation discovery. The presence of multiple sequencers and local computational servers for pipeline analysis ensures rapid turnaround times on projects. The genomics core space occupies 1,500 ft² laboratory space for library preparations, DNA and RNA sample preparation, benches sample handling laboratory work including centrifugations and extractions as needed, and a range of -80C freezers for short-term storage. Cubicles for genomic data processing are integral part of the laboratory space, in addition to a separate office for the genomics core manager.

(c) The UC Davis Genome Center's proteomics core directed by Dr Brett Phinney provides state-of-the-art analytical proteomic services with particular emphasis on label free quantitative proteomic profiling, the analysis of macromolecular complexes, the post-translational modification of their constituents and standard protein identification from complex protein mixtures. For high throughput identification, quantitation and characterization of proteins and peptides, the proteomics core utilizes two Thermo-Fisher Scientific Orbitrap Q-Exactive+ mass spectrometers, one ThermoFisher Fusion Orbitrap mass spectrometer and one Bruker timsTOF mass spectrometer. Each MS instrument is equipped with nanoLC-systems and autosamplers to perform automated multidimensional chromatography, sample fractionation, enrichment, collection and re-injection of peptide and protein samples. The core also performs classic amino acid analysis and Edman protein sequencing. The proteomics core space further extends to 1,500 ft² laboratory space for protein separations, column preparations, purifications, and benches for protein biochemistry laboratory work including performing gels, centrifugations and chemical modifications as needed. Cubicles for proteomic data processing are integral part of the laboratory space, and a dedicated proteomic conference room for up to eight scientists is used for meetings and project discussions.

(2) University of Michigan

The University of Michigan is one of the best public research universities in the United States and a leader in higher education. It was ranked the No. 1 U.S. public university by QS World University Rankings, 2022. Since 1817, the University has provided a national model of a complex, diverse, and comprehensive public institution of higher learning that supports excellence in research, provides outstanding undergraduate, graduate, and professional education, and demonstrates commitment to service through partnerships and collaborations that extend to the community, region, state, nation, and around the world.

The University's mission directs the institution "to serve the people of Michigan and the world through preeminence in creating, communicating, preserving and applying knowledge, art, and academic values, and in developing leaders and citizens who will challenge the present and enrich the future".

Permeating all 19 Schools and Colleges, research is central to the University of Michigan's mission and comprised over \$1.58 billion expenditures in FY2021. The University of Michigan is a strong advocate of promoting collaboration and interdisciplinary research initiatives that involve faculty and students across the campuses. The University of Michigan Office of Research's mission is to catalyze, support and safeguard the University's research. The University of Michigan Medical School currently has 3,855 faculty members teaching 692 medical students, 921 graduate students, 541 postdoctoral fellows, and over 1,300 interns and residents, as well as other groups of learners. In the 2022 U.S. News & World Report's Best Medical Schools rankings, the University of Michigan Medical School tie for No. 20 for primary care and No. 17 for research. The Medical School has twenty clinical and nine basic sciences departments, as well as the Unit for Laboratory Animal Medicine. Teaching, research, and clinical care often cross traditional departmental boundaries, particularly in the School's 50+ interdisciplinary research centers and institutes, including the Rogel Cancer Center, the Geriatrics Center, the Cardiovascular Center, and the Michigan Institute for Clinical and Health Research (MICHHR). Medical School faculty members lead research efforts in a broad scope of basic and clinical science areas. In FY2021, the Medical School received \$750 million in awards from all external sponsors. Annual federal grant funding awarded to the school's clinical researchers and biomedical scientists reached \$520.7 million in FY2021. Additionally, UMMS is recognized as 13th in medical schools across the nation in NIH grants awarded. In FY2021, Medical School researchers filed 205 reports of new inventions.

Evans laboratory at University of Michigan

Space: Dr. Charles Evans has approximately 100 square feet of private office space on the 6th floor of Brehm Tower. All Metabolomics Core faculty at the University of Michigan have ample desk space, computer, phone, and internet connectivity. All faculty and staff have access to sufficient clerical staff support. The Evans Metabolomics Core laboratories are housed at the Brehm Tower on the UM campus. The labs are fully outfitted, flexible, and were custom built for the metabolomics facility-intensive use. Two main laboratories occupy

approximately 3000 sq. ft. of laboratory space with walk in cold rooms and freezers, 4 chemical hoods, a room dedicated for sample processing, and a service corridor for nitrogen generators and pumps. There is shared space for equipment storage (freezers/refrigerators) and supply storage. The facility provides access to modern mass spectrometers and other state-of-the-art instrumentation to address a wide range of research projects requiring metabolomics. The focus of the laboratory is on the characterization and quantification of small molecules with special emphasis on biomarkers, drugs, and drug metabolites. The facility offers training for independent operation of instruments as well as sample analysis by facility personnel.

The metabolomics laboratories are part of the University of Michigan Biomedical Research Core Facilities which supports the core administratively, with staffing, instrumentation purchase and maintenance, billing for services, and other interactions with researchers at the University of Michigan as well as other institutions.

All metabolomics core faculty and staff have access to multiple Windows PCs (and Linux workstations when required), located in their offices, as personal laptops, and as workstations for instrumentation control and data analysis. All University of Michigan faculty and staff are provided with access to secure on-campus high performance computing clusters (HPC) and high-capacity data storage devices to enable high throughput, high quality data analysis. The core also maintains several dedicated high-performance data analysis servers of its own, accessible from anywhere with internet access via remote connection. These are used solely for metabolomics data analysis and are equipped with a wide variety of software to support this task. Other software available includes custom programming tools, routine non-targeted data analysis software (including LC-MS and GC-MS data), graphing, statistical analysis, image analysis, word processing and desktop publishing (for manuscript preparation). The hardware and software available are sufficient to carry out this project. Laboratory computers are linked to a local area network and the internet for access to a wide variety of scientific, data analysis and word processing programs. An automated data backup and recovery system is in place both on-site and at an off-site University server.

LIMS: In addition to instrument-specific and personal computers the untargeted metabolomics laboratory has a server-based Metabolomics Laboratory Information Management System (MetLIMS) that supports and integrates all analytical laboratory processes. MetLIMS is the Laboratory Information Management System around which all of our lab activities revolve. The objective of MetLIMS is to manage Compound and Sample inventory and to automate sample run and data processing while integrating samples, instrumentation and results. MetLIMS has various user level authentications and in general allows users to manipulate and reuse the client, project, experiment and sample data. It has been built to cater to the needs of the lab personnel and has user friendly interfaces which allow the users to add, modify and search sample and compound related data. A MetLIMS user can also create and run a sample worklist and document sample preparation details. MetLIMS controls the injection sequence of samples, it has an injection queue for each instrument on the server and knows which sample is currently being injected. The 'MetworksMessaging' module of MetLIMS specifically allows the integration of the mass spectrometers data output and the data processing servers with MetLIMS and makes it possible for data to be available in close to real time.

Other: The untargeted metabolomics laboratory also maintains a library of nearly 2000 authentic standard chemical compounds for identification, validation and quantitation of a wide range of biologically relevant small molecules.

(3) Pacific Northwest National Laboratory (PNNL)

Metz laboratory at PNNL

The facilities and other resources available to Dr. Metz and his research team at Pacific Northwest National Laboratory (PNNL) provide everything needed to undertake and complete the proposed research project successfully. Dr. Metz's team is part of the Integrative Omics (IO) group housed in the US Department of Energy (DOE)-sponsored National Scientific User Facility, the William R. Wiley Environmental Molecular Sciences Laboratory (EMSL), and the Biological Science Facility (BSF). The IO group research capabilities leverage the extensive infrastructure of the National Laboratory, as well as the supportive EMSL infrastructure. The EMSL is funded by the DOE's Office of Biological and Environmental Research, which supports world-class measurement capabilities and research in the biological, chemical, and environmental sciences to provide innovative solutions to the nation's environmental challenges as well as those related to energy production. The EMSL's distinctive focus on integrating computational and experimental capabilities as well as collaborating among disciplines yields a strong, synergistic scientific environment. Bringing together experts and state-of-the-art instruments critical to their research under one roof, EMSL has helped thousands of researchers use a multidisciplinary, collaborative approach to solve some of the most important national challenges in energy, environmental

sciences, and human health. Proteomics, metabolomics and lipidomics capabilities of the IO group including state-of-the-art instruments and sophisticated informatics and bioinformatics resources, have been developed and acquired in coordination with the EMSL and PNNL research efforts. Mass spectrometry (MS) and related instrumentation housed in these labs is described below. The IO group provides a large and skilled multidisciplinary team that continually extends the state of the art of capabilities in proteomics, metabolomics, lipidomics, and associated areas of systems biology research. Additional facilities and resources are located in the BSF building at PNNL with chemical synthesis, protein purification and characterization, molecular biology, cell culture, and live-cell fluorescent imaging capabilities. These facilities provide a scientific environment that is strongly supportive for the proposed research.

Space: Office space for researchers, collaborators, staff, and postdoctoral fellows is located within EMSL. Secretarial, word processing, and other support capabilities (e.g., graphics and photography) are available within the buildings and can be utilized on an as-needed basis. EMSL office space is provided to visiting scientists. Laboratory space available for the proposed research is located on the campus of PNNL in Richland, Washington. The primary facilities to be utilized are housed in the EMSL, a 200,000 sq. ft. DOE National Scientific User Facility. The project draws upon the broad set of facilities, capabilities, instrumentation and supportive infrastructure of a large center for bioanalysis at PNNL/EMSL. Laboratory space includes a High-Throughput Liquid Chromatography-Mass Spectrometry (LC-MS) 'Omics Characterization Lab (2100 ft²); a Fourier Transform Ion Cyclotron Resonance (FTICR) Mass Spectrometry Research Lab (2600 ft²); a Mass Spectrometry Development Lab (800 ft²) for development of fundamental mass spectrometry, ion source and interface, and vacuum system capabilities, and hardware; a Micro- and Nano-Scale Separation Research Lab (800 ft²) for development of advanced high pressure liquid chromatography (HPLC) separations, and automated sample clean-up; and a Biochemistry and Sample Preparation Lab (1200 ft²) for protein separations, cell growth, biological sample preparation, stable isotope labeling, protein purification, and wet chemistry. Additional laboratory spaces located in the BSF building at PNNL include three 'Omics Sample Preparation labs (4,300 ft²), providing 6 Biological Safety Cabinets for Biosafety Level 2 (BSL2) sample preparation, 5 analytical HPLC systems for fractionation of complex samples for LC-MS analysis, 2 automated liquid handler systems for large sample set preparations, 2 automated Solid Phase Extraction (SPE) robots for sample cleanup, along with sufficient standard equipment needed for a general biochemistry lab (e.g. centrifuges, balances, freezers, refrigerators, etc.). Additionally, a Metabolomics Sample Preparation lab (900 ft²) contains two fume hoods, one analytical HPLC for lipids analysis, balances, vacuum concentrators and other general equipment needed for volatile organic sample preparation; an Activity Based Protein Profiling lab (2200 ft²) for attaching chemical probes to protein to track their function contains all the necessary equipment an organic chemistry/chemical synthesis/molecular biology lab needs; both BSL2 Bacterial and Viral Culture labs (400 ft² each) to grow BSL2 organisms for experimental work. Access is also available to lab spaces located within BSF for protein purification and characterization, live cell fluorescent imaging, molecular biology, media preparation, large scale cell culture, binding kinetics and electrophoresis.

Computational Resources: PNNL has extensive computational capabilities and tools that are available to all aspects of research proposed for the project. Top-tier resources include: the dedicated PNNL Parallel Proteomics Processing Cluster (P4C), multiple supercomputers and ultra-high speed dedicated data networks, and the PNNL Scientific Data Archive, a hierarchical data storage management system specifically designed to store and archive the massive amounts of data produced by researchers and collaborators within PNNL. PNNL maintains a high-throughput omics data storage, management and processing infrastructure, which has been specifically developed and implemented for management, analysis and dissemination of large-scale proteomics, metabolomics and lipidomics data. Additional high-level capabilities include the Graphics and Visualization Laboratory; multimedia, and video production capabilities, and numerous UNIX and PC-based servers. A support staff is immediately available to help with issues such as proteomics database searching and quantification, neural net configuration, web-based content delivery, and micro-processor control of laboratory instrumentation. All laboratories and offices have full-time and direct access to dedicated computer resources. Additional resources are available as required.

Other Resources: *Scientific Library, Scientific Journals, Facilities, and Access.* The Hanford Technical Library (HTL), part of PNNL, provides a broad suite of library and information services. Many of the services are provided on-line directly to the offices and laboratories. HTL services include: on-line library catalog, electronic journal access, and on-line subscriptions-based database searching. Electronic journals access is through the DOE shared access to electronic journals; access is shared across PNNL, Los Alamos National Lab, Oak Ridge National Lab, and others. Many hundreds of current journals are available for immediate electronic access.

Library searching capabilities are provided either on-line or in person at the HTL located one-half mile from the main PNNL campus.

Electronics Shop and Custom Software. The Instrument Development Laboratory (IDL) is an in-house facility with a trained technical team dedicated to initial design, fabrication, testing, and deployment of research apparatus and tools. Software efforts in the IDL span the entire spectrum of EMSL research, including data acquisition, laboratory instrument control, experiment modeling, ion lens control, biomedical data visualization, and data mining.

Machine Shop and Scientific-Device Fabrication Support. Support in the EMSL is extensive, immediate, and multi-tiered. EMSL's full-capability research machine shop is staffed by four machinists with extensive experience and expertise in the design, fabrication, and assembly of custom assemblies for research. Scientific Device Fabrication is routinely performed for areas such as ultra high pressure chromatography (>20,000 psi) and ultra-high vacuum components.

On-Site Conference Capabilities: Research, Dissemination, and Training. The EMSL has onsite conference capabilities consisting of one amphitheater lecture auditorium (seating ~100 persons) with advanced video and projection capabilities, and three configurable meeting rooms, each with built-in projection capabilities. These meeting capabilities are used frequently for on-site conferences.

Additional Capabilities. The EMSL laboratories also include facilities for sample preparation and storage, chemical synthesis and derivatization, one-dimensional (1-D) and 2-D polyacrylamide gel electrophoresis, Gel-free Protein Fractionation System (Protein Discovery Inc.), CE and electrospray ionization (ESI)-MS, inductively Coupled Plasma (ICP)-MS, and related instrumentation development and experimentation, purifications, synthesis, and biological research activities. Four CE systems are available, including three Crystal systems that are optimal for MS interfacing. Other equipment includes an Applied Biosystems 270A integrated instrument and data system for CE with UV-Vis/absorbance detection and a Beckman P/ACE System 1000 CE instrument extensively modified for application with an ESI interface for mass spectrometry. Also located in EMSL are four custom CE systems with electrical isolation and interlock and equipped with robotic injectors.

Major Equipment

Fiehn laboratory, UC Davis

General equipment: The Fiehn laboratory of the UC Davis West Coast Metabolomics Center facility space uses two laboratories (in Sacramento OneHealth campus and on the Davis campus) for method development and research projects with the following general equipment.

- Computational equipment: Data from the Sacramento laboratory are transferred to the Genome Center via a secure dedicated line. All data are automatically backed up. The Fiehn laboratory has developed and continues to host 18 metabolomics databases and software resources: MassBank of North America for more than 2 million spectra (including all experimental MS/MS entries from MassBank, ReSpec, GNPS, HMDB, GolmDB, BinBase), LipidBlast, FiehnLib with mass spectra and retention indices of more than 1,200 authentic primary compounds, BinBase storing data for over 200,000 samples and more than 18,000 metabolites (including unknowns), miniX as LIMS system, the Chemical Translation Service DB to translate between more than 200 public databases of biochemical and chemical species, MetaMapp and ChemRICH as biochemical mapping software, and MetaBox (metDA) as genomics integration platform.
- Computer hardware calculations are performed via Amazon Web Service for the LC-BinBase production system. For in-house calculations we use a raw data file server consisting of 600 TB hard drive space with 24GB Ram on 16 cores. The file backup system runs on 1,200 TB hard drive space with 8 cores and 8 GB Ram. The Fiehnlab BinBase database runs on 8 cores with 16 GB Ram. The Fiehnlab cluster environment includes lambic with 16 cores/48GB and five nodes with 16-48 cores and 48-256 GB Ram. MassBank of North America runs on 48 Cores with 128 GB Ram. The miniX LIMS study design database server uses 24 Cores with 96GB Ram. The virtual machine vmware servers consist of four servers: 'sun' with 16 cores and 48GB Ram, 'pluto' with 12 Cores and 128GB Ram, 'uranus' with 8 Cores and 16 GB Ram and 'mars' with 8 cores and 16GB Ram. Through the Genome Center resources, Dr. Fiehn has access to two further computer clusters: a 110-node Linux cluster "Shiraz" with dual-socket, dual-core Opteron (including 32 nodes with 8GB RAM, rest 4GB) with a total of 440 CPU cores and 568GB of RAM and 10TB of storage; and a 104 node, 996 CPU core (mostly Opteron, some Intel) cluster "Merlot" with a total of 3,160GB of RAM and a total of 36TB of disk storage.
- *Davis campus facility:* four -20°C, three -80°C freezers, two microscale balances, GenoGrinder and Retsch homogenizers, a lyophilizer, six thermoshakers for derivatizations, vortexers, four centrifuges, three speedvacs for solvent removal (including for volatiles, aggressive solvents or chloroform) and ultrasonicators for instrument cleaning as well as dissolving standards or preparing cell extracts. For 96 well-plate extractions, we use a 96-well centrifuge with 96-well evaporators and 96-well plate sealer.
- *Sacramento campus facility:* two -20°C, one -80°C freezers, one microscale balance, two thermoshakers, vortexers, ultrasonicators for instrument cleaning as well as dissolving standards or preparing cell extracts. For 96 well-plate extractions, we have an additional 96-well centrifuge with 96-well evaporator and 96-well plate sealer.

Specialized equipment. The Fiehn laboratory of the UC Davis West Coast Metabolomics Center facility space uses *19 mass spectrometers* for method development and research projects.

Thirteen LC-based mass spectrometers:

- 6 ThermoFisher Scientific instruments (2 QExactive HF+ OrbiTrap MSMS instruments, 1 Exploris OrbiTrap MSMS instrument, 1 Tribrid Fusion Orbitrap MSMS instrument, 2 Altis triple quadrupole mass spectrometers),
- 4 Agilent instruments (two 6530 QTOF MS/MS instruments, one 6546 QTOF MS/MS and one 6550 QTOF MS/MS mass spectrometer),
- 2 Sciex instruments (one 6600 TripleTOF MS, one 6500+ QTRAP mass spectrometer),
- 1 Bruker instrument (timsTOF Pro2 Flex mass spectrometer with flexible use for LC-ion mobility MS/MS or MALDI-ion mobility-MSMS imaging).

All LC-MS instruments are equipped with either Agilent 1290 or ThermoFisher Vanquish UHPLC pumps and autosamplers.

Six GC-based mass spectrometers:

- 5 Leco mass spectromers (2 Pegasus III TOF MS, 2 Pegasus IV TOF MS, one Pegasus BT MS instrument),
 - 1 Agilent high resolution QTOF MS.
- All GC-instruments are equipped with Agilent GC ovens and Agilent autosamplers.

Evans laboratory, University of Michigan

General equipment: The metabolomics core possesses and maintains all equipment necessary for preparation, analysis, and storage of samples and operation and maintenance of its instrumentation. These include an Agilent Bravo automated sample handling station, multiple robotic fraction collectors, semi-preparative LC equipment, a Sorvall centrifuge, a Beckman Ultracentrifuge, several refrigerated benchtop centrifuges, a Precellys sample disruptor, sonic probe sample disruptors, benchtop vortexers and sample shakers, 4 nitrogen sample drying stations, 2 SpeedVac concentrators; 1 vacuum lyophilizer, multiple -80°C and -20°C freezers, a walk-in cold room and walk-in -20°C freezer, multiple freestanding refrigerators; a chromatography cabinet; a Thermo-Finnigan Spectrophotometer and multiple sets of electrophoretic equipment. We are also amply equipped with all needed autopipettes, glassware, and other consumables to carry out the proposed work.

Specialized equipment: The University of Michigan Metabolomics Core houses the following major instrumentation. The research team for this proposal has full access to the equipment listed below.

Fourteen LC-based mass spectrometers:

- Six Agilent quadrupole Time of Flight (Q-TOF) mass spectrometers (Four Agilent 6545s, one Agilent 6530, and one Agilent 6520 x1) coupled to Agilent 1290 Infinity II UHPLC instruments. The four Agilent 6545 instruments represent the metabolomics core lab's primary "data production" instruments for non-targeted metabolomics, delivering high throughput and built-in redundancy to allow work on multiple large projects in parallel. These instruments provide high sensitivity, high mass accuracy, high mass range, and rapid throughput ideal for metabolomic profiling. They also have MS/MS capabilities, making them suitable for compound identification and ID confirmation as well as relative and absolute quantitative measurements. They are configured with throughput in mind, having capacity to load and analyze hundreds of samples in sequence without additional human intervention (aside from routine data monitoring and QC during acquisition, which is an important component of our standard operating procedure).
- One Thermo Orbitrap ID-X mass spectrometer coupled to a Thermo Vanquish ultra-high pressure (1500 bar) LC system. This instrument provides unparalleled capabilities for small molecule detection, characterization, and identification. These capabilities include very high resolution ($R=500,000$) and mass accuracy (sub-1 ppm with internal reference ion correction enabled) for accurate molecular formula determination, multi-stage fragmentation (MS^n) for structural characterization, flexible experimental workflow including fragmentation by HCD or CID, and data collection either via orbitrap or linear ion trap. This instrument is primarily dedicated to compound identification-focused projects, but can be used as needed for data production and routine untargeted metabolomics sample analysis.
- Five Agilent Triple Quadrupole mass spectrometers (three Agilent 6490s, 2 Agilent 6410s) coupled with Agilent 1290 UHPLC instruments. These instruments offer the highest possible sensitivity for targeted metabolomics when operated in multiple reaction monitoring (MRM) mode.
- One ABSciex 5600 Triple TOF mass spectrometer coupled to a Shimadzu UHPLC. This instrument serves as our primary production platform for non-targeted lipidomics analysis. Our Agilent 6545 instruments and Orbitrap ID-X can also be used for lipidomics assays as needed.
- One ABSciex QTRAP 5500+ mass spectrometer equipped with a SeleXION differential mobility source, coupled to a Shimadzu UHPLC system. This instrument is useful for quantitative analysis of targeted metabolites and has the capability of differential ion mobility which can help resolve structural isomers and other isobaric species.

Three GC-based mass spectrometers:

- Three Agilent 6890 gas chromatograph (GC) instruments equipped with HP5973 mass -selective detectors. These GC/MS systems can be configured with electron impact ionization (EI) or chemical ionization (CI) sources and operated in positive or negative ion mode. These instruments are useful for analysis of volatile compounds, and can be used for more general metabolomics analysis after chemical derivatization of a sample to increase volatility of its otherwise non-volatile metabolites.

Metz laboratory, Pacific Northwest National Laboratory (PNNL)

Data storage, management and processing infrastructure

At PNNL, all mass spectrometers are connected on a common internal computer network with direct high-speed access to the in-house developed (and continually refined) Pan-Omics Research Information Storage and Management infrastructure. This infrastructure effectively captures, manages, stores, and retrieves the MS data. It also hosts a suite of commercial, open-source (e.g., FragPipe, MaxQuant), or PNNL-developed informatics tools (e.g., MSGF, MASIC, A-score) for automated, streamlined protein identification and quantification, and pre-processing steps for metabolomics and lipidomics data. These tools, plus several other infrastructure-related software tools and the implemented QA/QC performance metrics developed previously, allow the instrumentation staff and investigators to monitor and maintain the instrumentation to achieve a high degree of reliability and reproducibility. Currently, the PNNL omics data infrastructure is handling >5 terabytes of proteomics, metabolomics and lipidomics data per month as of January 2022, with the ability to easily release >200 terabytes of data from the archive. The open structure of the infrastructure also allows for implementing software updates or new informatics tools.

General lab equipment

This project will utilize extensive instrumentation capabilities and facilities already available and functioning in the Environmental Molecular Sciences Laboratory (EMSL) and Biological Sciences Facility (BSF), including a full range of facilities for cell culture and imaging, sample preparation, separations (liquid chromatography, capillary electrophoresis, capillary isoelectric focusing, size exclusion chromatography, 2D gel electrophoresis, etc.), storage, chemical synthesis, and derivatization, and wet chemistry including RT-PCR and western blot. Other relevant resources include the data processing, analysis and management infrastructure that were developed at PNNL to handle high-throughput acquisition and processing of separation and mass spectral data as well as their derived information, collect and maintains associated metadata, and provide cross platform web-based access to researchers.

Specialized lab equipment:

Liquid chromatography systems and capabilities

Twenty commercial nano-flow UPLC systems (Waters), five micro-flow LC systems (Agilent Technologies), six custom-built high-pressure LC systems, five LC fractionation systems with three high-flow (0.5-5 mL/min, Agilent Technologies) and two micro-flow (2-100 μ L/min, Waters) for biomolecular separations are also available. Advanced biological separation capabilities are available for producing unique and sophisticated biological separations systems for the following high-pressure régimes: high-pressure liquid chromatography (HPLC; ≤ 5000 psi); very high-pressure LC (VHPLC; $\sim 10,000$ psi); and ultra-high-pressure LC (UHPLC; $\sim 20,000$ psi). Additional capabilities are available for custom-packing LC columns in a wide range of lengths (10 to 2000 mm) and inner diameters (10 μ m to 1 mm) with two high-pressure packing systems ($\sim 20,000$ psi), and for a wide range of chemistries and separations modes (acidic and basic pH reversed-phase, ion-exchange, size-exclusion, affinity chromatography, etc.). LC systems for 1-D and 2-D proteomics separations, on-line sample clean-up, on-line concentration, and ultralow-flow nanoLC/MS analysis of proteolytic digests have been developed along with unique systems automation and integration capabilities.

Thirty-eight LC-based mass spectrometers:

The project has sufficient access to state-of-the-art validated platforms to meet the needs of the planned analyses in the study. The fully installed, validated commercial mass spectrometers at PNNL for metabolomics lipidomics and proteomics include:

- Two ThermoScientific Orbitrap Eclipse Tribrid mass spectrometers and three Orbitrap Fusion Lumos Tribrid mass spectrometers. The latest generation Orbitrap Eclipse instrument is the industry-leading high-performance mass spectrometer with enhanced sensitivity facilitated by a new high-capacity transfer tube, electrodynamic ion funnel, advanced quadrupole technology, advanced vacuum technology, electron-transfer dissociation HD, and real-time database search MS3. Both the Eclipse and Lumos instruments are designed to expand performance in advanced proteomics and lipidomics applications, including low level PTM or other modification analysis, data independent acquisition (DIA), off-set triggering parallel reaction monitoring (SureQuantTM), multiplexed quantitation using isobaric tags, and top-down proteomics. The equipped FAIMS Pro device makes the instrument even more effective in high-duty cycle, high-precision analysis using isobaric labeling such as the TMT reagents.
- Three ThermoScientific TSQ Altis, four ThermoScientific TSQ Vantage, and one Agilent Technologies triple quadrupole mass spectrometers. These instruments are primarily used for high throughput

- quantitative targeted proteomics measurements applying selected reaction monitoring (SRM) strategies.
- Seven ThermoScientific Q-Exactive mass spectrometers including two Q-Exactive HF and one Q-Exactive HF-X. These Q Exactive instruments combine high-performance quadrupole precursor selection with high-resolution, accurate-mass (HR/AM) Orbitrap detection. The superior quality of Q Exactive MS/MS data provides ultimate confidence for a wide range of qualitative and quantitative applications. Its high scan speed and spectral multiplexing capabilities make it fully compatible with UHPLC and fast chromatography techniques, ramping productivity to a new level. The Q Exactive instruments can easily handle routine applications in proteomics from bottom-up protein ID to semiquantitative isotopic labeling analyses to targeted quantification experiments, and their superior quality of their HR/AM data provides an improved degree of confidence for peptide sequencing.
- Five ThermoScientific LTQ-Orbitrap Velos hybrid mass spectrometers. These instruments combine the proven mass accuracy and ultra-high resolution of the Orbitrap mass analyzer, with the increased sensitivity and improved cycle time of the LTQ Velos. Each instrument has dual-pressure linear trap, HCD collision cell and multiple fragmentation techniques, including ETD, for optimal structural characterization.
- One Agilent Technologies 21 tesla (one of the three world's largest magnets) and two Bruker Solarix 15 tesla high-performance FTICR mass spectrometers. The ultrahigh magnetic field mass spectrometer is expected to result in significant improvement in mass accuracy, resolving power (or resolution), dynamic range, and mass range for characterization of larger intact proteins. A conventional Q-q front end adds a facile and efficient precursor ion selection and dissociation for on-the-fly data-dependent MS/MS analysis.
- Six Agilent Technologies Q-TOF mass spectrometers. The new generation of Q-TOF instruments with high mass accuracy offer higher resolving power and sensitivity that is on the order of five times higher than Agilent earlier Q-TOF instruments. The enhanced mass resolution and mass accuracy are achieved by new Agilent high voltage pulser and power supplies. New, fast autotune improves detection sensitivity, and new ion optics elements provides great robustness.
- Four Agilent Technologies Ion Mobility Q-TOF mass spectrometers. These ion mobility Q-TOF instruments coupled with UHPLC provide unrivaled separation and selectivity by combining chromatography, ion mobility and mass spectrometry. In addition to separation capacity, structural information that cannot be seen with traditional high-resolution LC-MS can be uncovered. The design features innovative electrodynamic ion funnel technology to dramatically improve sensitivity while maintaining an attractive low field drift tube design. This allows for direct calculation of collision cross section (CCS) as well as more complete control and preservation of labile targets.

All mass spectrometers are coupled to dedicated LC systems. The analyses using these LC and MS instruments, including sample analysis, data acquisition and data capture in the data infrastructure (described below), are fully automated, and can be carried out 24/7.

Two GC-based mass spectrometers:

- Two Agilent 5975C GC/MSD Gas Chromatography-Mass Selective Detector System. These mass spectrometers operate in electron ionization mode and are coupled with Agilent high performance 7890A GCs. These systems are routinely used for untargeted metabolomics measurements and can also be operated in SRM mode for targeted compound analyses. The NIST MS library and Agilent Fiehn Metabolomics Retention Time Locked Library are utilized with these instruments to facilitate compound identification.

RESEARCH & RELATED Senior/Key Person Profile (Expanded)

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			Fiehn_biosketch_R01_LC_BB_09_231041564443.pdf	
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			Wohlge-muth_biosketch_R01_LC_BB_9_231041564441.pdf	
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		Metz_R01_Letter_of_support1041564447.pdf		

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		File Name	
Attach Biographical Sketch*:		Evans_Biosketch_R01_LC_BinBase9_231041564439.pdf	
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BIOGRAPHICAL SKETCH

NAME: Oliver Fiehn

eRA COMMONS USER NAME OLIVERFIEHN

POSITION TITLE: Professor

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE	Completion Date	FIELD OF STUDY
Berlin Free University, Germany	Diploma	1993	Analytical Chemistry
Berlin Technical University, Germany	PhD	1997	Analytical Toxicology
Max-Planck Institute for Molecular Plant Physiology, Germany	Postdoc	1998-99	Metabolomics

A. Personal Statement

I am a faculty member of the UC Davis Genome Center since 2005. As investigator in metabolism, I have a long history in collaborating with biomedical researchers (see Contributions to Science) while advancing informatics and analytical chemistry. As Director of the UC Davis West Coast Metabolomics Center I enjoy improving metabolomics methods and collaborative activities, from study designs to data acquisition, data processing, compound identifications, databases, and discussing findings and disseminating our results to the broader community. I have worked for 10 years in the NIH Common Funds Metabolomics Consortium, including in tight collaborations with co-PIs Metz and Evans. Therefore, I am highly committed and experienced to interact and engage with diverse scientists and lead this multi-PI R01 project.

To best assist a diversity of biomedical and clinical researchers, I have focused on informatics and analytical chemistry advancements over the entire path of my career. I have published a range of studies using or advancing Machine Learning tools, including for retention time prediction (PMC8715951), ion mobility (PMID30096227), data normalization (PMC9652764, PMC10456436) and disease classifications. Today, I supervise a team of four software engineers, headed by team leader Gert Wohlgemuth with whom I work for over 20 years now. These staff maintain our diverse sets of databases and software that we have produced, and ensure that program environments are frequently updated and remain functional and interactive with extramural databases. Scientific coding and proof-of-concept software is developed by a team of cheminformatics researchers, led by Project Scientist Dr. Yuanyue Li, who leads both informatics and computational chemistry scientists (Dr. Huaxu Yu, Delilah Milner, Jesi Lee, and Fanzhou Kong). I am strongly committed to keep this balance between software and analytical chemistry developments, led by Project Scientist Dr. Tong Shen who oversees the mass spectrometry work by chemists (Dr. Min Liu, Dr. Uri Keshet, Lauren Bishop, Yendry Carvajal). I also oversee the West Coast Metabolomics Center service core with > 40,000 samples in 400 biomedical projects per year, giving ready access to input and needs by the clinical and biomedical community. I foster engagement and tight integration of cheminformatics, analytical chemistry and collaboration with biomedical scientists (for example, through the service core but also through my own collaborative grants) by weekly research seminars, RapidFire meetings, a large professional training course program and active engagement with the metabolomics community, including the Metabolomics Association of North America (MANA) and ASMS.

Ongoing and recently completed projects that I would like to highlight include:

NIH U2C ES030158 Fiehn, O (PI) 9/01/2018 – 6/30/2024 NCE

The West Coast Metabolomics Center

Role: PI

NIH R03 OD034497 Fiehn, O (PI) 09/01/2022-8/31/2024 NCE

Integrating metagenomics data into accurate mass stool metabolite identifications.

Role: PI

NIH U54DA049113 Olivier, M (PI) 09/01/19 to 10/30/2026

Integrated Omics Analysis of Pain: Omics Data Generation Center

Role: co-I

B. Positions and Honors

Positions and Employment

1994-1997	Research scientist, Technical University of Berlin, Germany
1998-1999	Postdoctoral researcher, Max-Planck Inst. Molecular Plant Physiology, Potsdam, Germany
1999	Visiting postdoctoral research scientist, Dept. Molecular Biol., University of Wash., Seattle, WA
1999-2004	Research group leader, Max-Planck Inst. Molecular Plant Physiol., Potsdam, Germany
2004-2009	Assoc. Professor, Dept. of Molecular & Cellular Biology, UC Davis Genome Center, CA
2009-present	Full Professor, Dept. of Molecular & Cellular Biology, UC Davis Genome Center, CA
2012-present	Director, UC Davis West Coast Metabolomics Center

Other Experience, Honors and Professional Memberships

2017-2022	Board of Directors, Treasurer, Metabolomics Association of North America
2014	Molecular & Cellular Proteomics Lecture Award
2014	Metabolomics Society Lifetime Achievement Award
2013-present	California EPA, Biomonitoring California, Scientific Guidance Panel member
2012	Phi Lambda Upsilon Rho –Chapter award, Lincoln, NE
2010-present	Editorial board member “Rapid Communication in Mass Spectrometry”
2009	Distinguished Scientist Award, CCDG, Chicago
2004-2010, 2013-15	Board of Directors, Metabolomics Society
2006-2011	Editorial Board ‘Journal of Biological Chemistry’
2008, 2009, 2015	Organizer, Metabolomics Society Conference, Boston, Edmonton, San Francisco
2004-present	Editorial board member ‘Metabolomics’

C. Contributions to Science

Early contributions 1994-1998 were geared at analytical chemistry and toxicology, specifically in finding unknown organics that exerted toxic effects. Since 2004, I have continued to lead the field of metabolomics, especially for cheminformatics, and its application in biology and medicine as Professor at UC Davis.

Lipidomics databases and lipid atlases. The Fiehn laboratory was the first to recognize and exploit the regular pattern in lipidomics MS/MS spectra as a general tool for compound identification, generating the LipidBlast library (Kind et al., 2013). We validated untargeted lipidomics as quantitative tool across nine different mass spectrometers (Cajka et al, 2018), merging targeted with untargeted approaches to increase robustness and quantification. We have used these resources in generating major atlases, including over 2 million lipid MS/MS spectra as part of our MS-DIAL software (Tsugawa et al., 2020) that we have integrated into MassBank of North America (MassBank.us). We have used these resources in major biological projects such as the mouse metabolome atlas (Ding et al., 2021)

1. Kind T, Liu K-H, Lee DY, DeFelice B, Meissen JK, **Fiehn O** (2013) LipidBlast in silico tandem mass spectrometry database for lipid identification. *Nature Methods* 10: 755-758 PMC3731409
2. Cajka T, Smilowitz JT, **Fiehn O**. (2017) Validating quantitative untargeted lipidomics across nine liquid chromatography–high-resolution mass spectrometry platforms. *Analytical Chemistry* 89(22):12360-8
3. Tsugawa H, Ikeda K, Takahashi M, Satoh A, Mori Y, Uchino H, Okahashi N, Yamada Y, Tada I, Bonini P, Higashi Y, Okazaki Y, Zhou Z, Zhu ZJ, Koelmel J, Cajka T, **Fiehn O**, Saito K, Arita M. (2020) A lipidome atlas in MS-DIAL 4. *Nature Biotechnology* 38(10):1159-63
4. Ding J, Ji J, Rabow Z, Shen T, Folz J, Brydges CR, Fan S, Lu X, Mehta S, Showalter MR, Zhang Y, Araiza R, Bower LR, Lloyd KC, **Fiehn O** (2021) A metabolome atlas of the aging mouse brain. *Nature Communications* 12(1):6021

Cheminformatics for compound identification. We have focused for a long time on accurate mass MS/MS based workflows for identifying the hundreds of unknown metabolic signals in cells, tissues or biofluids (Lai et al, 2018). Here, we now employ workflows that including machine-learning based of retention times (Bonini et al., 2020) and developed a new algorithm to improve False Discovery Rates in MS/MS based library matching (Li et al., 2021). We also showed that remaining unidentified compounds can be mapped to chemical structure classes using hybrid-search algorithms (Blaženović et al. 2019).

1. Lai Z, Tsugawa H, Wohlgemuth G, Mehta S, Mueller M, Zheng Y, Ogiwara A, Meissen J, Showalter M, Takeuchi K, Kind T, Arita M, **Fiehn O** (2018). Identifying metabolites by integrating metabolome databases with mass spectrometry cheminformatics. *Nature Methods* 15:53. PMC6358022

2. Bonini P, Kind T, Tsugawa H, Barupal DK, **Fiehn O.** (2020) Retip: retention time prediction for compound annotation in untargeted metabolomics. *Analytical Chemistry* 92(11):7515-22
3. Li Y, Kind T, Folz J, Vaniya A, Mehta SS, **Fiehn O.** (2021) Spectral entropy outperforms MS/MS dot product similarity for small-molecule compound identification. *Nature Methods* 18(12):1524-31
4. Blaženović I, Kind T, Sa MR, Ji J, Vaniya A, Wancewicz B, Roberts BS, Torbašinović H, Lee T, Mehta SS, Showalter MR, **Fiehn O.** (2019) Structure annotation of all mass spectra in untargeted metabolomics. *Analytical Chemistry*. 2019 Jan 4;91(3):2155-62

Improving reproducibility and interpretability of metabolomics. Our specific aims continue a trajectory of methods in data processing, normalization, statistics and genomics integration that we have followed over many years. We showed that residual technical errors in metabolomics and lipidomics datasets can be reduced to 3% RSD when using machine learning models on quality controls that are regularly measured in large cohort studies (Fan et al., 2019). We also generated public interfaces to translate chemical identifiers and chemical names to genomic database identifiers in batch mode (Wohlgemuth et al., 2010). To improve biological interpretations of metabolomic signatures we established chemical set enrichment software, ChemRICH, that empowers enrichment statistics for all identified metabolites instead of compounds found only in enzymatic databases (Barupal and Fiehn, 2017). Finally, we integrated our statistical and visualization tools into a combined toolbox for improved interpretation of multi-omics data (Wanichthanarak et al, 2017).

1. Fan S, Kind T, Cajka T, Hazen SL, Tang WW, Kaddurah-Daouk R, Irvin MR, Arnett DK, Barupal DK, **Fiehn O.** (2019) Systematic error removal using random forest for normalizing large-scale untargeted lipidomics data. *Analytical Chemistry*. 2019 Feb 13;91(5):3590-6.
2. Wohlgemuth G, Haldiya PK, Willighagen E, Kind T, **Fiehn O.** (2010) The Chemical Translation Service—a web-based tool to improve standardization of metabolomic reports. *Bioinformatics* 26(20): 2647-8.
3. Barupal DK, **Fiehn O** (2017) Chemical Similarity Enrichment Analysis (ChemRICH) as alternative to biochemical pathway mapping for metabolomic datasets. *Scientific Reports* 7:14567. PMC5673929
4. Wanichthanarak K, Fan S, Grapov D, Barupal DK, **Fiehn O.** (2017) Metabox: A toolbox for metabolomic data analysis, interpretation and integrative exploration. *PLoS one*. 12(1):e0171046

Large scale data processing and databases to study primary metabolism. Biomedical metabolomics in the Fiehn laboratory started with studies in primary metabolism in ovarian tumors (Denkert et al., 2006) which quickly led to a range of additional studies in diabetes, cardiovascular health and related fields. Early on, we recognized the need for computational infrastructure to compare data across studies. In 2005, we developed a schema to track all studies in the UC Davis metabolomics facility including their biological designs (species, organs treatments) in the miniX database (Scholz et al., 2005) and joined it with an automatic data processing and database approach for GC-TOF MS primary metabolism data, BinBase (Kind et al., 2009). We also took a leading role in formulating minimal standards for metabolomics databases and data reports by chairing the Metabolomics Standards Initiative (Fiehn et al., 2007). Today, this GC-MS (primary metabolism) centric BinBase database hosts more than 200,000 samples of more than 2,800 studies.

1. Denkert C, Budczies J, Kind T, Weichert W, Tablack P, Sehouli J, Niesporek S, Konsgen D, Dietel M, **Fiehn O.** (2006) Mass spectrometry–based metabolic profiling reveals different metabolite patterns in invasive ovarian carcinomas and ovarian borderline tumors. *Cancer Research*;66(22):10795-804
2. **Fiehn O**, Wohlgemuth G, Scholz M. (2005) Setup and annotation of metabolomic experiments by integrating biological and mass spectrometric metadata. In: *International Workshop on Data Integration in the Life Sciences*. (pp. 224-239). Springer, Berlin, Heidelberg
3. Kind T, Wohlgemuth G, Lee DY, Lu Y, Palazoglu M, Shahbaz S, **Fiehn O.** (2009) FiehnLib: mass spectral and retention index libraries for metabolomics based on quadrupole and time-of-flight gas chromatography/mass spectrometry. *Analytical Chemistry*;81(24):10038-48
4. **Fiehn O**, Robertson D, Griffin J, van der Werf M, Nikolau B, Morrison N, Sumner LW, Goodacre R, Hardy NW, Taylor C, Fostel J. et al. (2007) The metabolomics standards initiative (MSI). *Metabolomics*. 3:175-8

Collaboration with biomedical researchers. While my laboratory does not initiate biomedical studies, we frequently collaborate with diabetes, cardiovascular health, neurobiology, aging and cancer researchers at US-wide institutions. For example, we associated a peripheral gut microbiome-derived metabolite with altered neural processing and anxiety in humans, including its functional magnetic resonance imaging-based neurologic signature (Brydges et al. 2021). We also showed that that pancreatic islet autoantibodies against GAD or insulin in the development of type 1 diabetes were preceded by different patterns of dehydroascorbate, GABA, multiple amino acids, and fatty acids (Li et al. 2020). In cancer research, we collaborated with Ma et al. (2020) to show

that nanomedicine drugs were highly effective in inducing lysosomal disruption, lysosomal dysfunction and autophagy blockade. We interacted with a team of aging researchers to find that altered lipid metabolism is associated with biological aging and provide novel insights on the relationship between dyslipidemia, biological aging, and age-related diseases (Subedi et al. 2022)

1. Brydges CR, **Fiehn O**, Mayberg HS, Schreiber H, Dehkordi SM, Bhattacharyya S, Cha J, Choi KS, Craighead WE, Krishnan RR, Rush AJ, Dunlop BW, Kaddurah-Daouk R. Indoxyl sulfate, a gut microbiome-derived uremic toxin, is associated with psychic anxiety and its functional magnetic resonance imaging-based neurologic signature. *Scientific reports*. 2021; 11: 21011
2. Li Q, Parikh H, Butterworth MD, Lernmark Å, Hagopian W, Rewers M, She JX, Toppari J, Ziegler AG, Akolkar B, **Fiehn O**, Fan S, Krischer JP. Longitudinal metabolome-wide signals prior to the appearance of a first islet autoantibody in children participating in the TEDDY study. *Diabetes*. 2020 Mar 1;69(3):465-76
3. Ma Z, Li J, Lin K, Ramachandran M, Zhang D, Showalter M, De Souza C, Lindstrom A, Solano LN, Jia B, Urayama S, Duan Y, **Fiehn O**, Lin TY, Li M, Li Y. (2020) Pharmacophore hybridisation and nanoscale assembly to discover self-delivering lysosomotropic new-chemical entities for cancer therapy. *Nature Communications*. 11(1):1-2
4. Subedi P, Palma-Gudiel H, **Fiehn O**, Best LG, Lee ET, Howard BV, Zhao J. Lipidomics profiling of biological aging in American Indians: the Strong Heart Family Study. *GeroScience*. 2022: doi.org/10.1007/s11357-022-00638-9

Complete List of Published Work in MyBibliography: (h-index 119, i10-index 395)

<https://www.ncbi.nlm.nih.gov/myncbi/oliver.fiehn.1/bibliography/public/>

BIOGRAPHICAL SKETCH

NAME: Gert Wohlgemuth

eRA COMMONS USER NAME: GERTWOHLGEMUTH

POSITION TITLE: Applications Programmer IV

EDUCATION / TRAINING

INSTITUTION AND LOCATION	DEGREE	Completion Date	FIELD OF STUDY
Institute for Learning Systems, Hamburg, Germany	Java Programmer	2002	Java Programming
Lise-Meitner School, Berlin, Germany	Chemist	1998-2002	Chemistry

A. Personal Statement

I started working in cheminformatics with Prof. Oliver Fiehn back in Germany in 2002 and built the cheminformatics programs of the UC Davis West Coast Metabolomics Center from the ground up, starting in 2004 with the GC-BinBase automated data processing system. Today, I am leading a team of four professional programmers and engage in weekly scientific meetings with the Fiehn laboratory and the West Coast Metabolomic Center working groups. **Over the past three years, I have developed the AWS cloud data processing LC-BinBase database** that led to a major improvement in consistency and comparability of metabolomic datasets. Together with my team I have built 18 databases and resources over the years, all still in use, namely the MS-DIAL data processing software, the BinVestigate public utility, the MassBank.us public mass spectral resource, the ClassyFireBatch ontology tool, the Chemical Translation Service (CTS) resource and many other projects. I will now use that wealth of experience to improve code to integrate our software tools into a common pipeline, from library MS/MS reports (in MassBank.us) to retention time predictions (using Retip.app) and metabolite use in genomics DBs (in Reactome, rhea and KBase).

For the last 10 years, I have gained valuable management skills for leading a team of individuals and integrating different aspects of code, from unit testing to static and dynamic functionality tests and many more across several frameworks and programming languages. I am fluent in major programming languages like Java, Scala, Groovy, Python, C/CPP and their related major frameworks (Spring, Hibernate, spark, pandas, numpy, sci-py), as well as major database systems like Postgres, Redis and DynamoDB. I have more than 20 years of experience with distributed application (e.g. J2EE) and database design as well as being focused on horizontally scalable solutions, based on messaging protocols and interprocess communication techniques. I am fully committed to continue to oversee and integrate software developments by my team, specifically Nolan Guzman (MassBank.us), Mr. Pedrosa (CTS, MS-DIAL) and Mr. Rundle (front ends) and perform monthly code reviews. Working with Dr. Yuanyue Li, Dr. Tong Shen and Fanzhou Kong I will further integrate MassWiki and Retip 2.0 and newly developed algorithms into the LC-BinBase DB environment. I will be responsible for data and code sharing and implement consistent data standards and data security protocols. I am looking forward to interact with the teams led by Prof. Evans and Dr. Metz on data operability, and integrating new algorithms into LC-BinBase such as U Michigan's Binner and PNNL's ion mobility data processing approaches!

B. Positions and Honors**Positions and Employment**

1998-2002 Chemist, Schering AG, Berlin, Germany
 2002-2004 Programmer, Max-Planck Institute for Molecular Plant Physiology, Potsdam, Germany
 2004-2005 Programmer Junior Specialist I, UC Davis Genome Center, CA
 2006-2008 Programmer Junior Specialist II, UC Davis Genome Center, CA
 2008-2012 Programmer Assistant Specialist II, UC Davis Genome Center, CA
 2012-2014 Programmer Specialist III, UC Davis Genome Center, CA
 2015-2017 Lead Programmer III, UC Davis Genome Center, CA

Other Experience, Honors and Professional Memberships

2017- Member, Metabolomics Association of North America
2008- Member, Metabolomics Society
2012- Member, American Society for Mass Spectrometry
2002-2004 Max Planck Fellow, MPI-MP, Potsdam, Germany

C. Contributions to Science

Although my main role is in developing databases and data processing support for science, I enjoy working in collaborative team science. My current h-index is 24, and I have contributed to 26 papers with more than 10 citations each, yielding a total of 4,213 citations.

Development of databases for mass-spectrometry based metabolomics. The Fiehn laboratory has pioneered metabolomics since 1998. I have led the efforts to build automatic tools for peak annotations and assorting processed data into large databases, the GC-BinBase repository. Published in conjunction with our laboratory information management system miniX, this architecture was published in 2005 and is still in use today, holding more than 200,000 samples (Fiehn et al., 2005). I have soon noticed the need to link annotated metabolites to database identifiers from a variety of sources and built the Chemical Translation Service that cross-references identifiers for more than 200 chemical and biochemical databases using International Chemical Identifier keys (Wohlgemuth et al., 2010). The success of this service gave the idea to build similar identifier keys to facilitate a seamless exchange of mass spectra between international databases, the Spectral Hash Key SPLASH (Wohlgemuth et al., 2016). SPLASH keys are now the de-facto norm for all small molecule mass spectral libraries (Kind et al., 2018).

1. Fiehn O, **Wohlgemuth G**, Scholz M. (2005) Setup and annotation of metabolomic experiments by integrating biological and mass spectrometric metadata. *Data Integration in the Life Sciences, Proceedings*, 3615: 224.
2. **Wohlgemuth G**, Haldiya PK, Willighagen E, Kind T, Fiehn O. (2010) The Chemical Translation Service - a web-based tool to improve standardization of metabolomic reports. *Bioinformatics* 26: 2647
3. **Wohlgemuth G**, Mehta SS, Mejia RF, Neumann S, Pedrosa D, Pluskal T et al. (2016) SPLASH, a hashed identifier for mass spectra. *Nature Biotechnology* 34: 1099
4. Kind T, Hiroshi T, Cajka T, Ma Y, Lai Z, Mehta SS, **Wohlgemuth G**, Barupal DK, Showalter M, Fiehn O. (2018) Identification of small molecules using accurate mass MS/MS search, *Mass spectrometry Reviews* 37: 513

Development of high quality tools for compound identification and metabolite mapping. The success of metabolomics is not only based on data processing tools and organizing spectra into repositories, but also on amassing spectra for compound annotations and subsequent mapping to metabolite clusters. I was instrumental in building the high-quality Fiehnlib repository for over 1,000 authentic chemicals (Kind et al., 2009) that today has grown to the much larger public MassBank.us repository with over 650,000 mass spectra associated with identified metabolites. I have collaborated with guest scientist Hiroshi Tsugawa in building the BinVestigate public gateway into the BinBase database to prioritize unknowns and identify these compounds using accurate mass and substructure identification workflows that integrated MS-DIAL data processing with MS-FINDER software (Lai et al., 2018). I have built workflows for quality control in MS-based metabolomics that utilized for the first time the MSI-annotation confidence scheme proposed the Metabolomics Society (Fiehn et al. 2008). I have also helped developing a system to map unknown metabolites based on their mass spectral similarities in proximity with identified compounds, to cluster biochemical relationships in MetaMapp (Barupal et al. 2012). This idea laid the foundation for today's much enlarged GNPS network system built by the Dorrestein laboratory.

1. Kind T, **Wohlgemuth G**, Lee DY, Lu Y, Palazoglu M, Shahbaz S, Fiehn O. (2009) FiehnLib: Mass spectral and retention index libraries for metabolomics based on quadrupole and time-of-flight gas chromatography/mass spectrometry. *Analytical Chemistry* 81(24): 10038-10048.
2. Lai Z, Tsugawa H, **Wohlgemuth G**, Mehta SS, Mueller M, Zheng Y, Ogiwara A, Meissen J, Showalter M, Takeuchi K, Kind T, Arita M, Fiehn O. (2018) Identifying metabolites by integrating metabolome databases with mass spectrometry cheminformatics. *Nature Methods* 15:53. PMID: 29176591 PMCID: PMC6358022

3. Fiehn O, **Wohlgemuth G**, Scholz M, Kind T, Lee DY, Lu Y, Moon S, Nikolau B. (2008) Quality control for plant metabolomics: reporting MSI-compliant studies. *Plant Journal* 53(4): 691.
4. MetaMapp: mapping and visualizing metabolomic data by integrating information from biochemical pathways and chemical and mass spectral similarity. Barupal DK, Haldiya PK, **Wohlgemuth G**, Kind T, Kothari SL, Pinkerton KE, Fiehn O. (2012) *BMC Bioinformatics* 13:1-15

Collaborative work in clinical and biomedical team science projects. The success of metabolomics tools is measured in its applications. I have contributed to a wide range of projects that are currently focusing on large cohort projects (Miao *et al.* 2021), with several other collaborative papers with cancer investigators (Miyamoto *et al.*, 2015). For these studies, I generated data formats that were used to build cross-omics comparisons, including use of the Chemical Translation Service. Yet, apart from clinical studies I also contribute by data processing, quality control and generating tools to project for cell-based studies such as on mitochondrial respiration of long-chain fatty acids (Seifert *et al.*, 2010) or building community projects such as the large mouse knockout phenotyping database, together with the UC Davis mouse biology program (Barupal *et al.*, 2019). These examples demonstrate my interest and aptitude to work in team science, deploying code for programs that assist in data utilization and data interpretation.

1. Miao G, Zhang Y, Huo Z, Zeng W, Zhu J, Umans JG, **Wohlgemuth G**, Pedrosa D, DeFelice B, Cole SA, Fretts AM, [..], Fiehn O, Zhao J. (2021) Longitudinal Plasma Lipidome and Risk of Type 2 Diabetes in a Large Sample of American Indians With Normal Fasting Glucose: The Strong Heart Family Study. *Diabetes Care*, 44:2664
2. Miyamoto S, Taylor SL, Barupal DK, **Wohlgemuth G**, *et al.* (2015) Systemic metabolomic changes in blood samples of lung cancer patients identified by gas chromatography time-of-flight mass spectrometry. *Metabolites* 5 (2), 192-210, 2015
3. Seifert EL, Fiehn O, Bezaire V, Bickel DR, **Wohlgemuth G**, Adams SH, Harper ME. (2010) Long-chain fatty acid combustion rate is associated with unique metabolite profiles in skeletal muscle mitochondria. *PLoS One*, 5(3): e9834
4. Barupal DK, Zhang Y, Shen T, **Wohlgemuth G**, Roberts BS, *et al.* (2019) A comprehensive plasma metabolomics dataset for a cohort of mouse knockouts within the international mouse phenotyping consortium. *Metabolites* 9 (5), 101

Published Code in Github

I have generated public 51 repositories in Github (<https://github.com/berlinguyinca>) that have 52 active followers. As well as contributing and managing the official WCMC Github repo (<https://github.com/metabolomics-us>), which has 30 public repositories. I am an Arctic Code Vault Contributor. In the last year, I have made public 1,498 contributions in Github, 90% of which were commits. My contributions are listed in Github since 2008, including the six most popular projects SpectraHash in C++ with 11 forks (the international hash key to exchange mass spectra across databases) and four python-coded projects including pycuda, backtesting.py, and osra-docker (providing the osra structural recognition software).

Published Code in Bitbucket

Deposited in in <https://bitbucket.org/fiehnlab/workspace/repositories>, I have overseen the generation of 50 active repositories within 7 projects from 2015-2021 including Python REST client for OSSA OpenCL Similarity Search, Stasis result AGregAtor, MS-FLO (to improve the quality of feature lists/peak tables after initial processing), Chemical Translation Service (as RESTful architecture, spring-boot based, with javascript plugin frontend), MGF file parser for AngularJS, a Peak Annotation Software Benchmarking Utility, and MassBank of North America (MoNA; as metadata-centric, auto-curating public MS repository), and others.

Published Docker Images

Located at <https://gallery.ecr.aws/?searchTerm=fiehnlab>, I manage all public and private docker images, for different applications, developed for the WCMC over the last years. To provide the community and contributors with easy access to pre-build ready to run images of our applications.

BIOGRAPHICAL SKETCH**NAME: Tong Shen****eRA COMMONS USER NAME: TSSHEN****POSITION TITLE: Project Scientist****EDUCATION/TRAINING:**

INSTITUTION AND LOCATION	DEGREE	Completion Date	FIELD OF STUDY
University of California Davis	Postdoctoral	02/2019	Metabolomics
Nanyang Technological University, Singapore	Postdoctoral	11/2015	Metabolomics
National University of Singapore, Singapore	PhD	01/2015	Lipidomics, Biological Sciences
Beijing Normal University	BSs	08/2009	Biological Sciences

A. Personal Statement

My training and research interests are in the field of mass spectrometry-based lipidomics and metabolomics. Over the past 11 years, I have applied liquid chromatography-mass spectrometry (LC-MS) methods to deliver quantifiable measures to research projects, to “make measurable what is not so” (Galileo Galilei). That summarizes my two main directions of research interest—improving the state-of-the-art methodologies and applying novel metabolomics approaches to elucidate metabolic phenotypes. I extended my PhD expertise in lipid biochemistry and mass spectrometry at the UC Davis West Coast Metabolomics Center, first as postdoctoral scholar since 2016 and then as senior team leader for analytical chemistry and program coordinator of our NIH Common Funds metabolomics U2C grant since 2019. Through these activities, I am very aware of the needs and the benefits of data sharing, protocol sharing and general activities in developing common reporting formats and discussing and furthering common goals. For instance, I was one of the major drivers in the “Unknown Lipids” and “Unknown polar compound” projects of the NIH Common Funds metabolomics program, in collaboration with Prof. Charles Evans of University of Michigan. Therefore, I am highly committed to assisting the overall aims of this extension towards an automated LC-BinBase database in collaboration between the Fiehn, Evans and Metz laboratories.

In my own research within the Fiehn laboratory, I actively support compound identification through advancing mass spectrometry with novel technologies, for instance, ion mobility spectrometry, high-resolution quantitative parallel reaction monitoring assays, and ultraviolet photoionization. In 2020, I contributed to acquiring our new Bruker timsTOF Pro2 flex instrument through the NIH S10 award mechanism. This instrument now enables us to capture high resolution metabolomic and lipidomic changes in human plasma by giving us more chemical resolution in LC-MS/MS by its added ion mobility capabilities. I am thrilled to use my expertise and this new instrument to associate metabolite changes with differences in *in vitro* microbiota compositions, to oversee the development and application of assays for targeted and untargeted metabolomics data acquisitions, to guide our core staff with respect to designing assays and troubleshooting QA/QC problems, to review data reports and to identify novel compounds that become relevant in the planned studies.

Ongoing and recently completed projects I would like to highlight include:

NIH U2C ES030158

Fiehn, O (PI); Tantillo, D ; Wang, LP; Kind, T.; Shen, T. (co-I)

05/31/2018 – 06/30/2024NCE

The West Coast Metabolomics Center for Compound Identification

NIH S10 OD030253

Fiehn, O (PI); Shen, T. (co-I)

06/15/2021- 05/31/2022

Bruker timsTOF Pro liquid chromatography mass spectrometry (LC-MS) system

NIH R01 HL157535

Byambaa, E (PI), Fiehn, O; Shen, T. (co-I)

09/01/2021 – 08/31/2025

Lp(a) and Oxidized Phospholipids - Impact of Diets

B. Positions and Honors

Positions and Employment

2009-2014	PhD, Department of Biological Sciences, National University of Singapore, Singapore
2009-2011	Teaching Assistant, Department of Biochemistry, National University of Singapore, Singapore
2014-2015	Research Assistant, Life Science Institute, National University of Singapore, Singapore
2015-2016	Research Fellow, Singapore Phenome Center, Nanyang Technological University, Singapore
2016-2019	Postdoctoral Researcher, West Coast Metabolomics Center, University of California, Davis
2019-present	Assistant Project Scientist, West Coast Metabolomics Center, University of California, Davis
2020-present	Program Coordinator, West Coast Metabolomics Center, University of California, Davis

Other Experience and Professional Memberships

2013-	Member, Singapore Society for Mass Spectrometry
2015-	Member, American Society for Mass Spectrometry
2016-	Member, International Metabolomics Society
2018-	Member, Metabolomics Association of North America
2018-2019	Organizer, 4 th and 5 th Annual UC Davis Postdoctoral Research Symposium
2018-present	Manuscript reviews for the journals Nature Communication, Scientific Reports, Journal of Proteome Research, Chemical Research in Toxicology, Phytochemical Analysis, Frontiers Plant Science, Eurasian Journal of Analytical Chemistry, Acta Pharmaceutica Sinica B, GigaScience, and Science of the Total Environment

Honors, Awards, and Scientific Committees

2007	2nd prize in 15th Jingshi Award in Science and Technology, Beijing
2008	3rd prize in 16th Jingshi Award in Science and Technology, Beijing
2009-2013	NUS PhD Scholarship, Department of Education, Singapore
2014	Sing Hai Yi Scholarship, Singapore
2018-2020	UC Davis Postdoctoral Research Symposium organizing committee
2018-	Association of Women in Science member

C. Contributions to Science

During my early career publications, I focused on on proprotein convertase Furin and contributed to molecular mechanisms of cell migration and invasion in human reproductive system (1). My lipidomics studies provided large scale phenotyping to detect molecular modulation. I developed untargeted to targeted liquid chromatography-mass spectrometry methods to characterize phosphatidylethanol, a class of phospholipids produced when cells undergo ethanol stress (2). That methodology led to my first finding on the role of phosphatidylethanol in ethanol metabolism in *S. cerevisiae*. I utilized Matrix Assisted Laser Desorption/Ionization (MALDI) MS on 384-spot plates and achieved high throughput lipidomics screening over 1,000 yeast knockout strains (3).

- 1) Z Zhou, **T Shen**, BH Zhang, XY Lv, HY Lin, C Zhu, LQ Xue, HM Wang. The proprotein convertase furin in human trophoblast: Possible role in promoting trophoblast cell migration and invasion. *Placenta*. **2009**; 30(11):929-938. PMID: 19853298
- 2) T Shen, A Cazenave-Gassiot, M Wenk. Phosphatidylethanol in budding yeast fermentation. 5th International Singapore Lipid Symposium. Singapore. **2014**
- 3) T Shen, A Cazenave-Gassiot, LS Wong, M Wenk. A lipidomics study of two putative lysophosphatidic acid acyltransferases in *Saccharomyces cerevisiae*. 4th International Singapore Lipid Symposium. 4th International Singapore Lipid Symposium. Singapore. **2012**

Since moving to UC Davis, I have led new developments of methods in analytical chemistry, specifically for high resolution MS/MS and ion mobility. Specifically, I established a novel workflow for untargeted LC-ion mobility-MS metabolomics workflow using the Agilent 6560 drift tube ion mobility-QTOF mass spectrometer (1). Around

3-fold more mass/retention time features were detected in lipidomics assays after optimizing ion trapping in the drift cell. I have demonstrated that LC-IM-MS separated metabolites with subtle structural differences, e.g., functioning group-positional isomers of natural products. I contributed to essential findings such as CCS measurement accuracy and increased peak capacity with this technique. High-resolution MS/MS methods were also developed for a ring trial project that demonstrated how lipidomics and metabolomics data can be normalized and compared across laboratories (2). Similarly, I contributed to generating example data sets in mouse plasma KO genotypes to showcase best-in-class data sharing for subsequent data re-use (3), and helped validate sample preparation methods in freezing delay studies (4).

- 1) I Blaženović, **T Shen**, S Mehta, T Kind, J Ji, M Piparo, F Cacciola, L Mondello, O Fiehn. Increasing Compound Identification Rates in Untargeted Lipidomics Research with Liquid Chromatography Drift Time-Ion Mobility Mass Spectrometry. *Anal Chem*. 2018; 90 (18), 758-764. (co 1st author)
- 2) JW Thompson, KJ Adams, J Adamski, Y Asad, D Borts, JA Bowden, G Byram, V Dang, WB Dunn, F Fernandez, O Fiehn, DA Gaul, AF Huhmer, A Kalli, T Koal, S Koeniger, R Mandal, F Meier, FJ Naser, D O'Neil, A Pal, GJ Patti, H Pham-Tuan, C Prehn, FI Raynaud, **T Shen**, AD Southam, LS John-Williams, K Sulek, CG Vasilopoulou, M Viant, CL Winder, D Wishart, L Zhang, J Zheng, MA Moseley. International ring trial of a high resolution targeted metabolomics and lipidomics platform for serum and plasma analysis. *Analytical Chemistry* 91 (22), 14407-14416. 2019. PMID: 31638379
- 3) DK Barupal, Y Zhang, **T Shen**, S Fan, B Roberts, P Fitzgerald, B Wancewicz, L Valdiviez, G Wohlgemuth, G Byram, Y Choy, B Haffner, M R. Showalter, A Vaniya, C S Blois, J S Folz, T Kind, O Fiehn. A comprehensive plasma metabolomics dataset for a cohort of mouse knockouts within the international mouse phenotyping consortium. *Metabolites* 9 (5), 101. 2019 PMCID: PMC6571919
- 4) M La Frano, J Newman, S Carmichael, C Ma, M Hardley, **T Shen**, R Wong, L Rosales, K Borkowski, T Pedersen, G Shaw, Da Stevenson, O Fiehn. Impact of Post-Collection Freezing Delay on the Reliability of Serum Metabolomics in Samples Reflecting the California Mid-term Pregnancy Biobank. *Metabolomics*. 2018, 14: 151. PMID: 30830400

Since 2016 I became the lead analytical developer in the WCMC and steers the group of analytical chemists and graduate students at the WCMC as Project Scientist for novel technologies. I have exploited cutting-edge technologies to aid in metabolomics and collaborated in a range of biological projects using advanced methods in animal models, plant biology and protein-ligand projects. From ten published papers, here are specifically noteworthy examples of my collaborative efforts—deep mapping the metabolome of aging mouse brain (1), MS-based strategies to characterize lipids for probing ligand-protein interactions (2). I also validated chemical structures in mechanisms for diterpene synthases that led to large isoprenoid structures (3) and led the experimental work to establish lipidomic and metabolomics data generation for resolving complex mixtures in gut microbial samples that represent truly challenging matrix interference problems (4)

- 1) Ding J, Ji J, Rabow Z, **Shen T**, Folz J, Brydges CR, Fan S, Lu X, Mehta S, Showalter MR, Zhang Y, Araiza R, Bower LR, Lloyd KCK, Fiehn O. A metabolome atlas of the aging mouse brain. *Nat Commun*. 2021 Oct 15;12(1):6021.
- 2) J Park, ZJ Fu, A Frangaj, J Liu, L Mosyak, **T Shen**, VN Slavkovich, KM Ray, B Cao, Y Geng, H Zuo, Y Kou, R Grassucci, S Chen, Z Liu, W Rice, E Eng, RK Huang, RK Soni, B Kloss, Z Yu, C Potter, BO Carragher, PA Slesinger, WA Hendrickson, M Quick, J Graziano, H Yu, O Fiehn, R Henderson, OB Clarke, J Frank, QR Fan. Cryo-EM structure of human GABAB G protein-coupled receptor. *Nature*. (co 1st author) **2020**
- 3) PS Karunanithi, D Berrios, S Wang, J Davis, **T Shen**, O Fiehn, J Maloof, P Zerbe. The Foxtail Millet (*Setaria Italica*) Terpene Synthase Gene Family. *Plant Journal*. doi: 10.1111/tpj.14771. **2020** PMID: 32282967
- 4) B Gao, Y Duan, S Lang, D Barupal, L Valdiviez, B Roberts, Y Choy, **T Shen**, G Byram, Y Zhang, S Fan, B Wancewicz, Y Shao, K Vervier, Y Wang, T Wu, S Nath, R Loomba, JG Abraldes, R Bataller, X Tu, P Stärkel, TD Lawley, O Fiehn, B Schnabl. Functional microbiomics reveals alterations of the gut microbiome and host co-metabolism in alcoholic hepatitis patients. *Hepatology Communications*. **2020**

Complete List of Published Work in My Bibliography:

<https://www.ncbi.nlm.nih.gov/myncbi/1Llhi-2q3K5kv/bibliography/public/>

BIOGRAPHICAL SKETCHNAME: **Thomas O. Metz**

eRA COMMONS USER NAME : TOM_METZ

POSITION TITLE: Staff Scientist

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE	Completion Date	FIELD OF STUDY
Frostburg State University, Frostburg, MD	B.S.	1997	Biology
San Jose State University, San Jose, CA	B.S.	1999	Chemistry
University of South Carolina, Columbia, SC	Ph.D.	2003	Chemistry
Pacific Northwest National Laboratory, WA	Postdoctoral	2003-2005	Mass Spectrometry

A. Personal Statement

Throughout my scientific career, I have been most interested in the study of disease and the role that molecules play in promoting or resolving disease or healthy states. I have approached this broad interest through the application of mass spectrometry (MS) and primarily advanced MS-based metabolomics, lipidomics, and proteomics (collectively 'omics) techniques. After completing my PhD studies, I began a post-doc at Pacific Northwest National Laboratory (PNNL) where I was trained in advanced MS-based proteomics analyses and developed methods for MS-based metabolomics and lipidomics analyses. Then, I moved beyond chemical analyses and data acquisition towards scientific data analysis, chemometrics and computational prediction. Here, I led the project for using Machine Learning in protomer predictions as well as ISiCLE, a software for ML-based prediction of collisional cross sections in ion mobility.

Currently, I am a Principal Investigator of the Pacific Northwest Advanced Compound Identification Core within the NIH Common Fund Metabolomics Program Phase II, PI of the Proteomics Laboratory for The Environmental Determinants of Diabetes in the Young consortium, and the Lead of the PNNL-funded *m/q* Initiative. I am vested in the advancement of metabolomics and lipidomics technologies, specifically for the development of confidence scores for compound identification. Here, I have very much enjoyed the collaborations with Prof. Fiehn (UC Davis) and Prof. Evans (U Michigan) with respect to developing common standards in data acquisition and data processing. I strongly feel that the field needs harmonization of approaches, and ready-to-use informatics workflows and databases that can be employed by metabolomics laboratories across the United States. The expanse of mass spectrometry-based omics technical expertise and experience in my group combined with the state-of-the-art instrumentation at PNNL – including extensive ion mobility spectrometry-mass spectrometry (IMS-MS) resources – provides a unique environment for evaluating the contribution of IMS-MS in the development of confidence scores for compound identification and unified data processing. I am excited about expanding the collaborations with Fiehn and Evans towards developing and evaluating LC-BinBase, processing data from common materials provided by the Evans lab through LC-BinBase and compare the data to workflows that I have established in my own laboratory to improve overall accuracy and precision of the data analyses.

Ongoing and recently completed projects that I would like to highlight include:

U01 DK127786-01

Mirmira (contact), Eizirik, Evans-Molina, Metz, Ramanadham, Webb-Robertson (MPIs) 01/05/21-06/30/24

The Integrated Stress Response in Human Islets During Early T1D

This effort will provide a multi-omics evaluation of the hypothesis that the activation of the intracellular stress granules are an early cellular response initiating β cell stress in T1D that determines cell survival and can be monitored in pre- and early T1D individuals with minimal invasiveness

Role: PI (multi-PI)

TEDDY Consortium

Smith/Metz (PI)

08/12/14-03/31/22

Proteomics Laboratory for the Environmental Determinants of Diabetes in the Young

The goal of this project is to establish a proteomics laboratory to study human plasma samples from The Environmental Determinants of Diabetes in the Young (TEDDY) study in order to identify biomarkers of developing autoimmunity.

Role: PI

U2CES030170 Metz (PI) 09/01/18-06/30/23

Pacific Northwest Advanced Compound Identification Core

This compound identification core funded within the NIH Common Fund Metabolomics Program Phase II seeks to develop advanced quantum chemistry and machine learning predictions for properties measured by mass spectrometry and that can lead to 'reference-free' metabolomics.

Role: PI

U01 TR001395 Metz (contact), Koeller (MPIs) 09/20/15-04/30/19

Undiagnosed Diseases Network Metabolomics Core

This metabolomics core funded within the NIH Common Fund Undiagnosed Diseases Network (UDN) provided comprehensive analytical methods, analyses, technologies, and metabolomics expertise to the UDN to aid in clinical diagnoses and to investigate potential mechanisms underlying phenotypic changes in patients.

Role: PI

B. Positions, Scientific Appointments, and Honors

Positions and Employment

2014 – Present Metabolomics Team Lead, Integrative Omics Group, PNNL, Richland, WA

2012 – Present Adjunct Faculty, Oregon State University, Corvallis, OR

2005 – Present Staff Scientist, Integrative Omics Group, PNNL, Richland, WA

Other Experience and Professional Memberships

Member, American Chemical Society

Member, American Society for Mass Spectrometry

Member, Metabolomics Society

Founding Member, Metabolomics Association of North America

2021 – present Lead, *m/q* Initiative, PNNL

2021 – present President, Metabolomics Association of North America (MANA)

2021 – 2022 Chair, Environmental Molecular Sciences Laboratory (EMSL) User Executive Committee (UEC)

2020 Young Investigator Awards Committee, MANA 2nd Annual Conference

2020 – 2022 Co-Lead, MANA Microbiome Interest Group

2020 – 2021 Vice-Chair, EMSL UEC

2020 – present Member, EMSL, UEC

2019 – 2021 Chief Scientist, *m/q* Initiative, PNNL

2019 Member, Scientific Organizing Committee, MANA 1st annual conference

2019 – present Board Member, MANA

2018 – present Chair, Compound Identification Development Cores Sub-Committee, NIH Common Fund Metabolomics Program Phase II

2018 – present Member, Steering Committee, NIH Common Fund Metabolomics Program Phase II

2018 Member, Local Organizing Committee, 14th Annual Conference of the Metabolomics Society

2015 – 2018 Member, Steering Committee, Undiagnosed Diseases Network

2015 – 2018 Co-Chair, Metabolomics Working Group, Undiagnosed Diseases Network

2012 – present Scientific Advisory Board – Mass Spectrometry Facility, Oregon State University

2010 Membership Committee of the International Maillard Reaction Society

Honors and Awards

2022 Pathways to Excellence Award, PNNL

2021 Paper of the Year Award, Biological Sciences Division (BSD), PNNL

2019 Paper of the Year Award, BSD, PNNL

2018 Major Proposal Award, PNNL

2017 Paper of the Year Award, BSD, PNNL

2017 Major Proposal Award, PNNL

2017	Nominated for election to Metabolomics Society Board of Directors
2017	Keynote Speaker, Metabolomics in Health and Disease II, 13 th Annual Conference of the Metabolomics Society
2016	Patent Award, PNNL
2014	Patent Award, PNNL
2014	Software Creator Award, PNNL
2012	Environmental Molecular Sciences Laboratory Outstanding Performance Award, PNNL
2008	Fundamental and Computational Sciences Directorate Outstanding Performance Award (2), PNNL
2007	Fundamental and Computational Sciences Directorate Outstanding Performance Award, PNNL
2006	National Security Directorate Outstanding Performance Award, PNNL

C. Contributions to Science

I have authored >180 publications and the most relevant for this proposal are listed below. Co-authorships due to inclusion in study group author lists are not included. The asterisks designate corresponding authorship.

1. Development and application of advanced MS-based omics technologies in the study of type 1 diabetes mellitus. Despite decades of research aimed at identifying the cause of type 1 diabetes, the initiating event leading to autoimmunity and subsequent diabetes remains to be discovered. To address this gap, I have applied high throughput MS-based omics technologies in fundamental characterizations of the human islet in response to glucose stimulation, and in the search for novel biomarkers of type 1 diabetes mellitus. These efforts resulted in the most extensive characterization of the human islet proteome to date, and in the identification of novel protein markers of type 1 diabetes. Most recently, our collaborative work has led to the identification of growth differentiation factor 15 (GDF15) as a novel protective factor for human beta cells. Collectively, these efforts represent some of the first applications of MS-based omics in the study of diabetes mellitus.

Webb-Robertson BM, Nakayasu ES, Frohnert BI, Bramer LM, Akers SM, Norris JM, Vehik K, Ziegler AG, **Metz TO**, Rich SS, Rewers MJ. Integration of Infant Metabolite, Genetic and Islet Autoimmunity Signatures to Predict Type 1 Diabetes by 6 Years of Age. *J Clin Endocrinol Metab.* 2022, dgac225. PMID: 35468213.

Colli ML, Ramos-Rodríguez M, Nakayasu ES, Alvelos MI, Lopes M, Hill JLE, Turatsinze JV, Coomans de Brachène A, Russell MA, Raurell-Vila H, Castela A, Juan-Mateu J, Webb-Robertson BM, Krogvold L, Dahl-Jorgensen K, Marselli L, Marchetti P, Richardson SJ, Morgan NG, **Metz TO**, Pasquali L, Eizirik DL. An integrated multi-omics approach identifies the landscape of interferon- α -mediated responses of human pancreatic beta cells. *Nat Commun.* 2020, 11(1):2584. PMC7244579

Nakayasu ES, Syed F, Tersey SA, Gritsenko MA, Mitchell HD, Chan CY, Dirice E, Turatsinze JV, Kulkarni RN, Eizirik DL, Qian WJ, Webb-Robertson BM, Evans-Molina C, Mirmira RG, **Metz TO***. Comprehensive proteomics analysis of stressed human islets identifies GDF15 as a target for type 1 diabetes intervention. *Cell Metabolism.* 2020, 31(2):363-374.e6. PMID: 31928885.

Ramos-Rodríguez M, Raurell-Vila H, Colli ML, Alvelos MI, Subirana M, Juan-Mateu J, Norris R, Turatsinze JV, Nakayasu ES, Webb-Robertson BM, Inshaw J, Piemonti L, Esteller M, Todd JA, **Metz TO**, Eizirik DL, Pasquali L. The impact of pro-inflammatory cytokines on the β -cell regulatory landscape provides new insights into the genetics of type 1 diabetes. *Nature Genetics.* 2019, 51(11):1588-1595. PMC7040466.

2. Application of advanced MS-based omics technologies in the study of pathogenic viruses and bacteria. Beginning in 2010, I became involved in studies of pathogenic viruses, starting with hepatitis C. This work led to my involvement in a Proteomics and Metabolomics Core for a NIAID Systems Biology for Infectious Disease Research Center (2008-2013). I led the metabolomics efforts on a second NIAID-funded center that focused on the study of the enteropathogens *Salmonella enterica* serovar Typhimurium and *Yersinia pestis*. Subsequently, I was Co-PI for for a NIAID-funded center (2013-2018) that focused on the study of the human viruses influenza, West Nile, MERS, and Ebola. Research performed resulted in some of the first applications of omics in studies of the host response to infection, to elucidate the role of lipids in the host response.

Farley SE, Kyle JE, Leier HC, Bramer LM, Weinstein JB, Bates TA, Lee JY, **Metz TO**, Schultz C, Tafesse FG. A global lipid map reveals host dependency factors conserved across SARS-CoV-2 variants. *Nat Commun.* 2022, 13(1):3487. PMID: 35715395

Leier HC, Weinstein JB, Kyle JE, Lee JY, Bramer LM, Stratton KG, Kempthorne D, Navratil AR, Tafesse EG, Hornemann T, Messer WB, Dennis EA, **Metz TO**, Barklis E, Tafesse FG. A global lipid map defines a network essential for Zika virus replication. *Nat Commun.* 2020, 11(1):3652. PMID: 32694525 PMCID: PMC7374707

Kyle JE, Burnum-Johnson KE, Wendler JP, Einfeld AJ, Halfmann PJ, Watanabe T, Sahr F, Smith RD, Kawaoka Y, Waters KM, **Metz TO***. Plasma lipidome reveals critical illness and recovery from human Ebola virus disease. *PNAS*. 2019, 116(9):3919-3928. PMC6397561.

Einfeld AJ, Halfmann PJ, Wendler JP, Kyle JE, Burnum-Johnson KE, Peralta Z, Maemura T, Walters KA, Watanabe T, Fukuyama S, Yamashita M, Jacobs JM, Kim YM, Casey CP, Stratton KG, Webb-Robertson BJ, Gritsenko MA, Monroe ME, Weitz KK, Shukla AK, Tian M, Neumann G, Reed JL, van Bakel H*, **Metz TO***, Smith RD*, Waters KM*, N'jai A*, Sahr F*, Kawaoka Y*. Multi-platform 'Omics Analysis of Human Ebola Virus Disease Pathogenesis. *Cell Host Microbe*. 2017, 22(6):817-829. PMC5730472.

3. **Development of high throughput MS-based metabolomics and lipidomics approaches.** When I joined PNNL as a post-doctoral research associate in 2003, I was tasked with developing MS-based metabolomics capabilities by leveraging the extensive capabilities already in place for proteomics. It became immediately clear that only some of the proteomics capabilities would be useful for metabolomics. Therefore, I developed de novo metabolomics and lipidomics capabilities and worked to transform these into high throughput modalities. This led to those capabilities being adopted and offered to the scientific community by the PNNL national scientific user facility, the Environmental Molecular Sciences Laboratory (EMSL).

Colby SM, Chang CH, Bade JL, Nunez JR, Blumer MR, Orton DJ, Bloodsworth KJ, Nakayasu ES, Smith RD, Ibrahim YM, Renslow RS, **Metz TO***. DEIMoS: An Open-Source Tool for Processing High-Dimensional Mass Spectrometry Data. *Anal Chem*. 2022, 94(16):6130-6138. PMID: 35430813; PMCID: PMC9047447

Ma J, Casey CP, Zheng X, Ibrahim YM, Wilkins CS, Renslow RS, Thomas DG, Payne SH, Monroe ME, Smith RD, Teeguarden JG, Baker ES*, **Metz TO***. PIXiE: An Algorithm for Automated Ion Mobility Arrival Time Extraction and Collision Cross Section Calculation using Global Data Association. *Bioinformatics*. 2017, 33(17):2715-2722. PMC5860068.

Kyle JE*, Crowell KL, Casey CP, Fujimoto GM, Kim S, Dautel SE, Smith RD, Payne SH, **Metz TO***. LIQUID: an open source software for identifying lipids in LC-MS/MS-based lipidomics data. *Bioinformatics*. 2017, 33(11):1744-1746. PMC5860051.

Nakayasu ES, Nicora CD, Sims AC, Burnum-Johnson KE, Kim YM, Kyle JE, Matzke MM, Shukla AK, Chu RK, Schepmoes AA, Jacobs JM, Baric RS, Webb-Robertson BJ, Smith RD, **Metz TO***. MPLEx: a Robust and Universal Protocol for Single-Sample Integrative Proteomic, Metabolomic, and Lipidomic Analyses. *mSystems*. 2016, 1(3). pii: e00043-1. PMC5069757.

4. **Development and applications of computational and experimental approaches for 'reference-free' metabolomics.** Through innovations in computational chemistry and advanced ion mobility spectrometry (IMS) separations coupled with MS, we are overcoming a significant, long standing obstacle in the field of metabolomics: the absence of methods for accurate and comprehensive identification of metabolites without relying on data derived from analysis of authentic reference compounds. We use molecular properties that can be both predicted computationally with high accuracy and experimentally measured with high precision, without the need for reference libraries constructed through experimental analysis of authentic chemical standards. We are striving to arrive at a metabolomics paradigm that can be fully automated, including methods that estimate and control for metabolite identification false discovery rate.

Borges RM, Colby SM, Das S, Edison AS, Fiehn O, Kind T, Lee J, Merrill AT, Merz KM Jr, **Metz TO**, Nunez JR, Tantillo DJ, Wang LP, Wang S, Renslow RS. Quantum chemistry calculations for metabolomics. *Chem Rev*. 2021, 121(10):5633-5670. PMC8161423

Nielson FF, Colby SM, Thomas DG, Renslow RS*, **Metz TO***. Exploring the impacts of conformer selection methods on ion mobility collision cross section predictions. *Anal Chem*. 2021, 93(8):3830-3838. PMID: 33606495

Nuñez JR, Colby SM, Thomas DG, Tfaily MM, Tolic N, Ulrich EM, Sobus JR, **Metz TO***, Teeguarden JG*, Renslow RS*. Evaluation of in silico multifeature libraries for providing evidence for the presence of small molecules in synthetic blinded samples. *J Chem Inf Model*. 2019, 59(9):4052-4060. PMID: 31430141; PMCID: PMC7206898

Colby SM, Thomas DG, Nuñez JR, Baxter DJ, Glaesemann KR, Brown JM, Pirrung MA, Govind N, Teeguarden JG, **Metz TO***, Renslow RS*. ISiCLE: A quantum chemistry pipeline for establishing in silico collision cross section libraries. *Anal Chem*. 2019, 91(7):4346-4356. PMC6526953.

BIOGRAPHICAL SKETCH

NAME: Evans, Charles Robert

eRA COMMONS USER NAME : CHEVANS1

POSITION TITLE: Associate Research Scientist

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE	Completion Date	FIELD OF STUDY
Lake Forest College, Lake Forest, IL	B.A.	05/2002	Chemistry, Biology
University of North Carolina, Chapel Hill, NC	Ph.D.	12/2007	Analytical Chemistry
University of Michigan, Ann Arbor, MI	Postdoctoral	06/2009	Bioanalytical Chemistry

A. Personal Statement

As the metabolomics and the other 'omics sciences continue to advance in technical capabilities, it is essential that the research community work together to develop and adhere to consistent standards for data generation and reporting. This will allow our methods to be consistently and reliably used for fundamental and applied biological research and will facilitate direct comparison of metabolomics data across laboratories. In the case of metabolomics, this is no trivial task. It requires coordination between scientists throughout the community, who must discuss and agree upon consistent practices to generate reliable data, even as they work toward pushing the capabilities of the techniques they use forward. Such inter-laboratory harmonization requires us to develop, evaluate, and use coordinated analytical methods and, critically, software tools that allow different labs to process data in a consistent manner. This is the goal of the LC-BinBase software tool, which forms the centerpiece of this proposal. Together, in the context of LC-BinBase we will develop, implement and evaluate rigorous practices to collect and process metabolomics data across labs, focusing on the critical tasks of compound identification, quantification, and application to a range of biomedical samples. These aims are consistent with and will be enhanced by my previous research experience, described below.

The focus of my career has been to develop and apply the technique of metabolomics to study the metabolic underpinnings of human health and disease. As an analytical chemist with over 19 years of academic research experience in mass spectrometry and the 'omics sciences, I enjoy the challenge of developing and implementing novel methods to identify and quantitate biological molecules. As a biomedical researcher published in fields including metabolism, exercise science, obesity, and diabetes, I routinely apply these techniques to the study of conditions that directly impact human health. At present, I am a PI of the NIH-funded Michigan Compound Identification Development Core (CIDC). Oliver Fiehn and Tom Metz, the other PIs of this multi-PI proposal, are also heads of CIDC labs; in the context of the NIH Metabolomics Consortium we have had frequent opportunities to work together over the last 5 years. The aim of the CIDCs was to develop and implement novel computational and experimental approaches to tackle the long-standing challenge of identifying the high proportion of unidentified features typically detected in untargeted metabolomics data. I have led two inter-institutional collaborative working groups in this consortium, one pertaining to lipidomics and the other to polar metabolites. The aims of both groups are to assess the extent of overlap in identified and unknown features observed by different labs, and to develop inter-laboratory methods to improve data reproducibility and interoperability. Our findings, which we are in process of publishing, support innovation and harmonization for future metabolomics research, which are directly consistent with the aims of this grant application.

I am also a co-investigator in another multi-institution consortium, the Molecular Transducers of Physical Activity Consortium (MoTrPAC). Working with PIs Dr. Charles Burant and Dr. Jun Li, I am directing operations

of the analytical core of Michigan's Chemical Analysis Site in MoTrPAC. Our primary role in this study is to perform multi-platform metabolomics assays on thousands of samples from human subjects and animals undergoing both acute and long-term exercise. While the goals of MoTrPAC are predominantly biological rather than focused on technology development as in the current proposal, it has strengthened my view of the importance of inter-laboratory collaboration to solve the largest challenges facing science today.

Ongoing and recently completed projects that I would like to highlight include:

NIH U2C CES30164 Evans/Nesvizhskii (Multi-PI) 9/1/2018-5/31/2023

Michigan Compound Identification Development Core (MCIDC)

MCIDC is a member of a consortium of metabolomics labs using a combination of cutting-edge and novel computational and experimental approaches to systematically address the challenges of compound identification in metabolomics to benefit current and future metabolomics research.

Role: PI (Multi-PI)

NIH U24 DK112342 Burant/Li(Multi-PI) 12/8/2016 - 11/30/2023

Michigan MoTrPAC Chemical Analysis Site (MiCAS)

The focus of this consortium is to systematically analyze plasma and tissues collected from human participants and animals undergoing physical activity, developing a map of the molecular transducers of physical activity.

Role: Co-Investigator

NIH R01 DK118631 Jun (PI) 12/01/2019 – 12/01/2023

Translational Metabolomics in Critical Care

This project aims to: 1) increase understanding of the metabolic processes that underlie clinically challenging problems like sepsis; and 2) improve prediction of drug response (pharmacometabolomics), including adverse drug reactions.

Role: Co-Investigator

B. Positions, Scientific Appointments, and Honors

Positions and Employment

2013-	Research Assistant Professor, Department of Internal Medicine, Division of Metabolism, Endocrine and Diabetes, School of Medicine, University of Michigan
2009-2013	Research Investigator, Department of Internal Medicine, Division of Metabolism, Endocrine and Diabetes, School of Medicine, University of Michigan
2007-2009	Postdoctoral Research Fellow, Department of Chemistry, University of Michigan
2002-2007	Research Assistant, Department of Chemistry, University of North Carolina

Other Experience and Professional Memberships

2019-	Member, Metabolomics Association of North America
2014-	Member, American Society for Mass Spectrometry
2013-	Member, Metabolomics Society
2006-	Member, American Chemical Society

Honors

2008	NIH Kirschstein-NRSA Institutional Postdoctoral Training Grant in Endocrinology and Metabolism
2005	Pfizer Graduate Research Fellowship in Analytical Chemistry

C. Contributions to Science

1. Development of novel instrumentation, methodologies, and software tools to detect, identify and quantitate biomolecules and interpret complex data has been critical to enabling many major innovations in biomedical research made over the past decades. Specifically, the advent of 'omics methodologies has enabled more comprehensive and rapid assessment of biological changes associated with health or

disease than was previously possible with classical assays and biochemical techniques. Over the course of my career I have been involved in development and application of new analytical technologies, including refinements to chromatographic methods to enhance compound identification, multidimensional chromatographic separations and ultra-high pressure liquid chromatography, as well as software tools for normalization and meta-analysis of metabolomics data. All of these techniques have found continued application to biomedical research, including my own work and that of other researchers.

- a. Anderson BG, Raskind A, Habra H, Kennedy RT, Evans CR. Modifying Chromatography Conditions for Improved Unknown Feature Identification in Untargeted Metabolomics. *Anal Chem*. 2021 Dec 7;93(48):15840-15849. Epub 2021 Nov 18. PMID: 34794310.
- b. MetabCombiner: Paired Untargeted LC-HRMS Metabolomics Feature Matching and Concatenation of Disparately Acquired Data Sets. Habra H, Kachman M, Bullock K, Clish C, Evans CR, Karnovsky A. *Anal Chem*, 30 Mar 2021, 93(12):5028-5036. PMID: 33734799
- c. Thonusin C, IglayRager HB, Soni T, Rothberg AE, Burant CF, Evans CR. Evaluation of intensity drift correction strategies using MetaboDrift, a normalization tool for multi-batch metabolomics data. *Journal of Chromatography A* 2017, 1523, 265-274. PMCID: PMC5788449.
- d. Evans CR, Jorgenson JW. Multidimensional LC-LC and LC-CE for high resolution separations of biological molecules. *Anal Bioanalytical Chemistry* 2004, 378, 1952-1961.

2. Physical activity, and maximal oxidative capacity ($VO_2\text{max}$), are strongly correlated with health and longevity in humans, but the mechanisms of this association are incompletely understood. In my research, I have sought to elucidate this connection by investigating metabolic determinants of the health benefits of exercise and exercise capacity. Using metabolomic and proteomic profiling, I demonstrated that rats selectively bred for high exercise capacity (HCR) efficiently oxidize fatty acids (FAs) and branched-chain amino acids (BCAAs) compared to their low exercise capacity counterparts (LCR), sparing glycogen and reducing accumulation of short- and medium-chain acylcarnitines. HCR mitochondria have reduced acetylation of mitochondrial proteins at rest and rapid protein deacetylation occurs with exercise. Stable isotope tracer studies focused on valine catabolism demonstrated a functional role of differential protein acetylation in HCR and LCR. These data suggest that efficient FA and BCAA utilization contribute to high intrinsic exercise capacity and the health and longevity benefits associated with enhanced fitness.

I am currently involved in the NIH Molecular Transducers of Physical Activity Consortium (MoTrPAC) as a co-investigator of the Michigan Motrpac Chemical Analysis Site (MiCAS, PIs Charles Burant and Jun Li). As the managing director of the analytical core of MiCAS, I am directing analysis of thousands of human and animal tissues to help develop a comprehensive, multi-omics map of the metabolic effects of exercise.

Commonly-reported methods of tissue collection for metabolomics may allow degradation of labile metabolites due to exposure to hypoxic conditions. We performed a comprehensive study of the effect of several commonly-used methods of anesthesia and euthanasia on the rodent tissue metabolome. We developed a more optimal strategy for metabolomics of mammalian tissues and have made recommendations regarding these findings to the research community, including MoTrPAC investigators.

- a. Overmyer KA*, Evans CR*, Qi NR, Minogue CE, Carson JJ, Chermiside-Scabbo CJ, Koch LG, Britton SL, Pagliarini DJ, Coon JJ, Burant CF (2015). Maximal oxidative capacity during exercise is associated with skeletal muscle fuel selection and dynamic changes in mitochondrial protein acetylation. *Cell Metabolism*, 3 Mar 2015, 21(3), 468-478. PMCID: PMC4350023. (*co first-author)
- b. Overmyer KA, Thonusin C, Qi NR, Burant CF, Evans CR. Impact of Anesthesia and Euthanasia on Metabolomics of Mammalian Tissues: Studies in a C57BL/6J Mouse Model. *PLoS ONE*, 6 Feb. 2015, 10(2): e0117232. PMCID: PMC4319778
- c. Sandford JA et al., Molecular Transducers of Physical Activity Consortium (MoTrPAC): Mapping the Dynamic Responses to Exercise. *Cell*, 5 Jun 2020, 181(7): 1464-1474. PMID: 32589957 (C. Evans included in author list).

3. Through an ongoing collaboration with Dr. Scott Leiser, I have applied advanced metabolomics methodologies to study the biological function of flavin monooxygenase (FMO) enzymes. These conserved class of enzymes have classically been categorized as catalysts for oxidation of xenobiotic compounds, but our research using *C. elegans* as a model system has demonstrated that FMOs may have important regulatory functions in endogenous metabolism with relevance to stress resistance and aging. In a study focused on methodologies including metabolomics in *C. elegans*, we determined that using

paraformaldehyde-treated *E. coli* used as a food source allows study of *C. elegans* metabolism without the interference from bacterial enzymatic activity that is typically observed when either live or UV-killed bacteria are used.

- a. Choi C, Leiser S, Evans C, Beard D. Metabolic Regulation of Longevity by One-Carbon Metabolism and Flavin-Containing Monooxygenase. *Innov Aging*. 2020 Dec 16;4(Suppl 1):126. PMID: PMC7741500.
 - b. Beydoun S, Choi HS, Dela-Cruz G, Kruempel J, Huang S, Bazopoulou D, Miller HA, Schaller ML, Evans CR, Leiser SF. An alternative food source for metabolism and longevity studies in *Caenorhabditis elegans*. *Commun Biol*. 2021 Feb 26;4(1):258. PMID: PMC7910432.
 - c. Huang S, Howington MB, Dobry CJ, Evans CR, Leiser SF. Flavin-Containing Monooxygenases Are Conserved Regulators of Stress Resistance and Metabolism. *Front Cell Dev Biol*. 2021 Feb 12;9:630188. PMID: PMC7907451.
4. Sepsis, a complex systemic response to infection, is a substantial hazard to human health; its most severe form, septic shock, has a mortality rate of approximately 40%. Sepsis can be characterized as a metabolic crisis. It is associated with multiple alterations in metabolism including elevated serum lactate and alterations in serum amino acid and free fatty acid levels, but fundamental understanding of why these and other alterations occur is limited. Growing evidence supports disturbance of carnitine metabolism as playing a role in sepsis. Through an extended collaboration with Drs. Kathleen Stringer and Alla Karnovsky, I have used metabolomics to characterize metabolic perturbations in the lungs and plasma of sepsis patients. We have also investigated systemic metabolism in a recent clinical trial evaluating rapid administration of L-carnitine as a novel treatment for sepsis.
- a. Evans CR, Karnovsky A, Kovach MA, Standiford TJ, Burant CF, Stringer KA. Untargeted LC-MS of Bronchoalveolar Lavage Fluid Differentiates Acute Respiratory Distress Syndrome from Health. *Journal of Proteome Research*, 7 Feb 2014, 13, 640-649. PMID: PMC4068805
 - b. Puskarich MA, Evans CR, Karnovsky A, Das AK, Jones AE, Stringer KA. Septic Shock Nonsurvivors Have Persistently Elevated Acylcarnitines Following Carnitine Supplementation. *Shock*, Apr. 2018. 49, 412-419. PMID: PMC5847421
 - c. Evans CR, Karnovsky A, Puskarich MA, Michailidis G, Jones AE, Stringer KA. Untargeted Metabolomics Differentiates L-Carnitine Treated Septic Shock 1-Year Survivors and Nonsurvivors. *J Proteome Res*. 2019. 18, 2004-2011. PMID: PMC6501183
 - d. Jenaro TS, Puskarich MA, McCann MR, Gillies CE, Pai MP, Karnovsky A, Evans CR, Jones AE, Stringer KA. Using L-carnitine as a Pharmacologic Probe of the Interpatient and Metabolic Variability of Sepsis. *Pharmacotherapy*. 2020 Sep;40(9):913-923. PMID: PMC7722044
5. Metabolomics is applicable to study of a broad range of pathways and disease states, yet its potential has only begun to be realized within the biomedical research community. Collaborations are essential to bringing the potential of metabolomics research to fruition, and I have been fortunate to develop impactful collaborations with experts in multiple fields of biomedical research. Working with Dr. Charles Brenner, I developed a novel targeted metabolomic method which enabled simultaneous quantitation of all intermediates in the NAD⁺ biosynthesis and degradation pathways. We used this method to study the effects of caloric restriction and nicotinic acid supplementation in yeast and demonstrated that glucose restriction does not alter NAD⁺ or nicotinamide levels in ways that would increase sirtuin activity. I coordinated metabolomics and fluxomics components of a study which demonstrated that inflammatory breast cancer cells exhibit a variety of metabolic abnormalities compared to less aggressive breast cancer cells, including altered glutamine metabolism and production of n-acetyl aspartate, a metabolite predominantly associated with neuronal disorders and gliomas. In another study, I used metabolomics and metabolic flux analysis, including studies with stable-isotope labeled glucose, to help understand the metabolic role of CARKL, an orphan protein we identified as a sedoheptulose kinase, which we found to be essential to normal polarization of macrophages toward either the M1 or M2 phenotype. I have also aided in application of metabolomics to projects pertaining the effect of GPR40 receptor signaling and other signaling pathways in beta-cell metabolism.
- a. Evans CR, Bogan KL, Song P, Burant CF, Kennedy RT and Brenner C. Yeast NAD⁺ Metabolite Levels are Altered by Provision of Vitamins and Yeast Extract but not by Calorie Restriction. *BMC Chemical Biology*. 2010, 10:2. PMID: PMC2834649.

- b. Wynn ML, Yates JA, Evans CR, Van Wassenhove L, Wu ZF, Bridges S, Bao L, Fournier C, Ashrafzadeh S, Merrins MJ, Satin LS, Schnell S, Burant CF, Merajver SD. RhoC is a Potent Regulator of Glutamine Metabolism and N-acetylaspartate Production in Inflammatory Breast Cancer Cells. *The Journal of biological chemistry*. 2016, 291(26), 13715-29. PMCID: PMC4919454
- c. Haschemi A, Kosma P, Gille L, Evans CR, Burant CF, Starkl P, Knapp B, Haas R, Schmid JA, Jandl C, Amir S, Lubec G, Park J, Esterbauer H, Bilban M, Brizuela L, Pospisilik AJ, Otterbein LE, Wagner O. The Sedoheptulose Kinase CARKL Directs Macrophage Polarization Through Control of Glucose Metabolism. *Cell Metabolism*, 6 Jun. 2012, 15, 813–826. PMCID: PMC3370649
- d. El-Azzouny M, Evans CR, Treutelaar MK, Kennedy RT, Burant CF. Increased glucose metabolism and glycerolipid formation by fatty acids and GPR40 receptor signaling underlies the fatty acid potentiation of insulin secretion. *J Biol Chem*. 2014 May 9. 289, 3575-88. PMCID: PMC4036363

Complete List of Published Work in MyBibliography:

<http://www.ncbi.nlm.nih.gov/sites/myncbi/charles.evans.1/bibliography/41675809/public/?sort=date&direction=ascending>.



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September 12, 2023

Professor Oliver Fiehn, PhD
Director, UC Davis West Coast Metabolomics Center

Dear Oliver,

I am writing to enthusiastically indicate my willingness to collaborate with you on the Multi-PI R01 grant that you will lead and entitled "A reproducible database for untargeted metabolomics data processing across laboratories." This project has the exciting goal of standardizing untargeted metabolomics and lipidomics data processing to reproducibly acquire, compare and interpret biomedical metabolomics data within and across different mass spectrometry platforms, across laboratories, and for different types of matrices.

We will contribute to this project as follows:

- Perform ion mobility spectrometry-mass spectrometry (IMS-MS) measurements of pure standards and various animal and human biofluids and tissues, providing experimentally determined collision cross sections and "all ions" MS/MS spectra for detected molecules for entry into your LC-BinBase infrastructure.
- Assist in evaluating the suitability of IMS-MS data for providing additional confidence in compound identification.
- Evaluate MS-DIAL in the processing of LC-IMS-MS/MS data.
- We will also contribute to collaborative interpretation of the data and writing of manuscripts.

This project is a natural extension of our collaboration under the Unknown Lipids Working Group of the NIH Common Fund Metabolomics Program Phase II, and I look forward to continuing our productive work together.

Sincerely,

Thomas O. Metz

A handwritten signature in blue ink, appearing to read "T.O. Metz".

Metabolomics Team Lead

Biological Sciences Division

Pacific Northwest National Laboratory



MEDICAL SCHOOL UNIVERSITY OF MICHIGAN

Department of Internal Medicine
Biomedical Research Core Facilities
Metabolomics Core

Sep. 18, 2023

Dear Dr. Fiehn,

I am very pleased to participate with you and Dr. Thomas Metz in our R01 application entitled "A reproducible database for untargeted metabolomics data processing across laboratories". Through this important project we will develop, evaluate and promote use of harmonized software tools and analytical methods for metabolomics and lipidomics. These methods will improve data reproducibility and inter-laboratory consistency, contributing to enhanced biological impact and long-term reusability of metabolomics data for both basic research and clinical applications.

Over the past five years it has been my pleasure to collaborate with you on several studies assessing the feasibility of harmonizing untargeted metabolomics methods between labs. These studies have been carried out in collaboration with your lab and several others funded as Compound Identification Development Cores (CIDCs) through the NIH Common Funds Metabolomics Consortium. Our prototype work in this area included efforts to establish universal protocols with common internal standards for sample preparation, use common chromatographic columns and gradients, harmonize mass spectrometer data acquisition parameters, and perform common-pipeline metabolomics data processing. The methods we evaluated were designed to be compatible with a diverse range of accurate-mass liquid chromatography-mass spectrometry and gas chromatography-mass spectrometry instruments, reflecting the diversity of equipment deployed and used in the field of metabolomics. Compared to the status quo of lab-specific methods, the data from our preliminary studies demonstrated substantial improvements in inter-laboratory data reproducibility and enhanced ability to align features across datasets, including both identified and unknown compounds.

I and the researchers supported by this project at the University of Michigan will actively participate in all the aims of the application, which share the overarching goal of developing standardized untargeted metabolomics and lipidomics data collection and processing methods. We will coordinate with Dr. Fiehn and Dr. Metz to select and refine harmonized methods which we will implement over the course of the grant. We will participate in all data proposed data collection tasks, implementing and evaluating the methods according to procedures we develop as a team. Primary development of the LC-BinBase software tool will be managed by Dr. Fiehn and his team at UC-Davis, but our team will routinely evaluate its functionality, make recommendations, and assist with implementing these recommendations when appropriate. We anticipate that our most substantial contribution to the project will be to lead careful review and analysis of the data generated by all three contributing labs, using software tools we and others have developed to evaluate feature degeneracy, data consistency, and feature alignment accuracy between labs, much as we have done in the pilot projects described above. We will share all data we generate in the context this project with the laboratories of Dr. Fiehn and Dr. Metz, and we will participate in and/or lead efforts to disseminate our results to the research community through preparation of publications in scientific journals and giving presentations at national and international conferences.

As you are aware, the University of Michigan Metabolomics Core Facility runs tens of thousands of samples per year for clients both local and worldwide. We are equipped with ample instrumentation and, importantly, sufficient data processing hardware, software, and personnel with expertise to operate these tools, as will be required to carry out the collaborative research we proposed. I look forward to working with you to writing the next chapter in the development of metabolomics, as we move it toward a truly standardized, harmonized and informative technique!

Sincerely,

A handwritten signature in black ink that reads "Charles R. Evans". The signature is written in a cursive, flowing style.

Charles R. Evans, PhD

Associate Research Scientist, University of Michigan, Department of Internal Medicine
Faculty Director, Metabolomics Core, University of Michigan Biomedical Research Core
Facilities

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

* UEI: TX2DAGQPENZ5

* Budget Type: ☒ Project ☐ Subaward/Consortium

Enter name of Organization: The Regents of the University of California (Davis)

* Start Date: 07-01-2024 * End Date: 06-30-2025 Budget Period: 1

A. Senior/Key Person												
Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary	Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested (\$)
						(\$)	Months	Months	Months	Salary (\$)	Benefits (\$)	
1.	Dr.	Oliver	Fiehn	Ph.D	PD/PI	212,100.00	2.4			42,400.00	3,858.00	46,258.00
2.		Gert	Wohlgemuth		Programmer	118,740.00	6			59,370.00	31,525.00	90,895.00
3.		Tong	Shen		Project Scientist	79,300.00	3			19,825.00	8,089.00	27,914.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:		Mime Type:		Total Senior/Key Person					165,067.00

B. Other Personnel													
* Number of		* Project Role					Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested	
Personnel							Months	Months	Months	Salary (\$)	Benefits	(\$)	
1	Post Doctoral Associates						12			60,000.00	13,680.00	73,680.00	
	Graduate Students												
	Undergraduate Students												
	Secretarial/Clerical												
1	Asst Project Scientist						3			19,825.00	8,089.00	27,914.00	
2	Total Number Other Personnel									Total Other Personnel		101,594.00	
							Total Salary, Wages and Fringe Benefits (A+B)					266,661.00	

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

* **UEI:** TX2DAGQPENZ5

* **Budget Type:** ☒ Project ☐ Subaward/Consortium

Enter name of Organization: The Regents of the University of California (Davis)

* **Start Date:** 07-01-2024

* **End Date:** 06-30-2025

Budget Period: 1

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment Item

* Funds Requested (\$)

Total funds requested for all equipment listed in the attached file

Total Equipment

Additional Equipment:

File Name:

Mime Type:

D. Travel

Funds Requested (\$)

1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)

3,000.00

2. Foreign Travel Costs

Total Travel Cost

3,000.00

E. Participant/Trainee Support Costs

Funds Requested (\$)

1. Tuition/Fees/Health Insurance

2. Stipends

3. Travel

4. Subsistence

5. Other:

Number of Participants/Trainees

Total Participant/Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 1

* UEI: TX2DAGQPENZ5

* Budget Type: ☒ Project ☐ Subaward/Consortium

Enter name of Organization: The Regents of the University of California (Davis)

* Start Date: 07-01-2024

* End Date: 06-30-2025

Budget Period: 1

F. Other Direct Costs		Funds Requested (\$)
1. Materials and Supplies		27,338.00
2. Publication Costs		2,000.00
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		304,121.00
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Data Management and Sharing		1,000.00
Total Other Direct Costs		334,459.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	604,120.00

H. Indirect Costs				
Indirect Cost Type		Indirect Cost Rate (%)	Indirect Cost Base (\$)	* Funds Requested (\$)
1. On Campus Org Research MTDC		61	349,999.00	213,499.00
			Total Indirect Costs	213,499.00
Cognizant Federal Agency		HHS Jeffrey Warren 415-437-7820		
(Agency Name, POC Name, and POC Phone Number)				

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	817,619.00

J. Fee	Funds Requested (\$)
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K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	817,619.00

L. * Budget Justification	File Name:	Mime Type: application/pdf
	BudgJust_Fiehn_R01_LC_09_21_231041563648.pdf	
	(Only attach one file.)	

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 2

* UEI: TX2DAGQPENZ5

* Budget Type: ☒ Project ☐ Subaward/Consortium

Enter name of Organization: The Regents of the University of California (Davis)

* Start Date: 07-01-2025 * End Date: 06-30-2026 Budget Period: 2

A. Senior/Key Person												
Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary	Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested (\$)
						(\$)	Months	Months	Months	Salary (\$)	Benefits (\$)	
1.	Dr.	Oliver	Fiehn	Ph.D	PD/PI	212,100.00	2.4			42,400.00	3,986.00	46,386.00
2.		Gert	Wohlgemuth		Programmer	118,740.00	6			59,370.00	32,475.00	91,845.00
3.		Tong	Shen		Project Scientist	79,300.00	3			19,825.00	8,327.00	28,152.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:		Mime Type:		Total Senior/Key Person				166,383.00	

B. Other Personnel													
* Number of		* Project Role					Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested	
Personnel							Months	Months	Months	Salary (\$)	Benefits	(\$)	
1	Post Doctoral Associates						12			60,000.00	14,100.00	74,100.00	
	Graduate Students												
	Undergraduate Students												
	Secretarial/Clerical												
1	Asst Project Scientist						3			19,825.00	8,327.00	28,152.00	
2	Total Number Other Personnel								Total Other Personnel		102,252.00		
							Total Salary, Wages and Fringe Benefits (A+B)					268,635.00	

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 2

* **UEI:** TX2DAGQPENZ5

* **Budget Type:** ☒ Project ☐ Subaward/Consortium

Enter name of Organization: The Regents of the University of California (Davis)

* **Start Date:** 07-01-2025

* **End Date:** 06-30-2026

Budget Period: 2

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment Item

* Funds Requested (\$)

Total funds requested for all equipment listed in the attached file

Total Equipment

Additional Equipment:

File Name:

Mime Type:

D. Travel

Funds Requested (\$)

1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)

3,000.00

2. Foreign Travel Costs

Total Travel Cost

3,000.00

E. Participant/Trainee Support Costs

Funds Requested (\$)

1. Tuition/Fees/Health Insurance

2. Stipends

3. Travel

4. Subsistence

5. Other:

Number of Participants/Trainees

Total Participant/Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 2

* UEI: TX2DAGQPENZ5

* Budget Type: ☒ Project ☐ Subaward/Consortium

Enter name of Organization: The Regents of the University of California (Davis)

* Start Date: 07-01-2025

* End Date: 06-30-2026

Budget Period: 2

F. Other Direct Costs		Funds Requested (\$)
1. Materials and Supplies		25,364.00
2. Publication Costs		2,000.00
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		306,774.00
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Data Management and Sharing		1,000.00
Total Other Direct Costs		335,138.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	606,773.00

H. Indirect Costs				
Indirect Cost Type		Indirect Cost Rate (%)	Indirect Cost Base (\$)	* Funds Requested (\$)
1. On Campus Org Research MTDC		61	299,999.00	182,999.00
			Total Indirect Costs	182,999.00
Cognizant Federal Agency		HHS Jeffrey Warren 415-437-7820		
(Agency Name, POC Name, and POC Phone Number)				

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	789,772.00

J. Fee	Funds Requested (\$)
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K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	789,772.00

L. * Budget Justification	File Name:	Mime Type: application/pdf
	BudgJust_Fiehn_R01_LC_09_21_231041563648.pdf	
	(Only attach one file.)	

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 3

* UEI: TX2DAGQPENZ5

* Budget Type: ☒ Project ☐ Subaward/Consortium

Enter name of Organization: The Regents of the University of California (Davis)

* Start Date: 07-01-2026 * End Date: 06-30-2027 Budget Period: 3

A. Senior/Key Person												
Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary	Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested (\$)
						(\$)	Months	Months	Months	Salary (\$)	Benefits (\$)	
1.	Dr.	Oliver	Fiehn	Ph.D	PD/PI	212,100.00	2.4			42,400.00	4,113.00	46,513.00
2.		Gert	Wohlgemuth		Programmer	118,740.00	6			59,370.00	33,425.00	92,795.00
3.		Tong	Shen		Project Scientist	79,300.00	2.4			15,860.00	6,867.00	22,727.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:		Mime Type:		Total Senior/Key Person				162,035.00	

B. Other Personnel													
* Number of		* Project Role					Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested	
Personnel							Months	Months	Months	Salary (\$)	Benefits	(\$)	
1	Post Doctoral Associates						12			60,000.00	14,520.00	74,520.00	
	Graduate Students												
	Undergraduate Students												
	Secretarial/Clerical												
1	Asst Project Scientist						2.4			19,825.00	8,584.00	28,409.00	
2	Total Number Other Personnel								Total Other Personnel		102,929.00		
							Total Salary, Wages and Fringe Benefits (A+B)					264,964.00	

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 3

* **UEI:** TX2DAGQPENZ5

* **Budget Type:** ☒ Project ☐ Subaward/Consortium

Enter name of Organization: The Regents of the University of California (Davis)

* **Start Date:** 07-01-2026

* **End Date:** 06-30-2027

Budget Period: 3

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment Item

* Funds Requested (\$)

Total funds requested for all equipment listed in the attached file

Total Equipment

Additional Equipment:

File Name:

Mime Type:

D. Travel

Funds Requested (\$)

1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)

3,000.00

2. Foreign Travel Costs

Total Travel Cost

3,000.00

E. Participant/Trainee Support Costs

Funds Requested (\$)

1. Tuition/Fees/Health Insurance

2. Stipends

3. Travel

4. Subsistence

5. Other:

Number of Participants/Trainees

Total Participant/Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 3

* UEI: TX2DAGQPENZ5

* Budget Type: ☒ Project ☐ Subaward/Consortium

Enter name of Organization: The Regents of the University of California (Davis)

* Start Date: 07-01-2026

* End Date: 06-30-2027

Budget Period: 3

F. Other Direct Costs	Funds Requested (\$)
1. Materials and Supplies	29,035.00
2. Publication Costs	2,000.00
3. Consultant Services	
4. ADP/Computer Services	
5. Subawards/Consortium/Contractual Costs	306,709.00
6. Equipment or Facility Rental/User Fees	
7. Alterations and Renovations	
8. Data Management and Sharing	1,000.00
Total Other Direct Costs	338,744.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	606,708.00

H. Indirect Costs				
	Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	* Funds Requested (\$)
1.	On Campus Org Research MTDC	61	299,999.00	182,999.00
			Total Indirect Costs	182,999.00
Cognizant Federal Agency		HHS Jeffrey Warren 415-437-7820		
(Agency Name, POC Name, and POC Phone Number)				

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	789,707.00

J. Fee	Funds Requested (\$)
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K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	789,707.00

L. * Budget Justification	File Name:	Mime Type: application/pdf
	BudgJust_Fiehn_R01_LC_09_21_231041563648.pdf	
	(Only attach one file.)	

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 4

* UEI: TX2DAGQPENZ5

* Budget Type: ☒ Project ☐ Subaward/Consortium

Enter name of Organization: The Regents of the University of California (Davis)

* Start Date: 07-01-2027 * End Date: 06-30-2028 Budget Period: 4

A. Senior/Key Person												
Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary	Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested (\$)
						(\$)	Months	Months	Months	Salary (\$)	Benefits (\$)	
1.	Dr.	Oliver	Fiehn	Ph.D	PD/PI	212,100.00	2.4			42,400.00	4,240.00	46,640.00
2.		Gert	Wohlgemuth		Programmer	118,740.00	6			59,370.00	34,435.00	93,805.00
3.		Tong	Shen		Project Scientist	79,300.00	2.4			15,860.00	7,074.00	22,934.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:		Mime Type:		Total Senior/Key Person				163,379.00	

B. Other Personnel													
* Number of		* Project Role					Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested	
Personnel							Months	Months	Months	Salary (\$)	Benefits	(\$)	
1	Post Doctoral Associates						12			60,000.00	14,940.00	74,940.00	
	Graduate Students												
	Undergraduate Students												
	Secretarial/Clerical												
1	Asst Project Scientist						2.4			19,825.00	8,842.00	28,667.00	
2	Total Number Other Personnel								Total Other Personnel		103,607.00		
							Total Salary, Wages and Fringe Benefits (A+B)					266,986.00	

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 4

* **UEI:** TX2DAGQPENZ5

* **Budget Type:** ☒ Project ☐ Subaward/Consortium

Enter name of Organization: The Regents of the University of California (Davis)

* **Start Date:** 07-01-2027

* **End Date:** 06-30-2028

Budget Period: 4

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment Item

* Funds Requested (\$)

Total funds requested for all equipment listed in the attached file

Total Equipment

Additional Equipment:

File Name:

Mime Type:

D. Travel

Funds Requested (\$)

1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)

3,000.00

2. Foreign Travel Costs

Total Travel Cost

3,000.00

E. Participant/Trainee Support Costs

Funds Requested (\$)

1. Tuition/Fees/Health Insurance

2. Stipends

3. Travel

4. Subsistence

5. Other:

Number of Participants/Trainees

Total Participant/Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 4

* UEI: TX2DAGQPENZ5

* Budget Type: ☒ Project ☐ Subaward/Consortium

Enter name of Organization: The Regents of the University of California (Davis)

* Start Date: 07-01-2027

* End Date: 06-30-2028

Budget Period: 4

F. Other Direct Costs	Funds Requested (\$)
1. Materials and Supplies	27,013.00
2. Publication Costs	2,000.00
3. Consultant Services	
4. ADP/Computer Services	
5. Subawards/Consortium/Contractual Costs	306,188.00
6. Equipment or Facility Rental/User Fees	
7. Alterations and Renovations	
8. Data Management and Sharing Costs	1,000.00
Total Other Direct Costs	336,201.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	606,187.00

H. Indirect Costs				
	Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	* Funds Requested (\$)
1.	On Campus Org Research MTDC	61	299,999.00	182,999.00
			Total Indirect Costs	182,999.00
Cognizant Federal Agency		HHS Jeffrey Warren 415-437-7820		
(Agency Name, POC Name, and POC Phone Number)				

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	789,186.00

J. Fee	Funds Requested (\$)
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K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	789,186.00

L. * Budget Justification	File Name:	Mime Type: application/pdf
	BudgJust_Fiehn_R01_LC_09_21_231041563648.pdf	
	(Only attach one file.)	

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - Cumulative Budget

	Totals (\$)	
Section A, Senior/Key Person		656,864.00
Section B, Other Personnel		410,382.00
Total Number Other Personnel	8	
Total Salary, Wages and Fringe Benefits (A+B)		1,067,246.00
Section C, Equipment		
Section D, Travel		12,000.00
1. Domestic	12,000.00	
2. Foreign		
Section E, Participant/Trainee Support Costs		
1. Tuition/Fees/Health Insurance		
2. Stipends		
3. Travel		
4. Subsistence		
5. Other		
6. Number of Participants/Trainees		
Section F, Other Direct Costs		1,344,542.00
1. Materials and Supplies	108,750.00	
2. Publication Costs	8,000.00	
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs	1,223,792.00	
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Other 1	4,000.00	
9. Other 2		
10. Other 3		
11. Other 4		
12. Other 5		
13. Other 6		
14. Other 7		
15. Other 8		
16. Other 9		
17. Other 10		
Section G, Direct Costs (A thru F)		2,423,788.00
Section H, Indirect Costs		762,496.00
Section I, Total Direct and Indirect Costs (G + H)		3,186,284.00
Section J, Fee		
Section K, Total Costs and Fee (I + J)		3,186,284.00

BUDGET JUSTIFICATION

Institute: University of California, Davis

PI: Oliver Fiehn, Ph.D

KEY PERSONNEL

Prof. Oliver Fiehn, Ph.D. (Principal Investigator), will commit 2.4 calendar months (20.0% efforts) to coordinate the work for this project, including study designs, curation and validation of MS/MS data, integration of genomic and biochemical metadata, and engagement with the metabolomics and cancer communities. Prof. Fiehn will be responsible for overall project oversight, reporting, and data dissemination of all findings throughout the project. Specifically, Prof. Fiehn will interact with lead database engineer Gert Wohlgemuth for all aspects of software integration for compound annotations and metadata import, as well as usability of export results. Prof. Fiehn will personally merge, curate and integrate external database metabolite identifiers for updating LC-BinBase compounds for usability in biomedical and systems biology projects. Prof. Fiehn will interact with Dr. Yuanyue Li and Fanzhou Kong on all aspects of FDR calculations of metabolite and lipid annotations, and integration of retention time modeling and use of QCs and internal standards for quantifications. Prof. Fiehn will engage with his team in weekly project meetings to ensure that timelines and expectations are met and to discuss optimal data validation and code deployment. Prof. Fiehn will coordinate the work with external partner laboratories Metz and Evans in monthly team meetings. He will oversee the work by Dr. Tong Shen on results of specimen testing between the three laboratories and comparison between manual and automatic data processing. Prof. Fiehn will initiate and oversee manuscript writing and data and code sharing.

Senior software- and database developer **Gert Wohlgemuth** will commit 50% efforts in year 01-04 to the overall program. Gert Wohlgemuth joined the Fiehn laboratory in 2002 and has been responsible for all major database and software developments in the West Coast Metabolomics Center for the past 21 years, ranging from the original GC-TOF MS focused BinBase database and miniX study design database to the CTS chemical translation service and the public MassBank.us database with its more than 2 million mass spectra of authentic metabolites. Over the past three years he has programmed the prototype Amazon Web Service-based LC-BinBase software that automatically aligns chromatograms and generates reports for untargeted lipidomic and metabolomic assays. For this project, Mr. Wohlgemuth will develop code to integrate a suite of software tools developed by Dr. Yuanyue Li and Dr. Huaxu Yu into a common pipeline, from library MS/MS reports (in MassBank.us) to retention time predictions (using Retip.app) and adding genomics metadata (from Reactome to BioPAN). Mr. Wohlgemuth will use his excellent expertise in Python, JavaScript, Node.js, MongoDB, Java Springboot, Express.js, and Oracle SQL programming environments and coding languages for tool integration, unit testing, code reviews and overall functionality tests. He will also be responsible for code sharing with external communities and implementing QA/QC reports, consistent data standards and data security protocols. Gert Wohlgemuth will work with informatics team members in the Metz and Evans laboratories for integrating code on ion mobility and adduct formations into the LC-BinBase routines.

Dr. Tong Shen, Project Scientist, is the lead analytical chemist in the Fiehn laboratory. She will commit 25% efforts in years 01-02 and 20% efforts in years 03-04 to this project to oversee the experimental analyses in specific aim 01 and the implementation of BioPAN lipid pathway identifiers and increasing usability of lipidomics data for specific aim 03. She will also oversee the pilot tests of LC-BinBase with external lipidomics and metabolomics laboratories and perform quality controls of metabolite and lipid annotations in LC-BinBase. Dr. Shen has joined the UC Davis West Coast Metabolomics Center in 2016 after obtaining her PhD in Singapore in 2014 in the lipidomics group of Prof. Markus Wenk. Since 2018, Dr. Shen leads the analytical chemistry unit of the UC Davis West Coast Metabolomics Center in all aspects, including throughput, quantification, compound identifications and quality control. She will acquire mass spectrometry data for validation specimen sent by the Evans laboratory and compare manually processed data to LC-BinBase

reports. She will advise the data processing informatics team on issues in result reports and parameters of data processing such as m/z accuracy windows, retention index tolerances, compound identification MSI-annotations and upper and lower QC warning and intervention limits. She will engage with the Metz and Evans laboratories on protocol transfers, analytical chemistry optimizations, and compound annotation confidence as well as on comparison of result data between the laboratories for the different cells, organs and biofluids.

OTHER PERSONNEL

Dr. Yuanyue Li, Project Scientist, is the lead cheminformatics expert in the Fiehn laboratory. He will commit 25% efforts in years 01-04 to this project. He joined the West Coast Metabolomics Center in December 2019 and has since developed the Entropy Similarity MS/MS algorithm that was published in *Nature Methods* in 2021 and FlashEntropy published in *Nature Methods* in 2023. He was promoted to Project Scientist in 2022. Dr. Li was trained in multi-omics analyses, including proteomics and genomics, at the European Molecular Biology Laboratory (EMBL) at Peer Bork's group in Heidelberg, Germany. He had obtained his PhD in 2015 at Xiamen University (China) on analyzing high-throughput sequencing data for RNA-seq and for analyzing interactome changes across cellular phenotypes. He will now use his expertise to analyze LC-BinBase high resolution MS/MS spectra using his Entropy Similarity algorithm and integrate this tool with the results of retention time predictions, exact mass distributions and biological likelihood estimates to form a metabolomic confidence score. He will oversee the addition and use of cancer-relevant genomic and biochemical database identifiers to improve the usability of LC-BinBase data reports. Together with Prof. Fiehn, he will advise cheminformatics postdoctoral scholar Huaxu Yu on algorithm development, testing and validation. Dr. Li will report to Prof. Fiehn and write first drafts of the outcome manuscripts. He will engage with the informatics team members in the Evans and Metz laboratories with respect to confidence measures in metabolite annotations, and with respect to LC-BinBase metadata and identifiers for systems biology use.

Dr. Huaxu Yu, postdoctoral scholar in cheminformatics, will commit 100% allowable time to the project in years 01-04. Before joining UC Davis in Spring 2023, he worked at University of British Columbia, Vancouver/Canada, where he was lead author on publications on metabolomics and cheminformatics projects including use of maximal density fold changes, feature specific data transformation, nonlinear electrospray ionization responses in mass spectrometry, patterned signal ratio biases, and computational variation in automated data processing in untargeted metabolomics. Hence, he is well versed in all aspects of metabolomics data curation, normalization, and data transformation for hypothesis-based and discovery-driven informatics data analyses. For specific aim 1, Dr. Huaxu Yu oversee data acquisitions and analysis of exported results from LC-BinBase. For specific aim 2, he will implement and apply his MS/MS autocuration code on automated mass spectral library building for LC-BinBase by cleaning mass spectra from noise ions using fragmentation trees. Together with Dr. Li, he will work on building probability models to integrate retention index, accurate mass and MS/MS similarities into a unified compound ID confidence score. He will also work with Gert Wohlgemuth to test and implement retip2.0 code into the LC-BinBase environment and assist Dr. Yuanyue Li to assess metabolomics and lipidomics data from the Metz and Evans laboratories for integrating code on ion mobility and adduct formations into the LC-BinBase routines.

FRINGE BENEFITS

Fringe benefits are calculated per the Colleges and Universities Rate Agreement approved on 6/28/2021.

OTHER COSTS

Consumables.

For improving, testing and validating the *LC-BinBase* database, using recurrent analyses of LC-MS/MS data, we will Amazon Web Services, for example for testing automatic parameter optimization in raw data processing. Based on the experience over the past three years, costs of \$12,000 per year are requested.

Metabolomics and lipidomics analyses will be performed in the Fiehn laboratory for raw data acquisition of biological specimens (organs, body fluids and cells) purchased from BioIVT. Metabolomics costs for consumables are requested for \$15,338 in year 01, \$13,364 in year 02, \$17,035 in year 03 and \$15,013 in year 04 for LC-MS/MS raw data acquisitions. These costs include BioIVT sample costs, fractional costs for instrument use and maintenance that include consumables such as vials, plates, liners, nuts, ferrules, septa, valves, columns, guard columns, syringes, fractional costs for computer soft- and hardware, chemicals such as internal standards, reference compounds, buffers solvents, gases, 96-well plates and plastic ware, tubing, paper, and printer necessities.

Travel

For travel to the annual ASMS or MANA conferences, we request \$3,000 per year to cover economy transport, lodging and registrations for two scientists (total) per year.

Publication costs

To defray publication costs, including costs for online access, we request \$2000 per year.

Data Management and Sharing Justification

For data curation and the development of related documentation, the project is requesting \$ 1,000 per year to offset hourly staff costs for data transfer. These funds will allow us to prepare data for sharing including deidentification of data, the incorporation of metadata to ensure discoverability and the data transfer process to the MetabolomicsWorkbench.org repository for preservation and access.

Subaward Costs

To enable external metabolomics partners to develop and integrate algorithms for metabolite annotation confidence testing, and to beta-test and validate the LC-BinBase database for biomedical projects, costs for subcontracts are requested for the Metz laboratory (PNNL) and for the Evans laboratory (U Michigan) for \$100,000 per year (from years 01-04) direct costs each, plus corresponding indirect costs applicable to their institutions. These costs will cover activities for both informatics efforts and mass spectrometry work at these labs. Specifically, the Evans and Metz laboratories will perform raw data acquisitions for experimental robustness tests in Specific Aim 1. Both laboratories will acquire data for subsequent validation studies, process the data using their in-house routines, and upload and compare data processed through LC-BinBase. Both the Evans and Metz laboratories will also actively support overall informatics efforts to optimize LC-BinBase workflows, including the integration of adduct formation software from the U Michigan lab (Binner), and integration of ion mobility calculation software from the PNNL lab (ISiCle and autoCCS). Both laboratories will engage in common data evaluation, manuscript writing, code testing and dissemination of results.

FACILITIES AND ADMINISTRATION

F&A Costs were estimated in accordance with UC Davis's approved indirect cost rate agreement with the Department of Health and Human Services (DHHS), dated June 28, 2021, the Federal Cognizant Agency for UC Davis. Indirect costs are approved with a Modified Total Direct Costs (MTDC) base. The costs of any proposed equipment, capital expenditures, graduate fees and tuition, rental costs of off-site facilities, scholarships, fellowships, participant support costs, and the portion of each sub-award in excess of \$25K are excluded from the MTDC base.

R&R SUBAWARD BUDGET ATTACHMENT(S) FORM

				Subaward IDC Ceiling: 25,000	
Period	Subaward Direct Costs	Subaward Indirect Costs	Subaward Costs	Allocated To IDC Base	
1. PNNL					
1	100,000	48,121	148,121	25,000	
2	100,000	50,774	150,774	0	
3	100,000	50,709	150,709		
4	100,000	50,188	150,188		
All	400,000	199,792	599,792	25,000	
2. University of Michigan					
1	100,000	56,000	156,000	25,000	
2	100,000	56,000	156,000		
3	100,000	56,000	156,000		
4	100,000	56,000	156,000		
All	400,000	224,000	624,000	25,000	
All Subawards					
1	200,000	104,121	304,121	50,000	
2	200,000	106,774	306,774	0	
3	200,000	106,709	306,709	0	
4	200,000	106,188	306,188	0	
All	800,000	423,792	1,223,792	50,000	

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

* UEI: CWKJEXDG79A7

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: Pacific Northwest National Laboratory

* Start Date: 07-01-2024 * End Date: 06-30-2025 Budget Period: 1

A. Senior/Key Person												
Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary	Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested (\$)
						(\$)	Months	Months	Months	Salary (\$)	Benefits (\$)	
1.	Thomas		Metz		PD/PI	0.00	0.6			10,061.00	2,742.00	12,803.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:		Mime Type:		Total Senior/Key Person				12,803.00	

B. Other Personnel													
* Number of		* Project Role					Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested	
Personnel							Months	Months	Months	Salary (\$)	Benefits	(\$)	
1	Post Doctoral Associates						9.3			69,432.00	12,567.00	81,999.00	
	Graduate Students												
	Undergraduate Students												
	Secretarial/Clerical												
1	Total Number Other Personnel										Total Other Personnel		81,999.00
							Total Salary, Wages and Fringe Benefits (A+B)					94,802.00	

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

* UEI: CWKJEXDG79A7

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: Pacific Northwest National Laboratory

* Start Date: 07-01-2024

* End Date: 06-30-2025

Budget Period: 1

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment Item

* Funds Requested (\$)

Total funds requested for all equipment listed in the attached file

Total Equipment

Additional Equipment:

File Name:

Mime Type:

D. Travel

Funds Requested (\$)

1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)

1,824.00

2. Foreign Travel Costs

Total Travel Cost

1,824.00

E. Participant/Trainee Support Costs

Funds Requested (\$)

1. Tuition/Fees/Health Insurance

2. Stipends

3. Travel

4. Subsistence

5. Other:

Number of Participants/Trainees

Total Participant/Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 1

* UEI: CWKJEXDG79A7

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: Pacific Northwest National Laboratory

* Start Date: 07-01-2024

* End Date: 06-30-2025

Budget Period: 1

F. Other Direct Costs		Funds Requested (\$)
1. Materials and Supplies		3,042.00
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Procurment Svc Chgs		158.00
9. Waste Disposal Svc Ctr		174.00
Total Other Direct Costs		3,374.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	100,000.00

H. Indirect Costs			
	Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$) * Funds Requested (\$)
1. Other		12.76	94,802.00 12,097.00
2. Other		3.9	22,279.00 869.00
3. Other		26.92	109,537.00 29,487.00
4. Other		3.96	143,119.00 5,668.00
Total Indirect Costs			48,121.00
Cognizant Federal Agency			
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	148,121.00

J. Fee	Funds Requested (\$)
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K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	148,121.00

L. * Budget Justification	File Name:	Mime Type: application/pdf
	7__81394_Budget_Justification1041563837.pdf	
	(Only attach one file.)	

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 2

* UEI: CWKJEXDG79A7

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: Pacific Northwest National Laboratory

* Start Date: 07-01-2025 * End Date: 06-30-2026 Budget Period: 2

A. Senior/Key Person												
Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary	Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested (\$)
						(\$)	Months	Months	Months	Salary (\$)	Benefits (\$)	
1.	Thomas		Metz		PD/PI	0.00	0.6			10,072.00	2,679.00	12,751.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:		Mime Type:		Total Senior/Key Person				12,751.00	

B. Other Personnel												
* Number of		* Project Role					Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested
Personnel							Months	Months	Months	Salary (\$)	Benefits	(\$)
1	Post Doctoral Associates						8.8			71,030.00	12,857.00	83,887.00
	Graduate Students											
	Undergraduate Students											
	Secretarial/Clerical											
1	Total Number Other Personnel										Total Other Personnel	83,887.00
							Total Salary, Wages and Fringe Benefits (A+B)					96,638.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 2

* UEI: CWKJEXDG79A7

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: Pacific Northwest National Laboratory

* Start Date: 07-01-2025

* End Date: 06-30-2026

Budget Period: 2

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment Item

* Funds Requested (\$)

Total funds requested for all equipment listed in the attached file

Total Equipment

Additional Equipment:

File Name:

Mime Type:

D. Travel

Funds Requested (\$)

1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)

2. Foreign Travel Costs

Total Travel Cost

E. Participant/Trainee Support Costs

Funds Requested (\$)

1. Tuition/Fees/Health Insurance

2. Stipends

3. Travel

4. Subsistence

5. Other:

Number of Participants/Trainees

Total Participant/Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 2

* UEI: CWKJEXDG79A7

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: Pacific Northwest National Laboratory

* Start Date: 07-01-2025

* End Date: 06-30-2026

Budget Period: 2

F. Other Direct Costs		Funds Requested (\$)
1. Materials and Supplies		3,180.00
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Procurement Svc Chgs		165.00
9. Waste Disposal Svc Cntr		17.00
Total Other Direct Costs		3,362.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	100,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	* Funds Requested (\$)
1. Other	12.53	94,802.00	11,879.00
2. Other	4.4	20,430.00	899.00
3. Other	29.3	109,278.00	32,018.00
4. Other	4.11	145,443.00	5,978.00
Total Indirect Costs			50,774.00
Cognizant Federal Agency			
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	150,774.00

J. Fee	Funds Requested (\$)
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K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	150,774.00

L. * Budget Justification	File Name:	Mime Type: application/pdf
	7__81394_Budget_Justification1041563837.pdf	
	(Only attach one file.)	

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 3

* UEI: CWKJEXDG79A7

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: Pacific Northwest National Laboratory

* Start Date: 07-01-2026 * End Date: 06-30-2027 Budget Period: 3

A. Senior/Key Person												
Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary	Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested (\$)
						(\$)	Months	Months	Months	Salary (\$)	Benefits (\$)	
1.	Thomas		Metz		PD/PI	0.00	0.6			10,072.00	2,679.00	12,751.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:		Mime Type:		Total Senior/Key Person				12,751.00	

B. Other Personnel												
* Number of		* Project Role					Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested
Personnel							Months	Months	Months	Salary (\$)	Benefits	(\$)
1	Post Doctoral Associates						8.6			69,498.00	12,579.00	82,077.00
	Graduate Students											
	Undergraduate Students											
	Secretarial/Clerical											
1	Total Number Other Personnel										Total Other Personnel	82,077.00
							Total Salary, Wages and Fringe Benefits (A+B)					94,828.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 3

* UEI: CWKJEXDG79A7

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: Pacific Northwest National Laboratory

* Start Date: 07-01-2026

* End Date: 06-30-2027

Budget Period: 3

C. Equipment Description		
List items and dollar amount for each item exceeding \$5,000		
Equipment Item		* Funds Requested (\$)
Total funds requested for all equipment listed in the attached file		
Total Equipment		
Additional Equipment:	File Name:	Mime Type:

D. Travel	Funds Requested (\$)
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	1,909.00
2. Foreign Travel Costs	
Total Travel Cost	1,909.00

E. Participant/Trainee Support Costs	Funds Requested (\$)
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees	Total Participant/Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 3

* UEI: CWKJEXDG79A7

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: Pacific Northwest National Laboratory

* Start Date: 07-01-2026

* End Date: 06-30-2027

Budget Period: 3

F. Other Direct Costs		Funds Requested (\$)
1. Materials and Supplies		2,936.00
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. procurement svcs chgs		153.00
9. Waste Disposal Svc Ctr		174.00
Total Other Direct Costs		3,263.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	100,000.00

H. Indirect Costs			
	Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$) * Funds Requested (\$)
1. Other		12.35	94,802.00 11,708.00
2. Other		4.4	22,239.00 979.00
3. Other		29.3	109,376.00 32,047.00
4. Other		4.11	145,389.00 5,975.00
Total Indirect Costs			50,709.00
Cognizant Federal Agency			
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	150,709.00

J. Fee	Funds Requested (\$)
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K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	150,709.00

L. * Budget Justification	File Name:	Mime Type: application/pdf
	7__81394_Budget_Justification1041563837.pdf	
	(Only attach one file.)	

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 4

* UEI: CWKJEXDG79A7

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: Pacific Northwest National Laboratory

* Start Date: 07-01-2027 * End Date: 06-30-2028 Budget Period: 4

A. Senior/Key Person												
Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary	Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested (\$)
						(\$)	Months	Months	Months	Salary (\$)	Benefits (\$)	
1.	Thomas		Metz		PD/PI	0.00	0.6			10,040.00	2,671.00	12,711.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:		Mime Type:		Total Senior/Key Person				12,711.00	

B. Other Personnel												
* Number of		* Project Role					Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested
Personnel							Months	Months	Months	Salary (\$)	Benefits	(\$)
1	Post Doctoral Associates						8.2			68,891.00	12,469.00	81,360.00
	Graduate Students											
	Undergraduate Students											
	Secretarial/Clerical											
1	Total Number Other Personnel										Total Other Personnel	81,360.00
							Total Salary, Wages and Fringe Benefits (A+B)					94,071.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 4

* UEI: CWKJEXDG79A7

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: Pacific Northwest National Laboratory

* Start Date: 07-01-2027

* End Date: 06-30-2028

Budget Period: 4

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment Item

* Funds Requested (\$)

Total funds requested for all equipment listed in the attached file

Total Equipment

Additional Equipment:

File Name:

Mime Type:

D. Travel

Funds Requested (\$)

1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)

1,952.00

2. Foreign Travel Costs

Total Travel Cost

1,952.00

E. Participant/Trainee Support Costs

Funds Requested (\$)

1. Tuition/Fees/Health Insurance

2. Stipends

3. Travel

4. Subsistence

5. Other:

Number of Participants/Trainees

Total Participant/Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 4

* UEI: CWKJEXDG79A7

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: Pacific Northwest National Laboratory

* Start Date: 07-01-2027

* End Date: 06-30-2028

Budget Period: 4

F. Other Direct Costs		Funds Requested (\$)
1. Materials and Supplies		3,615.00
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Procurement Svc Chgs		188.00
9. Waste Disposal Svc Cntr		174.00
Total Other Direct Costs		3,977.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	100,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	* Funds Requested (\$)
1. Other	12.07	94,802.00	11,443.00
2. Other	4.4	22,976.00	1,011.00
3. Other	29.3	108,461.00	31,779.00
4. Other	4.11	144,882.00	5,955.00
Total Indirect Costs			50,188.00
Cognizant Federal Agency			
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	150,188.00

J. Fee	Funds Requested (\$)
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K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	150,188.00

L. * Budget Justification	File Name:	Mime Type: application/pdf
	7__81394_Budget_Justification1041563837.pdf	
	(Only attach one file.)	

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - Cumulative Budget

	Totals (\$)	
Section A, Senior/Key Person		51,016.00
Section B, Other Personnel		329,323.00
Total Number Other Personnel	4	
Total Salary, Wages and Fringe Benefits (A+B)		380,339.00
Section C, Equipment		
Section D, Travel		5,685.00
1. Domestic	5,685.00	
2. Foreign		
Section E, Participant/Trainee Support Costs		
1. Tuition/Fees/Health Insurance		
2. Stipends		
3. Travel		
4. Subsistence		
5. Other		
6. Number of Participants/Trainees		
Section F, Other Direct Costs		13,976.00
1. Materials and Supplies	12,773.00	
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Other 1	664.00	
9. Other 2	539.00	
10. Other 3		
11. Other 4		
12. Other 5		
13. Other 6		
14. Other 7		
15. Other 8		
16. Other 9		
17. Other 10		
Section G, Direct Costs (A thru F)		400,000.00
Section H, Indirect Costs		199,792.00
Section I, Total Direct and Indirect Costs (G + H)		599,792.00
Section J, Fee		
Section K, Total Costs and Fee (I + J)		599,792.00

BUDGET JUSTIFICATION

Senior/Key Personnel

Thomas O. Metz, Ph.D., PD/PI, (.6 calendar months in all years) Dr. Metz has over 19 years of experience in project management and in administration of large research efforts and is the author/co-author of >165 scientific publications in the fields of metabolomics, lipidomics, and proteomics. His research efforts involve the development of high-throughput mass spectrometry-based methods for metabolomics and lipidomics applications, and application of these together with proteomics in systems biology studies of disease. Dr. Metz has many years of experience in applying omics measurements in the study of both chronic and infectious disease. Currently, he is the PI of the Proteomics Laboratory for The Environmental Determinants of Diabetes in the Young consortium (TEDDY), PI of a project within the Consortium on Beta Cell Death and Survival within the Human Islet Research Network, and PI of the NIH Metabolomics Common Fund Metabolomics Program-funded Pacific Northwest Advanced Compound Identification Core. Previously, he was PI of the Metabolomics Core for the NIH Common Fund Undiagnosed Diseases Network. Dr. Metz will have primary responsibility for selecting the staff and directing the research conducted at PNNL. He will also participate, together with Drs. Meehan and Webb-Robertson overall study design and execution, as well as analysis and interpretation of results.

Other Personnel

Post-doctoral Fellows, Ph.D. (Yr. 1 9.3 in calendar months, 8.8 in calendar months Yr. 2, 8.6 calendar months in Yr., and 4 8.2 calendar months in year 4) will be responsible for conducting specific mentored research projects including variously applications of PTM enrichment or discovery efforts, targeted verification efforts, data analysis and integration using inference and pathway tools, statistical analysis of resulting data under the supervision the appropriate project leaders and with training of the Center team members. The post-doctoral fellow will have important responsibilities in developing presentations and manuscripts.

Supplies (Yr. 1 -\$3042, Yr. 2 - \$3,179, Yr. 3 - \$2,936, Yr. 4 - \$3,615)

The supply request includes general and specialized laboratory supplies for sample preparation and instrument operation and maintenance. General and specialized laboratory supplies related to sample preparation include: chemicals, organic solvents for metabolite extraction, chemical derivatization reagents for GC-MS analysis, chemicals to extract and denature proteins, trypsin, buffers, solid phase extraction columns, chemical labeling reagents for peptides (e.g. iTRAQ or TMT reagents), autosampler glass vials with snap caps and low volume inserts, syringes, disposable gloves, disposable plastic ware such as pipette tips and sterile, siliconized microcentrifuge tubes. General and specialized laboratory supplies related to instrument operation and maintenance include: high pressure LC valves and seals, LC column material such as column packing materials and capillary fused silica for in-house high pressure capillary column packing, GC columns, mobile phase reagents including HPLC-grade acetonitrile, methanol, isopropanol and HPLC-grade water, as well as MS-grade ammonium acetate, vacuum pump repair supplies, and routine maintenance supplies.

Travel (Yr. 1 -\$1,824, Yr. 3 - \$1,909, Yr. 4 - \$1952)

Travel includes two trips in year 1 for three staff members to participate in the semi-annual PI meetings. In years 2-5 travel is included for two staff to attend the annual PI meeting as well as one additional meeting per year for 2 staff members to participate in other relevant scientific meetings.

Other Direct Costs (Yr. 1 -\$333, Yr. 2 - \$183, Yr. 3 - \$327, Yr. 4 - \$362)

Other expenses include procurement service charges and waste management fees. **Procurement Service Charge:** The procurement service charge is support costs for acquisition of goods and services that are recovered by applying the appropriate rate to all cost objectives. **Waste Management Fee:** As part of this proposal, we estimated the amount of waste that will be generated while performing the research. Waste costs are project specific and are viewed as direct project costs. These estimates are consistent with historical research costs at the Laboratory.

Indirect Costs

Rates shown in the Indirect Cost section will not match our rate notification that are submitted to DOE. PNNL's disclosed accounting practice is to apply organizational overhead on a dollar per hour basis. Each directorate calculates an organizational overhead based on the specific needs of that directorate. For this proposal, we have combined those organizational overheads into one amount and converted that to a weighted percentage rate and base that spans multiple fiscal years. In addition, the budget years of this proposal cross multiple fiscal years and in many cases our forecasted

indirect rates change between fiscal years. We have calculated a weighted composite rate that crosses multiple fiscal years for the rates and base of costs incurred that fall within the same budget year for the purpose of this proposal.

The indirect costs compound upon each other and have been calculated based upon the following:

Organizational Overhead

Organizational Overhead for technical organizations represents costs for management, supervision, and administration of technical departments. Organizational Overhead for each respective research organization also includes costs for building and utilities, small tools, lab supplies, laundry, maintenance, and expenses associated with equipment unless the equipment is assigned to a specific equipment center. In addition, Organizational Overhead includes allocation from the Minor Construction and Equipment Pool (MCE) allocated as an hourly rate to a base of research labor hours. This pool includes the cost of minor construction and equipment benefiting multiple research organizations.

Program Development and Management (PDM)

The Program Development and Management (PDM) pool is used to accumulate the costs associated with business development and program integration activities. PDM is allocated to objectives by applying the appropriate rate to value-added (excluding PDM) costs, plus materials and subcontracts (excluding SEE, IWO and ILA) costs.

General and Administrative Expenses (G&A)

The allocation base for G&A expenses is value added to final cost objectives. The value-added base includes labor, travel, service and equipment centers, organizational overhead, Program Development and Management, building and utility costs and other direct costs. Excluded from the base costs includes procurements, subcontracts, Science Engineering and Education (SEE), and other Hanford contractor services.

Contract Performance Fee

Contract Performance Fee is a cost-based charge that recovers, on behalf of the Department of Energy (DOE), award paid to the Management and Operations (M&O) contractor of Pacific Northwest National Laboratory. Costs are allocated on a fixed pool at the rate necessary to recover DOE's total estimated costs, as well as variances between estimated and actual cost, which are adjusted at the end of each fiscal year.

Safeguards and Security

Safeguards and Security (SAS) program is designed to ensure appropriate security measures are in place to support the mission requirements and to protect critical assets within the laboratory. This program also provides physical controls that will mitigate possible risks to employees, nuclear and special materials, classified and sensitive information, and facilities. The program also provides funding for cybersecurity for the laboratory's information technology systems. DOE direct funds SAS expenses for their work so this rate is applied to all Strategic Partnership Program (SPP), Department of Homeland Security (DHS), and Agreements to Commercialize Technology (ACT) projects. SAS is allocated to final objectives by applying the rate to the value-added base. The value-added base includes labor, travel, service and equipment centers, organizational overhead, program development and management, building and utility costs and other direct costs. Excluded from the base costs: procurements, subcontracts, Science and Engineering Education (SEE), and Other Hanford Contractor (OHC) services.

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

* UEI: GNJ7BBP73WE9

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: The Regents of the University of Michigan

* Start Date: 07-01-2024 * End Date: 06-30-2025 Budget Period: 1

A. Senior/Key Person												
Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary	Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested (\$)
						(\$)	Months	Months	Months	Salary (\$)	Benefits (\$)	
1.	Charles		Evans		PD/PI	0.00	1.8			19,732.00	6,314.00	26,046.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:		Mime Type:		Total Senior/Key Person				26,046.00	

B. Other Personnel												
* Number of		* Project Role					Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested
Personnel							Months	Months	Months	Salary (\$)	Benefits	(\$)
1	Post Doctoral Associates						6			28,440.00	9,101.00	37,541.00
	Graduate Students											
	Undergraduate Students											
	Secretarial/Clerical											
1	analyst						1.2			10,726.00	3,432.00	14,158.00
2	Total Number Other Personnel									Total Other Personnel		51,699.00
							Total Salary, Wages and Fringe Benefits (A+B)					77,745.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

* UEI: GNJ7BBP73WE9

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: The Regents of the University of Michigan

* Start Date: 07-01-2024

* End Date: 06-30-2025

Budget Period: 1

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment Item

* Funds Requested (\$)

Total funds requested for all equipment listed in the attached file

Total Equipment

Additional Equipment:

File Name:

Mime Type:

D. Travel

Funds Requested (\$)

1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)

2,500.00

2. Foreign Travel Costs

Total Travel Cost

2,500.00

E. Participant/Trainee Support Costs

Funds Requested (\$)

1. Tuition/Fees/Health Insurance

2. Stipends

3. Travel

4. Subsistence

5. Other:

Number of Participants/Trainees

Total Participant/Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 1

* UEI: GNJ7BBP73WE9

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: The Regents of the University of Michigan

* Start Date: 07-01-2024

* End Date: 06-30-2025

Budget Period: 1

F. Other Direct Costs		Funds Requested (\$)
1. Materials and Supplies		19,755.00
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
Total Other Direct Costs		19,755.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	100,000.00

H. Indirect Costs				
	Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	* Funds Requested (\$)
1. MTDC		56	100,000.00	56,000.00
Total Indirect Costs				56,000.00
Cognizant Federal Agency		DHHS, Olulola Oluborode 301-492-4065		
(Agency Name, POC Name, and POC Phone Number)				

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	156,000.00

J. Fee	Funds Requested (\$)
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K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	156,000.00

L. * Budget Justification	File Name: Justification_UM1041563841.pdf	Mime Type: application/pdf
(Only attach one file.)		

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 2

* UEI: GNJ7BBP73WE9

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: The Regents of the University of Michigan

* Start Date: 07-01-2025 * End Date: 06-30-2026 Budget Period: 2

A. Senior/Key Person												
Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary	Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested (\$)
						(\$)	Months	Months	Months	Salary (\$)	Benefits (\$)	
1.	Charles		Evans		PD/PI	0.00	1.8			19,732.00	6,314.00	26,046.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:		Mime Type:		Total Senior/Key Person				26,046.00	

B. Other Personnel													
* Number of		* Project Role					Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested	
Personnel							Months	Months	Months	Salary (\$)	Benefits	(\$)	
1	Post Doctoral Associates						6			28,440.00	9,101.00	37,541.00	
	Graduate Students												
	Undergraduate Students												
	Secretarial/Clerical												
1	Analyst						1.2			10,726.00	3,432.00	14,158.00	
2	Total Number Other Personnel								Total Other Personnel			51,699.00	
							Total Salary, Wages and Fringe Benefits (A+B)						77,745.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 2

* UEI: GNJ7BBP73WE9

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: The Regents of the University of Michigan

* Start Date: 07-01-2025

* End Date: 06-30-2026

Budget Period: 2

C. Equipment Description		
List items and dollar amount for each item exceeding \$5,000		
Equipment Item		* Funds Requested (\$)
Total funds requested for all equipment listed in the attached file		
Total Equipment		
Additional Equipment:	File Name:	Mime Type:

D. Travel	Funds Requested (\$)
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	2,500.00
2. Foreign Travel Costs	
Total Travel Cost	2,500.00

E. Participant/Trainee Support Costs	Funds Requested (\$)
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees	Total Participant/Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 2

* UEI: GNJ7BBP73WE9

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: The Regents of the University of Michigan

* Start Date: 07-01-2025

* End Date: 06-30-2026

Budget Period: 2

F. Other Direct Costs	Funds Requested (\$)
1. Materials and Supplies	19,755.00
2. Publication Costs	
3. Consultant Services	
4. ADP/Computer Services	
5. Subawards/Consortium/Contractual Costs	
6. Equipment or Facility Rental/User Fees	
7. Alterations and Renovations	
Total Other Direct Costs	19,755.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	100,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	* Funds Requested (\$)
1. MTDC	56	100,000.00	56,000.00
		Total Indirect Costs	56,000.00
Cognizant Federal Agency			
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	156,000.00

J. Fee	Funds Requested (\$)
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K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	156,000.00

L. * Budget Justification	File Name: Justification_UM1041563841.pdf	Mime Type: application/pdf
	(Only attach one file.)	

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 3

* UEI: GNJ7BBP73WE9

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: The Regents of the University of Michigan

* Start Date: 07-01-2026 * End Date: 06-30-2027 Budget Period: 3

A. Senior/Key Person												
Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary	Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested (\$)
						(\$)	Months	Months	Months	Salary (\$)	Benefits (\$)	
1.	Charles		Evans		PD/PI	0.00	1.8			19,732.00	6,314.00	26,046.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:		Mime Type:		Total Senior/Key Person				26,046.00	

B. Other Personnel											
* Number of		* Project Role				Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested
Personnel						Months	Months	Months	Salary (\$)	Benefits	(\$)
1	Post Doctoral Associates					6			28,440.00	9,101.00	37,541.00
	Graduate Students										
	Undergraduate Students										
	Secretarial/Clerical										
1	analyst					1.2			10,726.00	3,432.00	14,158.00
2	Total Number Other Personnel								Total Other Personnel		51,699.00
Total Salary, Wages and Fringe Benefits (A+B)											77,745.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 3

* **UEI:** GNJ7BBP73WE9

* **Budget Type:** ☐ Project ☒ Subaward/Consortium

Enter name of Organization: The Regents of the University of Michigan

* **Start Date:** 07-01-2026

* **End Date:** 06-30-2027

Budget Period: 3

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment Item

* Funds Requested (\$)

Total funds requested for all equipment listed in the attached file

Total Equipment

Additional Equipment:

File Name:

Mime Type:

D. Travel

Funds Requested (\$)

1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)

2,500.00

2. Foreign Travel Costs

Total Travel Cost

2,500.00

E. Participant/Trainee Support Costs

Funds Requested (\$)

1. Tuition/Fees/Health Insurance

2. Stipends

3. Travel

4. Subsistence

5. Other:

Number of Participants/Trainees

Total Participant/Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 3

* UEI: GNJ7BBP73WE9

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: The Regents of the University of Michigan

* Start Date: 07-01-2026

* End Date: 06-30-2027

Budget Period: 3

F. Other Direct Costs	Funds Requested (\$)
1. Materials and Supplies	19,755.00
2. Publication Costs	
3. Consultant Services	
4. ADP/Computer Services	
5. Subawards/Consortium/Contractual Costs	
6. Equipment or Facility Rental/User Fees	
7. Alterations and Renovations	
Total Other Direct Costs	19,755.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	100,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	* Funds Requested (\$)
1. Other	56	100,000.00	56,000.00
Total Indirect Costs			56,000.00
Cognizant Federal Agency			
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	156,000.00

J. Fee	Funds Requested (\$)
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K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	156,000.00

L. * Budget Justification	File Name: Justification_UM1041563841.pdf	Mime Type: application/pdf
(Only attach one file.)		

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 4

* UEI: GNJ7BBP73WE9

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: The Regents of the University of Michigan

* Start Date: 07-01-2027 * End Date: 06-30-2028 Budget Period: 4

A. Senior/Key Person												
Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary	Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested (\$)
						(\$)	Months	Months	Months	Salary (\$)	Benefits (\$)	
1.	Charles		Evans		PD/PI	0.00	1.8			19,732.00	6,314.00	26,046.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:		Mime Type:		Total Senior/Key Person				26,046.00	

B. Other Personnel													
* Number of		* Project Role					Cal.	Acad.	Sum.	* Requested	* Fringe	* Funds Requested	
Personnel							Months	Months	Months	Salary (\$)	Benefits	(\$)	
1	Post Doctoral Associates						6			28,440.00	9,101.00	37,541.00	
	Graduate Students												
	Undergraduate Students												
	Secretarial/Clerical												
1	Analyst						1.2			10,726.00	3,432.00	14,158.00	
2	Total Number Other Personnel									Total Other Personnel		51,699.00	
							Total Salary, Wages and Fringe Benefits (A+B)					77,745.00	

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 4

* UEI: GNJ7BBP73WE9

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: The Regents of the University of Michigan

* Start Date: 07-01-2027

* End Date: 06-30-2028

Budget Period: 4

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment Item

* Funds Requested (\$)

Total funds requested for all equipment listed in the attached file

Total Equipment

Additional Equipment:

File Name:

Mime Type:

D. Travel

Funds Requested (\$)

1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)

2,500.00

2. Foreign Travel Costs

Total Travel Cost

2,500.00

E. Participant/Trainee Support Costs

Funds Requested (\$)

1. Tuition/Fees/Health Insurance

2. Stipends

3. Travel

4. Subsistence

5. Other:

Number of Participants/Trainees

Total Participant/Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 4

* UEI: GNJ7BBP73WE9

* Budget Type: ☐ Project ☒ Subaward/Consortium

Enter name of Organization: The Regents of the University of Michigan

* Start Date: 07-01-2027

* End Date: 06-30-2028

Budget Period: 4

F. Other Direct Costs	Funds Requested (\$)
1. Materials and Supplies	19,755.00
2. Publication Costs	
3. Consultant Services	
4. ADP/Computer Services	
5. Subawards/Consortium/Contractual Costs	
6. Equipment or Facility Rental/User Fees	
7. Alterations and Renovations	
Total Other Direct Costs	19,755.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	100,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	* Funds Requested (\$)
1. MTDC	56	100,000.00	56,000.00
Total Indirect Costs			56,000.00
Cognizant Federal Agency		DHHS Olulola Oluborode 301-492-4065	
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	156,000.00

J. Fee	Funds Requested (\$)
---------------	-----------------------------

K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	156,000.00

L. * Budget Justification	File Name: Justification_UM1041563841.pdf	Mime Type: application/pdf
(Only attach one file.)		

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - Cumulative Budget

	Totals (\$)	
Section A, Senior/Key Person		104,184.00
Section B, Other Personnel		206,796.00
Total Number Other Personnel	8	
Total Salary, Wages and Fringe Benefits (A+B)		310,980.00
Section C, Equipment		
Section D, Travel		10,000.00
1. Domestic	10,000.00	
2. Foreign		
Section E, Participant/Trainee Support Costs		
1. Tuition/Fees/Health Insurance		
2. Stipends		
3. Travel		
4. Subsistence		
5. Other		
6. Number of Participants/Trainees		
Section F, Other Direct Costs		79,020.00
1. Materials and Supplies	79,020.00	
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Other 1		
9. Other 2		
10. Other 3		
11. Other 4		
12. Other 5		
13. Other 6		
14. Other 7		
15. Other 8		
16. Other 9		
17. Other 10		
Section G, Direct Costs (A thru F)		400,000.00
Section H, Indirect Costs		224,000.00
Section I, Total Direct and Indirect Costs (G + H)		624,000.00
Section J, Fee		
Section K, Total Costs and Fee (I + J)		624,000.00

Budget Justification – University of Michigan

Personnel

Charles Evans, PhD. Principal Investigator (1.8 calendar months)

Alongside PIs Oliver Fiehn and Thomas Metz, Dr. Evans will contribute to scientific and technical leadership of this study. He has significant expertise in metabolomics and lipidomics, including instrumental analysis and data processing. Dr. Evans will oversee, work closely with, and provide training for a Postdoctoral Research Fellow regarding tasks required by the study, which will include sample preparation, instrumental analysis, and data analysis. He will also oversee and regularly interact with programmer Sasha Raskind regarding his contributions to this project. When appropriate, Dr. Evans will also directly contribute to data collection and data analysis tasks. Dr. Evans will be primarily responsible for managing data reporting and will ensure data generated at UM is promptly shared with collaborators at the other participating institutions. He will also oversee contributions to publications and presentations to be prepared regarding data collected during this project.

In addition to these scientific roles, Dr. Evans will also oversee all administrative aspects of the portion of the grant housed at the University of Michigan. These tasks will include budget management, personnel hiring, timeline management and RPRR reporting.

Alexander (Sasha) Raskind, PhD. Programmer (1.2 calendar months)

Dr. Raskind is an expert data analyst and software programmer. He is proficient in with multiple programming languages including C++, Java, Python, and R. In coordination with Dr. Evans, Dr. Raskind will contribute to data analysis efforts. He will also be responsible for software development when needed, with the goals of 1) automating data analysis tasks to be performed at University of Michigan, and 2) to contributing to development of the main software tool, LC BinBase, which will be managed by UC-Davis.

Postdoctoral Research Fellow (TBD), (6 calendar months)

A postdoctoral research fellow will be recruited to contribute to completion of the aims of this project; we will use our network of candidates with peer metabolomics laboratories to seek out a candidate with prior experience in metabolomics and/or liquid chromatography/mass spectrometry and/or metabolomics data analysis. To whatever extent required or requested, the postdoctoral fellow will receive training in metabolomics methods from Dr. Evans. His or her primary contributions will pertain to sample preparation, instrumental analysis, and data analysis, especially with respect to the detailed review of data acquired by all participating labs. The postdoctoral fellow will be expected and encouraged to attend meetings and participate in discussions with collaborators in the project, to contribute to authorship of scientific publications regarding findings, and to present research results at national or international meetings.

Travel

We have budgeted \$2500/year to help support the PI and/or research staff's attendance of and presentation of research findings at national or international conferences. Such conferences include, but are not limited to, the American Society for Mass Spectrometry Conference, the Metabolomics Society Conference, and the Metabolomics Society of North America Conference.

Materials and Supplies

We have budgeted approximately \$19,755 annually for necessary supplies and maintenance items to carry out extraction and LC-MS-based metabolomics analysis of the samples to be analyzed at the University of Michigan. These include chromatographic columns, autosampler vials and caps, instrument consumables, solvents and chemical additives, instrument gases, pipette tips, gloves, disposable vials and tubes, authentic standard compounds and stable isotope labeled internal standards.

Data management and Sharing Justification

Data sharing is an important component of our proposed work. We have budgeted support effort required to curate and share both raw and processed data. We intend to upload all raw and processed data generated through this project to public data archive sites, including Metabolomics Workbench. Tasks required to upload data to such archives include curation of data into required tabular format, completion of forms and metadata tables, standardization of metabolite identifier information, and review of data submissions for accuracy and

completeness. As required, funds in this category may also be used in support of open-access journal publication fees (ensuring all journals are PubMed indexed and compliant with NIH data access policies). We will also coordinate with other participating labs to coordinate data sharing format and, if and when appropriate, to compile and share data in a single archive rather than an individual archive from each site.

PHS 398 Cover Page Supplement

1. Vertebrate Animals Section

Are vertebrate animals euthanized?

☐ Yes

☐ No

If "Yes" to euthanasia

Is method consistent with American Veterinary

Medical Association (AVMA) guidelines?

☐ Yes

☐ No

If "No" to AVMA guidelines, describe method and
provide scientific justification

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

☐ Yes

☒ No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period

*Anticipated Amount (\$)

*Source(s)

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells?

☐ Yes

☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: https://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, please check the box indicating that one from the registry will be used:

Cell Line(s):

☐ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions?

☐ Yes

☒ No

If "Yes" then provide the HFT Compliance Assurance: File Name: Mime Type:

If "Yes" then provide the HFT Sample IRB Consent Form File Name: Mime Type:

PHS 398 Cover Page Supplement

5. Inventions and Patents Section (RENEWAL)

*Inventions and Patents: ☐ Yes ☐ No

If the answer is "Yes" then please answer the following:

*Previously reported: ☐ Yes ☐ No

6. Change of Investigator / Change of Institution Section

☐ Change of Project Director / Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

☐ Change of Grantee Institution

*Name of former institution:

PHS 398 Modular Budget, Periods 1 and 2

OMB Number: 0925-0001

Budget Period: 1				
Start Date:		07/01/2024	End Date: 06/30/2025	
A. Direct Costs			Funds Requested (\$)	
			* Direct Cost less Consortium Indirect (F&A)	500,000.00
			Consortium Indirect (F&A)	104,121.00
			* Total Direct Costs	604,121.00
B. Indirect (F&A) Costs				
	Indirect (F&A) Type	Indirect (F&A) Rate (%)	Indirect (F&A) Base (\$)	* Funds Requested (\$)
1.	On Campus Org Research MTDC	61.00	349,999.00	213,499.00
2.				
3.				
4.				
Cognizant Agency (Agency Name, POC Name and Phone Number) HHS Jeffrey Warren 415-437-7820				
Indirect (F&A) Rate Agreement Date		06/28/2021	Total Indirect (F&A) Costs	213,499.00
C. Total Direct and Indirect (F&A) Costs (A + B)			Funds Requested (\$) 817,620.00	
Budget Period: 2				
Start Date:		07/01/2025	End Date: 06/30/2026	
A. Direct Costs			Funds Requested (\$)	
			* Direct Cost less Consortium Indirect (F&A)	500,000.00
			Consortium Indirect (F&A)	106,774.00
			* Total Direct Costs	606,774.00
B. Indirect (F&A) Costs				
	Indirect (F&A) Type	Indirect (F&A) Rate (%)	Indirect (F&A) Base (\$)	* Funds Requested (\$)
1.	On Campus Org Research MTDC	61.00	299,999.00	182,999.00
2.				
3.				
4.				
Cognizant Agency (Agency Name, POC Name and Phone Number) HHS Jeffrey Warren 415-437-7820				
Indirect (F&A) Rate Agreement Date		06/28/2021	Total Indirect (F&A) Costs	182,999.00
C. Total Direct and Indirect (F&A) Costs (A + B)			Funds Requested (\$) 789,773.00	

PHS 398 Modular Budget, Periods 3 and 4

OMB Number: 0925-0001

Budget Period: 3			
Start Date: 07/01/2026		End Date: 06/30/2027	
A. Direct Costs			Funds Requested (\$)
* Direct Cost less Consortium Indirect (F&A)			500,000.00
Consortium Indirect (F&A)			106,709.00
* Total Direct Costs			606,709.00
B. Indirect (F&A) Costs			
Indirect (F&A) Type	Indirect (F&A) Rate (%)	Indirect (F&A) Base (\$)	* Funds Requested (\$)
1. On Campus Org Research MTDC	61.00	299,999.00	182,999.00
2.			
3.			
4.			
Cognizant Agency (Agency Name, POC Name and Phone Number) HHS Jeffrey Warren 415-437-7820			
Indirect (F&A) Rate Agreement Date 06/28/2021		Total Indirect (F&A) Costs 182,999.00	
C. Total Direct and Indirect (F&A) Costs (A + B)			Funds Requested (\$) 789,708.00
Budget Period: 4			
Start Date: 07/01/2027		End Date: 06/30/2028	
A. Direct Costs			Funds Requested (\$)
* Direct Cost less Consortium Indirect (F&A)			500,000.00
Consortium Indirect (F&A)			106,188.00
* Total Direct Costs			606,188.00
B. Indirect (F&A) Costs			
Indirect (F&A) Type	Indirect (F&A) Rate (%)	Indirect (F&A) Base (\$)	* Funds Requested (\$)
1. On Campus Org Research MTDC	61.00	299,999.00	182,999.00
2.			
3.			
4.			
Cognizant Agency (Agency Name, POC Name and Phone Number) HHS Jeffrey Warren 415-437-7820			
Indirect (F&A) Rate Agreement Date 06/28/2021		Total Indirect (F&A) Costs 182,999.00	
C. Total Direct and Indirect (F&A) Costs (A + B)			Funds Requested (\$) 789,187.00

PHS 398 Modular Budget, Period 5 and Cumulative

OMB Number: 0925-0001

Budget Period: 5			
Start Date:		End Date:	
A. Direct Costs			Funds Requested (\$)
* Direct Cost less Consortium Indirect (F&A)			
Consortium Indirect (F&A)			
* Total Direct Costs			
B. Indirect (F&A) Costs			
Indirect (F&A) Type	Indirect (F&A) Rate (%)	Indirect (F&A) Base (\$)	* Funds Requested (\$)
1.			
2.			
3.			
4.			
Cognizant Agency (Agency Name, POC Name and Phone Number)			
Indirect (F&A) Rate Agreement Date		Total Indirect (F&A) Costs	
C. Total Direct and Indirect (F&A) Costs (A + B)			Funds Requested (\$)
Cumulative Budget Information			
1. Total Costs, Entire Project Period			
* Section A, Total Direct Cost less Consortium Indirect (F&A) for Entire Project Period		\$	2,000,000.00
Section A, Total Consortium Indirect (F&A) for Entire Project Period		\$	423,792.00
* Section A, Total Direct Costs for Entire Project Period		\$	2,423,792.00
* Section B, Total Indirect (F&A) Costs for Entire Project Period		\$	762,496.00
* Section C, Total Direct and Indirect (F&A) Costs (A+B) for Entire Project Period		\$	3,186,288.00
2. Budget Justifications			
Personnel Justification			
Consortium Justification			
Additional Narrative Justification			

Attachments

PersonnelJustification_attDataGroup0

File Name

Mime Type

ConsortiumJustification_attDataGroup0

File Name

Mime Type

AdditionalNarrativeJustification_attDataGroup0

File Name

Mime Type

PHS 398 Research Plan

Introduction

1. Introduction to Application
(or Resubmission and Revision applications)

Research Plan Section

2. Specific Aims Specific_Aims__R01_DB__09_26_20231041564387.pdf
3. Research Strategy* Research_Strategy_R01_DB_9_26_231041564448.pdf
4. Progress Report Publication List

Other Research Plan Section

5. Vertebrate Animals
6. Select Agent Research
7. Multiple PD/PI Leadership Plan Leadership_plan_R011041564388.pdf
8. Consortium/Contractual Arrangements CA_R01_Fiehn_Evans_Metz_02_20231041306878.pdf
9. Letters of Support
10. Resource Sharing Plan(s) Resource_Sharing_Plan_Fiehn_9_231041306896.pdf
11. Other Plan(s) DM_DS_Plan_R01_Fiehn_Evans_Metz_9_231041306895.pdf
12. Authentication of Key Biological and/or Chemical Resources

13. Appendix

12. Appendix

SPECIFIC AIMS

Metabolomics has matured over the past 20 years. It is now time to get to the next stage: make it standardized, harmonized and informative. Unlike for genetics and transcriptomics (and to some extent, proteomics), biomedical researchers remain uneasy about the types of metabolomics data they would receive for their studies, how fast they would get these data, how they could merge and compare data across different studies, and how they could interpret the data in line with disease phenotypes, animal models or cellular characteristics.

The three investigators at UC Davis, Pacific Northwest National Laboratory (PNNL) and University of Michigan have prototyped standardized methods towards unified data processing and compound identification in a manual way. This data processing approach combines a range of informatics and machine learning tools that we developed over the past 20 years, including MS-DIAL data processing software, ISICLE and autoCCS ion mobility software, MS-FINDER compound identification, and MassBank.us public mass spectral repositories. We will now advance beyond this manual data processing workflow by developing LC-BinBase, a new automated data processing pipeline and associated database, and showcasing its use in metabolomics pilot studies.

We will test our driving hypothesis that we can standardize untargeted metabolomics and lipidomics data processing to reproducibly acquire, compare and interpret biomedical metabolomics data within and across different mass spectrometry platforms, across laboratories, and for different types of matrices. We will advance automatic annotation of metabolites and lipids by integrating retention times, positive/negative electrospray precursor accurate masses, ion mobility collision cross sections and MS/MS fragmentations into calculated confidence scores. We will further develop tools for improved data interpretation of metabolomics and lipidomics results by integrating standardized naming and database identifiers for downstream pathway analyses.

These objectives will be met by the following Specific Aims:

Specific Aim 1: Develop LC-BinBase with standardized peak reports across assays and matrices.

We will develop improved schemas for untargeted metabolomics data processing by using internal standards-based retention time normalization instead of classic study-based LC-MS/MS retention alignments. We will optimize data acquisition parameters, number and type of internal standards to yield optimal retention error residuals for different assays and matrices using test/validation splits of internal standards. For this purpose, we will acquire data for commercial mouse organs, cell types and different biofluids (including saliva, fecal water, urine, plasma and amniotic fluid). We will test robustness of these methods for fixed versus variable sets of internal standards, and residual error windows across laboratories and across different instrumentation setups. We will further test how well such retention time normalization methods can be extended to similar, but not identical, chromatographic methods. We will then compile data distributions from the alpha-version of our cloud-based data processing database, LC-BinBase, to obtain error probability distributions of normalized retention times, accurate precursor masses, and MS/MS entropy similarity scores.

Specific Aim 2: Develop LC-BinBase with calculated metabolite annotation confidence scores.

Currently, researchers receive untargeted metabolomics data without evidence of plausibility for compound identities. We will develop and implement an automatic MS/MS denoising algorithm based on elemental formula calculations. We will further refine MS/MS based annotations using collision cross section data and retention time predictions to remove outliers outside confidence and prediction intervals. We will then combine probability distributions of accurate mass errors, MS/MS entropy similarity scores and normalized retention times to automatically calculate confidence levels for all annotated metabolites. We will verify these predictions using at least 1,000 newly available commercial standards. Using FlashEntropy scoring on 1 billion publicly available mass spectra, we will classify remaining unknown compounds by relative distance to clusters of confidently annotated compounds. We will release public interfaces to utilize these tools beyond LC-BinBase annotations.

Specific Aim 3: Improve and test the biological usability and scope of LC-BinBase data reports.

LC-BinBase has only been prototyped in interactions between the three partner laboratories. The long-term plan for this work is to generate a national resource and repository using standardized nontargeted methods for open-access data. We will therefore invite interested laboratories to test LC-BinBase to benchmark their own analytical methods against using the standardized LC-BinBase workflows. All metabolite results reported by LC-BinBase include unified mass spectral hash keys to enable merging and comparing data across studies. Biological interpretation of result data for identified metabolites and lipids will be enhanced by adding RefMet, BioPAN, KEGG, ClassyFire, and other domain-based identifiers to utilize data in biomedical studies. We will invite the scientific community to open workshops and online surveys to validate the use of LC-BinBase. These events will also serve to obtain feedback on data interoperability, integration and FAIRness.

RESEARCH STRATEGY

Our research team will focus on the ***following, overarching hypothesis***: We can overcome the current state of metabolomics, which currently depends on error-prone manual data processing. Instead, we will standardize data acquisition and automate data processing into a public database, LC-BinBase, to reproducibly compare and interpret metabolomics data across laboratories, studies and biomedical specimens. Our aims respond to PAR-22-127: *Focused Technology Research and Development* to explicitly develop technologies with existing proof-of-concept but remaining significant technical challenges that prevent full implementation and broad utility. Grant applications must not test specific biological questions.

We will validate LC-BinBase as a public data processing platform to enable metabolomics researchers at-large to process data using standardized untargeted liquid chromatography/mass spectrometry (LC-MS/MS) methods. Key to our approach (for both QTOF- and orbital ion trap mass spectrometers) is retention time locking by dozens of stable-isotope internal standards, unlike current approaches that all rely on total chromatogram similarities. We will further enhance LC-BinBase by calculating the confidence in metabolomic structure annotations by matching predicted to experimental properties (LC retention times¹, accurate mass²⁻⁵, MS/MS entropy similarities⁶ and ion mobility collision cross-sections, CCS)⁷⁻⁹. For all structurally annotated compounds we will add systems biology identifiers to enable biomedical researchers to efficiently use the data in downstream (external) computational software such as pathway mapping or enrichment statistics.

Over the past 10 years, the three co-investigators Fiehn (UC Davis), Metz (PNNL) and Evans (U Michigan) have collaboratively worked towards these aims by developing software for compound identification, by improving data processing workflows, and by applying LC-MS/MS methods to a range of health-related studies. We collaborate with clinical and biomedical scientists to obtain untargeted metabolomic data for hundreds of biomedical projects using a wide array of specimens, including all types of tissues, plasma, serum, other body fluids, and cells. At UC Davis alone, we process more than 40,000 samples for over 400 studies per year in our satellite fee-for-service core for providing untargeted and targeted metabolomics and lipidomics data. Result data are uploaded to the NIH-funded MetabolomicsWorkbench database after publication¹⁰, but a range of fundamental obstacles remain to be solved to enable biomedical researchers to reap the full benefits and promise of metabolomic and lipidomics studies: **consistency, interpretability, throughput, and reproducibility**. At present, more than 100 metabolomics centers and cores across North America offer analytical services¹¹, in addition to commercial providers like Metabolon, Biocrates and Metwarebio. All these cores and companies use liquid chromatography/mass spectrometry (LC-MS/MS) at the heart of their services, but lack cross-referencing of methods, data processing strategies or even consistent identifiers. The Metabolomics Workbench DB uses a stopgap measure (RefMet)¹² to reference annotated metabolites that are uploaded to their repository, but this method does not support identifiers for unknown metabolites, does not validate confidence scores in uploaded data, and does not even require MS/MS data to support claims of compound annotations.

Building on our experience in metabolomic data processing and cheminformatics over the past 15 years^{1-3, 13-28}, the collaborating laboratories have devised common protocols and strategies for unifying data reports across studies and organs, despite using different mass spectrometers. Key to this strategy was to use identical chromatographic conditions and the same set of internal standards that enabled us to align small differences in retention times, and enumerate all identifiable and unknown compounds based on a triad of metadata: **normalized retention time¹, accurate precursor mass³ and MS/MS spectral similarity⁶**. We have used a common data processing pipeline devised at UC Davis, MS-DIAL software¹³⁻¹⁴, and used the largest freely available MS/MS library to identify metabolites, also developed and hosted by the Fiehn laboratory at UC Davis, MassBank of North America (MassBank.us)²⁹. MassBank.us hosts more than 2 million MS/MS spectra that are linked to identified chemical structures (through International Chemical Identifiers, InChI). With LipidBlast^{15-16, 30} included in MassBank.us and MS-DIAL, we semi-quantified 8,051 lipids¹⁴. We have advanced and automated this strategy further into a common pipeline for data processing, data annotation, and metadata reporting, called LC-BinBase. At its heart, LC-BinBase re-uses the successful strategy that we implemented in continuous operation since 2005 for the more straightforward data processing schema in gas chromatography-mass spectrometry (GC-TOF MS), BinBase^{18, 24-25, 31} with its online tool BinDiscover³². BinBase relies on primary peak picking and deconvolution reports based on a specific vendor software, but it was not universally useful across different vendor instruments and data formats. In contrast, MS-DIAL is completely vendor-neutral and works on any type of LC-MS/MS data. We now have re-written the MS-DIAL software code for use in Amazon Web Services (AWS) cloud computing, relieving much code overhead that was used for visualizing data and simultaneously increasing throughput by parallelizing computations. Results are then accumulated in LC-BinBase and reported back to the user to perform statistics and biological interpretations.

SIGNIFICANCE

Metabolomics and lipidomics are now widely-used in biomedical research. Over 12,000 metabolomic studies were reported by PubMed in 2022, and result data from more than 2,000 studies have been uploaded to the MetabolomicsWorkbench database. However, due to lack of standardization, it is difficult to compare data across studies. In addition, it is unclear how to evaluate the quality of reported data, including the accuracy of reported metabolite and lipid annotations. Significant bottlenecks exist in the field that lead to **hesitancy among biomedical biologists to embrace metabolomics data** in a similar way as they have done for genomics data:

(a) Due to a *lack of standardization across the field*, validated untargeted metabolomics databases are missing. Covering the full metabolome requires integration of different techniques. Many different assays (and data processing methods) are used across metabolomic data providers. As there are no generally accepted benchmark data, **biomedical scientists are left uninformed about data coverage, data quality and reliability** when planning metabolomics studies. As the field is expanding to cover exogenous chemicals (the exposome), this problem is exacerbated. A partial solution was offered by MASST/GNPS, which focuses solely on MS/MS data comparison³³⁻³⁴, but ignores isomers that can be separated by liquid chromatography and ion mobility data. This MS/MS-only focused approach led to new discoveries for large-scale meta-analyses of studies³⁵⁻³⁶. However, MS/MS spectra differ between instrument types and data acquisition parameters, limiting the reliability of such analyses. Recent efforts to include retention time as parameter in GNPS analyses³³ remained at the level of isolated studies. The MetabolomicsWorkbench¹⁰ and MetaboLights³⁷ databases offer cross-referencing of metabolite names by RefMet and ChEBI identifiers^{12, 38} to link data across study reports but omit unknowns and do not provide compound ID confidence scores or quality control reports, despite mandating raw instrument data uploads. Efforts such as Galaxy workflows³⁹ have not amounted to public data repositories.

(b) Only about **10-30% of all detected metabolites are currently annotated by chemical structures** in nontargeted metabolomics reports⁴⁰, with currently up to 1,000-3,000 named metabolites per study⁴¹⁻⁴³. Classic confidence levels proposed by the Metabolomics Standards Initiative⁴⁴ are mainly based on subjective reporting, not on calculated probabilities or FDR calculations⁶, unlike to proteomics⁴⁵. Decoy databases⁴⁶ are less readily applicable than in proteomics as the metabolome chemical space is much less predictable due to the interactions of endogenous enzymes with microbial metabolism, food and other exposome sources. Newly discovered compound classes had large impact on biomedical research but are not covered by commercial providers, e.g., FAHFA lipids that impact diabetes^{26, 47}, inflammatory sulfonolipids inflammation⁴⁸⁻⁴⁹, and the large number of new modified bile acids and amino acids that are transformed by gut microbiomes and impact human physiology via transport and binding to cellular receptors⁵⁰⁻⁵¹. Here, we will introduce *calculated confidence scores for identified metabolites* to standardize reports across laboratories, while unknown compounds will be assigned to compound classes that are useful in downstream chemical set enrichment statistics.

Advancing the LC-BinBase database system will have **different types of significant impact**:

We will establish unified, standardized metabolomic methods and data reports by using **LC-BinBase as a reliable and open-access resource** for report generation. Publishing standardized metabolomic methods will also guide metabolomic method developments by providing benchmarking studies.

- (1) **LC-BinBase** will be the first open-data processing repository for LC-MS/MS-based metabolomics data and will be opened for beta-testing during the project period. If successful, we aim to expand this repository toward a national scale, to enable mining processed data across studies and perform meta-analyses.
- (2) LC-BinBase will serve as open reference library of standardized MS/MS, retention time and CCS data to be used and benchmarked by all laboratories and methods.
- (3) We will integrate different technologies for compound identification to obtain **calculated confidence scores in metabolite annotations** from experimental data.
- (4) We will dramatically accelerate the **turnaround time and throughput** of biomedical metabolomics by automating the most time-consuming bottleneck steps, compound annotation, currently performed manually.
- (5) The **re-use of metabolomics data** will be expanded to both known and structurally unknown compounds by using unique database identifiers for all variables, and by establishing open-access portals.
- (6) We will greatly improve the **comparability of metabolomic data reports across studies** and across biological specimens by using standardized study descriptors and universal spectral hash keys.
- (7) We will significantly improve the **usability of metabolomic and lipidomic data reports** by reporting a large range of identifiers useful in external bioinformatics programs.
- (8) We will generate benchmarked metabolomic and lipidomic data sets for often used biospecimens that can be used as cornerstones for generating **metabolomic atlases of organs, tissues and cell types** across biomedical phenotypes and developmental stages.

B. INNOVATION

This proposal is conceptually and technologically innovative for the following reasons:

- ✓ Standardized, cross-laboratory, and rapid **high resolution untargeted LC-MS/MS data processing** for both hydrophilic interaction chromatography (HILIC) and lipophilic (reversed phase, RP) chromatography methods on orbital ion trap- and QTOF-type accurate mass spectrometers.
- ✓ Sample-by-sample matching to enable peak alignment **independent of the biological matrix**. This innovation means that both identified and unknown metabolites can be compared across tissues. It also means that data processing can be parallelized for cloud computing, speeding up data reporting >100-fold.
- ✓ Automatic integration of AWS-processed datasets from the LC-BinBase platform **with biological metadata**, to form a standardized basis for cross-study comparisons of metabolomic results.
- ✓ **Harmonized compound data reporting** for both chemically identified and unknown metabolites, for FAIR traceable and re-usable data across biomedical metabolomic reports using spectral hash keys.
- ✓ Open-access, public **compound identification tool MassWiki** that combines the use of all public MS/MS libraries including retention time scoring. MassWiki feeds user annotations and comments into LC-BinBase to form a harmonized consensus on identifying and naming compounds.
- ✓ Automatic annotation of unknown compounds with **elemental compositions** and compound classification.
- ✓ Instead of using subjective and hard-coded thresholds, we will **calculate the degree of confidence** in compound identifications by integrating error probability models directly from experimental data⁶.
- ✓ Harmonized compound naming, for better comparability of data reports across studies and between databases. Key database identifiers are automatically exported by LC-BinBase for cross-referencing between databases and for use in external **bioinformatics software**.
- ✓ Thorough **validation of data processing results** by cross-laboratory and cross-instrument data acquisition of sets of samples from biofluids, animal tissues and cells.

C. RESEARCH APPROACH

Overview: We will develop, optimize, deploy and validate the new prototype LC-BinBase data processing and database tool for use in biomedical metabolomic projects. We envision LC-BinBase to be robust, fast, scalable and informative enough to be eventually extendable on a U.S. national scale. LC-BinBase has been developed at UC Davis over the past four years as a result of the NIH metabolomics common fund projects on ‘*Compound Identification*’⁵². In a major break from classic data processing software, **LC-BinBase relies on retention time normalization by internal standards instead of sample-alikeness**. This means that data processing is independent of sample matrices, and for the first time, data can be standardized and compared between different tissues, cells and biofluids. We will thoroughly validate the quality of data processing by comparing LC-BinBase results for both orbital ion trap and QTOF instruments to classic, in-house MS-DIAL software use and alternative peak picking tools by studying a range of biological specimens, from urine and plasma to mouse organs and cell types. **Secondly, we will establish computed compound identification confidence scores**. Here, we will add Retip 2.0 retention time models¹ into a MassWiki user frontend to automatically improve confidence scores by matching experimental to theoretical accurate masses, MS/MS spectra and retention times. We will add ion mobility into the LC-BinBase data processing schema to yield a fourth, independent parameter for compound identifications. **Third, we will largely improve data usability** for biomedical researchers by integrating a large range of database identifiers, from BiGG, KEGG, BioCyc and LIPIDMAPS to standardized RefMet names, InChI keys and chemical classes. This cross referencing will largely alleviate the use of data in downstream bioinformatics software and overall improve the comparability between studies.

Specific Aim 1: Develop LC-BinBase with standardized peak reports across assays and matrices.

Preliminary Data. To ensure that LC-BinBase is truly suitable for automatic and routine interlaboratory use, we have engaged in a tight collaboration between three leading metabolomic laboratories at UC Davis (Fiehn), PNNL (Metz) and University of Michigan (Evans). To test the ability to detect the same known and unknown compounds across test samples, we analyzed a set of mammalian tissue and biofluid reference samples that were extracted at the Evans lab and sent for LC-MS/MS data acquisition using the UC

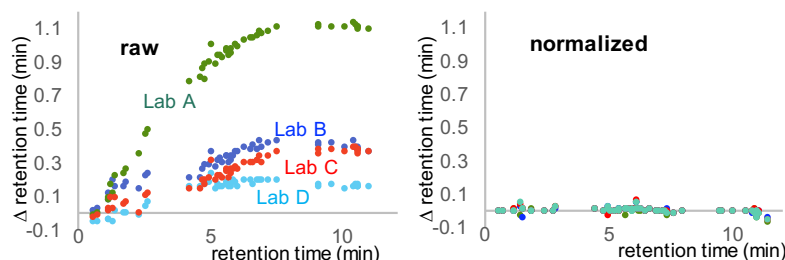


Fig. 1. Retention time normalization between Fiehn, Evans, Metz laboratories on lipidomics LC-MS/MS data acquisition of mouse organs and human plasma samples using 76 internal standards.

Davis chromatographic and data analysis methods. While identical solvent compositions, buffers and columns were prescribed for each laboratory, instrumentation used differed between laboratories, including UPLC pump setups and types of high resolution mass spectrometers. Accordingly, we found absolute (raw) retention time differences up to 1.1 minutes between laboratories compared to the default UC Davis method (**Figure 1**, Lab A using a quaternary LC pump). Using polynomial regressions on 76 stable isotope internal standards, raw input data were corrected to normalized retention times that resulted in <0.1min (6s) relative retention differences (**Figure 1**). In subsequent tests with LC-BinBase, we used data from >14,000 human cohort plasma samples that we have acquired under identical conditions in HILIC-MS/MS. Despite months of operation, we found median peak shifts of <2s with few outliers at 4s shifts (**Figure 2**).

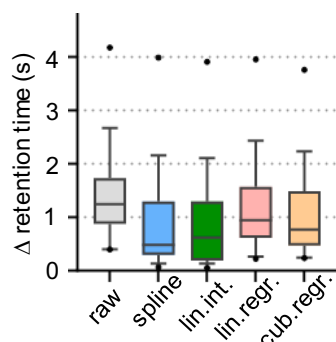


Fig. 2. HILIC retention time normalization for >14,000 human plasma samples.

When comparing cubic spline, linear interpolation, linear regression, and cubic polynomial regression for normalizing these retention time drifts using 32 internal standards, we reduced the retention time differential window to <2.1s for >99.7% of all data (3 σ confidence interval). Testing different peak picking algorithms, additional preliminary data showed that centWave (in XCMS) produces too many false positive peak detections when testing repeated analyses of pool plasma QC samples (**Figure 3**). While the number of true positives (blue, <20%RSD) was similar between algorithms, centWave gave too many false positives detections (red). It also did not remove isotopes efficiently and missed peak detections of internal standards and isomer pairs. Together, these error windows gave us input into further optimization and data robustness aims outlined below.

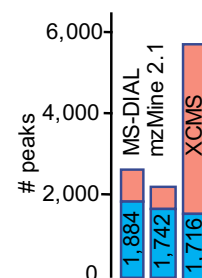


Fig. 3. Peak picking comparison.

(SA1a) Approach to reduce residual errors in data acquisition

Understanding the extent of variance in retention time and MS/MS data acquisition within-series, between-series and between laboratories is critical to achieve high accuracy of data normalization in LC-BinBase. We will test experimental robustness for our standardized lipidomics (reversed phase) and HILIC chromatography procedures. *For experimental robustness*, we will test subtle unintentional errors in following our standard operating procedures as they may happen in routine operation and when implementing protocols across laboratories. *(i)* Using NIST SRM1950 human plasma extracts, we will explore the extent of **absolute retention time deviations** based on errors in LC solvent compositions ($\pm 2\%$ variation), buffer compositions ($\pm 5\%$ variation), deviations in injection resuspension solvents ($\pm 2\%$ variation), deviations in chromatography temperatures ($\pm 3^\circ\text{C}$ variation), differences in LC-flow rates ($\pm 5\%$ variation), injection volumes ($\pm 10\%$ variation), differences in LC-gradient compositions ($\pm 2\%$ variation), differences in tubing lengths and tubing diameters between LC and MS instruments ($\pm 20\%$ variation), and errors in maximal start back-pressure readings that may indicate clogging or column overuse ($\pm 5\%$ variation). *(ii)* We will repeat these tests three times in all three laboratories, 3 months apart to test for **between-series and between-laboratory robustness** to define figures of merit of the overall analytical method (precision, accuracy, sensitivity, linear dynamic range, detection limit, and selectivity), using dilution curves of the labeled internal standards to estimate molar concentrations. *(iii)* We will then extend methods to **similar, but not identical, chromatography procedures** by changing column lengths from 5-10 cm, altering flow rates from 500-800 $\mu\text{L}/\text{min}$ and testing 3 different C18-based columns for lipidomics. *(iv)* Differences in **mass spectrometry parameters** will be tested for four types of instruments (Orbitraps, QTOFs, timsTOF and tripleTOF), specifically with respect to data acquisition speed in MS1 (from 4-40 Hz), mass resolving power ($R=10,000\text{-}50,000$) and MS/MS parameters (precursor ion inclusion windows, data dependent vs. iterative exclusions, CID energies, ESI modes and MS/MS duty cycles).

Once experimental robustness tests are completed using the NIST1950 plasma matrix, additional reference sample types including commercial mouse organs, cell types and different biofluids from BioIVT and NIST (including saliva, fecal water, urine, plasma and amniotic fluid) will be analyzed by the Fiehn/Metz/Evans labs to help evaluate successful operation of LC-BinBase in a wider variety of sample matrices

(SA1b) Approach to reduce residual errors in peak picking and retention time normalization

Similarly, analyzing the impact of software parameters on data processing is key to achieve highly confident data results for different biological matrices in LC-BinBase. We will test computational robustness for our standardized reversed phase and HILIC chromatography procedures. *(i)* First, we will test both **peak picking/peak smoothing algorithms**. We will implement the code from open-source software as variants of our current MS-

DIAL based peak picking algorithm in LC-BinBase, namely MZmine3.0 and Alphapept. Using default versus software-specific optimized parameters, we will compare data processing reports for plasma lipidomics and metabolomics with respect to peak missingness, accuracy of distinguishing isomer pairs, false positives for low abundant metabolites or tailing peaks, accuracy of deisotoping, precision of peak height reports in pool QC samples, and finally, accuracy, sensitivity and specificity for peak finding and peak quantitations using blinded raw datasets specifically prepared for software comparisons, provided by Prof. Corey Broeckling (see letter of support). These datasets will be tested offline to compare MS-DIAL against centWave (used in XCMS), GridMass (used in MZmine3), ADAP (used in MZmine3), EI-MAVEN, and FeatureFinderMetabo (used in OpenMS). (ii) Next, we will optimize the **number, type and distribution of internal standards** used as chemical anchors for accurate retention time corrections for both lipidomics and HILIC (polar metabolite) chromatography. At current, we use 76 labeled internal standards for lipidomics and 42 internal standards for HILIC chromatography. Search windows for internal standards will be automatically narrowed for every study by quality control blank samples that are mandated to be run for LC-BinBase data processing and that lack any matrix interference. We will use nonlinear retention time corrections by six transformation functions including cubic spline, linear interpolation, linear regression, s-spline, locally estimated scatterplot smoothing, and cubic polynomial regression. We will also study if using post-hoc Tukey M-estimators further reduce residual errors. In addition, we hypothesize that internal standards may not always be correctly detected in all organs, biofluids or cell types due to co-eluting interferences, ion suppression or impact of matrix components on peak shapes and relative peak positions, or even ambiguous MS/MS spectra compared to isobaric ion species detected in biological samples. We will utilize the extensive datasets from our experimental robustness tests to computationally determine how accurately retention times can be normalized if fewer internal standards are used for retention time model building. Specifically, we will test if a single set of manually validated internal standards will perform better than dynamic model building of suitable standards by automatic selection of training and model validation internal standards. We will also test if there are specific methods that could better address HILIC versus reversed-phase retention time models, or specific subclasses of compounds that perform worse than others. We will then query LC-BinBase to error probability distributions of normalized retention times, accurate precursor masses, and MS/MS entropy similarity scores to be used in confidence scoring in Aim 2.

Pitfalls and Alternative Strategies. If an offline peak picking/smoothing algorithm works significantly better than either MS-DIAL, Alphapept or MZmine3.0 (determined by specificity and sensitivity), we will incorporate this algorithm into the cloud-based LC-BinBase environment to benchmark its performance against all experimental datasets. If our experimental studies find that retention times for specific compounds or even compound classes shift relative to the internal standards between matrices, instrument setups or laboratories, we will determine additional weights for corresponding class-related internal standards to correct for such errors.

Specific Aim 2: Develop LC-BinBase with calculated metabolite annotation confidence scores.

Summary of Research Approach. Confidence in compound identification remains a difficult task in untargeted metabolomics and lipidomics. At current, both academic and commercial data sets lack confidence scores when annotating chemical structures from LC-MS/MS data. The Metabolomics Society Initiative suggested confidence scores⁴⁴ that lacked computational implementations, unlike FDR calculations for proteomic assessments. We will develop mathematical and computational tools to combine error models of independent experimental observations (**Figure 3**): differences in observed to theoretical masses (ΔM), differences in observed to predicted

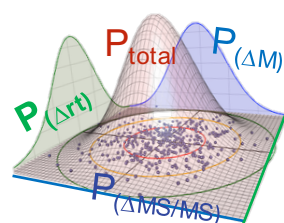


Figure 3. Uncorrelated total compound ID probability.

$$P_{total} = P_{(\Delta M)} * P_{(\Delta rt)} * P_{(\Delta MS/MS)}$$

retention times (Δrt) and differences in observed MS/MS library matching similarities compared to similarities expected from database-wide calculations ($\Delta MS/MS$). We will advance error models to include signal abundance as a confounding parameter, given the fact that highly abundant compounds yield better ion statistics for ΔM and $\Delta MS/MS$ calculations. We will explore ways to best use additional error models in overall computed confidence in metabolomic identifications, including ion mobility as an orthogonal and predictable parameter (ΔCCS). We will implement methods that automatically calculate elemental compositions for unidentified compounds, and classify unknowns based on entropy similarity against all publicly available mass spectra. We will use the elemental composition calculations to automatically denoise MS/MS spectra to further improve compound annotation confidence.

Preliminary Data. Compound identification has progressed greatly over the past 10 years^{21, 53} to the point that we routinely annotate more than 1,000 unique metabolites and lipids per study⁴¹⁻⁴³. The largest advance was

made by increasing the size of authenticated MS/MS spectra in public libraries, namely MassBank.us²⁹ (hosted by the Fiehn lab), but also by GNPS and the licensed NIST20 library. MassBank.us currently holds 2.05m MS/MS spectra (of 650,000 molecules), while NIST23 hosts 2.4m MS/MS spectra (of 51,000 compounds), and GNPS downloads have 0.5m spectra (for 24,500 annotated compounds). At UC Davis, LC-BinBase has already processed over 20,000 human cohort samples for both HILIC and lipidomics high resolution mass spectrometry samples. For MS/MS spectral matching, we use our new spectral entropy similarity scores that improves FDR rates and outperforms classic dot-score similarity matching⁶. Additionally, we will automatically denoise all MS/MS spectra by calculating elemental compositions and subtracting ions that cannot be logically deduced from the intact precursor molecules. In this way, spectra for lower abundant compounds (with higher levels of noise ions) can be confidently annotated (preliminary data **Figure 4**) by yielding better MS/MS similarity scores.

#	scan	splash	ret.time	m/z	entropy	Δ MS/MS	Δ M	open MS/MS
286	14720	splash10-0u...	7.991	100.076	1.28	0.937	0.001	0.760
2275	93555	splash10-0u...	25.392	240.232	2.53	0.363	13.979	0.892

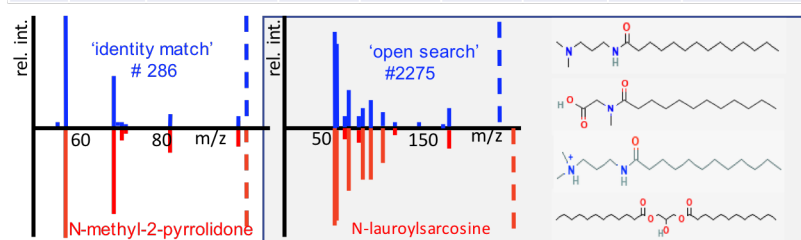


Figure 5. Preliminary MassWiki annotation of compounds in LC-BinBase. **Top:** LC-BinBase entries #286, #2275, tracked by Splash keys. **Bottom:** Δ MS/MS matching with Δ MS precursor m/z as dotted line. **Left:** Identification of #286 by Δ M 0.001 Da and Δ MS/MS entropy similarity 0.937. **Right:** No correct hit for #2275 due to Δ M > 0.005 Da and Δ MS/MS < 0.7. Open MS/MS (ignoring Δ M) gives Δ MS/MS > 0.8 for similar chemicals. Top-4 molecule structures support chemical class annotation as N-acyl amide.

authentic chemicals). Curation GUIs for LC-BinBase include adding metadata such as external database IDs (see *Aim 3*). Using accurate mass and MS/MS annotations alone, LC-BinBase currently contains 2,542 compounds with unique InChI key structures, normalized retention time, accurate mass and MS/MS for lipidomics and polar metabolites, including 753 exposome compounds such as drugs, food compounds, household chemicals, and environmental toxicants. We have recently improved our published retention time prediction model *retip*¹ to version 2.0 by decomposing all MassBank.us structures into 1,613 2D molecular descriptors via a Mordred library. We then built and evaluated ensemble models generated by the AutoGluon AutoML library using bagging (bootstrap aggregation) and stacking (optimal model combinations). In *retip* 2.0 software, we improved our retention prediction accuracy to a mean absolute error (MAE) of 7.1s. Hence, while previously we could exclude 68% of all candidate structures as likely false positives by retention prediction alone¹, implementing *retip*2.0 into LC-BinBase will be highly beneficial for annotating total probability confidence scores. Additionally, false positives can be excluded by ion mobility CCS values (**Figure 6**). Using a powerfit regression of CCS versus mass for the confirmed triacylglycerides in the Fiehn-Evans-Metz ring trial, we generated a prediction interval that included 99% of all confirmed triacylglycerides and excluded non-matching lipids by experimental CCS data (**Figure 6**).

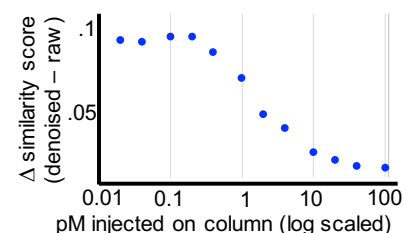


Figure 4. Improving MS/MS similarity scores by spectral denoising. Dilution series of 250 metabolite standards.

We will then optimize our preliminary online tool MassWiki (**Figure 5**) to rank all MS/MS spectra in LC-BinBase by experimental factors, including all publicly available MS/MS libraries. Importantly, LC-BinBase not only yields direct identity matches (**Figure 5**, left) but also substructure based compound class annotations using open-, neutral loss- and hybrid similarity searches⁵⁴ (**Figure 5**, right). Such complex queries are facilitated even against very large databases such as MASST with up to 1 billion spectra by our recently published FlashEntropy algorithm⁵⁵. Users submit annotations to LC-BinBase where database curators promote suggested annotations for LC-BinBase reports, or demote annotations based on new information (such as measured retention times of

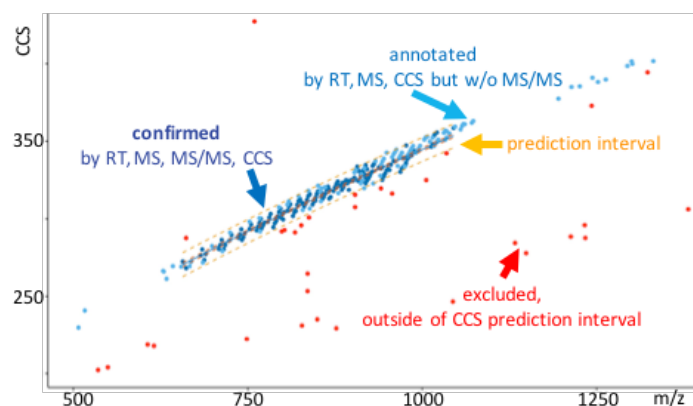


Figure 6. Preliminary data: Excluding false positive annotations of triacylglycerides in the Fiehn-Evans-Metz ring trial by CCS values from LC-ion mobility MS/MS lipidomics data.

(SA2a) Integrating new software and algorithms into the MassWiki and LC-BinBase framework

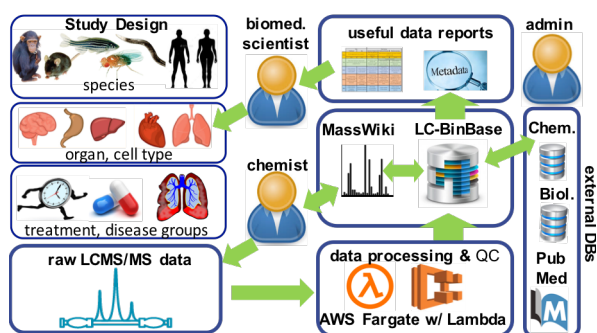


Figure 7. From study design to LC-BinBase reports. Biomedical study designs are captured by the miniX LIMS system that governs chemical data acquisitions. Cloud-based data processing is stored in LC-BinBase. New algorithms and external DBs feed into improved MassWiki compound metadata for better data reports.

At present, LC-BinBase generates unique ‘*accurate mass_{MS/MS} retention time*’ triplets (Bins) that are acquired from LC-MS/MS assays of biological samples (**Figure 7**). LC-BinBase uses a Postgresql/Aurora Amazon relational database service (RDS) for storing computational data and compounds, combined with a DynamoDB key-value storage for metadata and tracking information. Python language was picked for rapid prototyping and visualization, and to query the system via REST APIs. We have rewritten the code for the main MS-DIAL data processing algorithms¹³ to enable Amazon Web Services (AWS). LC-BinBase converts all vendor-specific data formats into mzML open data formats, successfully tested for ThermoFisher, Agilent, Bruker, and Sciex high resolution mass spectrometers. Data processing is executed on a AWS Fargate cluster to process hundreds of samples simultaneously. Using docker images enables quick transitioning from processing data in a

cloud environment (AWS) to a local system and back, for example, for developing and testing database functionalities or data processing parameters. Once all files of a specific biological study have been acquired, processed and exported, the Eventbridge system automatically detects completion and starts an aggregation task in ECS/Fargate to fetch all study metadata data from the DynamoDB Key/Value store.

The modular architecture of LC-BinBase and MassWiki, especially the API Gateway and its associated lambdas, will be used to improve data quality and data usefulness for biomedical scientists throughout all Specific Aims. (1) First, we will add an **automatic hyperparameter optimization layer** for the primary data processing module. For new methods, an optimization-based tuning algorithm will be tested and deployed. As we will use different mass spectrometers (from the three laboratories Fiehn/Metz/Evans), and because LC-BinBase supports both HILIC- and RPLC-MS/MS methods, raw data will be imported by LC-BinBase with different kinds of data quality (including peak shapes, ion intensities, noise values). While the main data types are defined by the mass spectrometer in a hierarchical fashion (such as ‘orbitrap’ and ‘QTOF’), application chemists will be able to select specific algorithms and configurations through the underlying combination of spring boot and yaml in the LC-BinBase central configuration server. Internally, we will use a versioning system to ensure that parameters (such as mass spectrometers operating at 1 vs. 5 mDa mass errors) are automatically detected and map to the same compound table, to generate unified reports over dozens of configurations. (2) Second, LC-BinBase data processing will improve to automatically map and merge compounds that are detected in both positive and negative electrospray mode (+/- ESI). As LC-BinBase relies on normalized retention times by internal standards, the same retention time windows apply in both ESI modes. (3) Third, accurate mass differences between +/- ESI modes within the retention time windows will be automatically interpreted by adduct lists (+/- H, Na⁺, Cl⁻, K⁺, NH₄⁺, OAc⁻, neutral in-source losses and rare adducts such as dimers) by adding code from software such as Binner at U Michigan^{56,57}. (4) Next, we will merge this Binner automatic detection of adducts with the new BUDDY **elemental formula algorithm** from the Tao Huan lab⁵⁸. BUDDY merged the MS-FINDER tool² with bottom-up integration of accurate mass and MS/MS fragment ions like Sirius⁵⁹, but it was faster and more accurate. Moreover, the dual verification of compounds detected in both +/- ESI modes will add confidence for elemental formula annotations. Hence, this tool will ensure that all structurally unannotated compounds in LC-BinBase will at least be reported with the starting information for compound identification, elemental compositions. (5) We will also deploy the **mzQuality** R packages from the Hankemeier lab⁶⁰ for monitoring technical variation in mass spectrometry measurements. It provides various metrics by utilizing pooled QC samples, blank samples and calibration lines. We will add these QC reports into the final data and metadata reports for biological scientists.

(SA2b) Total probability assessments for compound annotations in LC-BinBase reports.

Currently, MS-DIAL software calculates total similarity by equal weights for each attribute as $total\ score = \frac{\Delta MSMS + \Delta M + \Delta rt + 0.5 \Delta isotope\ ratio}{3.5}$. Individual equations to calculate mass and retention time differences in MS-DIAL (and LC-BinBase) use hard threshold settings defined by users (e.g., 10 ppm mass accuracy). We will replace user-defined hard thresholds of error windows with empirical distributions from LC-BinBase data, or by predicted and theoretical values where algorithms exist that enable structure-to-property calculations. **These probability distributions will serve to calculate total probabilities as given in Figure 3**, improving confidence in LC-BinBase reports. Conservatively, we estimate that we will have processed more than 50,000 samples through

LC-BinBase at project start, as we operate ten accurate mass LC-MS/MS instruments at UC Davis for routine profiling of more than 40,000 samples per year in > 400 studies. Hence, at project start we expect to have a sufficiently large data set to calculate distributions of Δ MS/MS, Δ MS¹ and Δ rt for all Bins in LC-BinBase (**Fig.8**).

(1) Retention times, Δ rt. Our preliminary data show that our current accuracy for shifts of normalized retention times (after correction by internal standards) lies within a 2.1s retention time window (**Figure 2**). This rt window accounts for >99.7% of all detections of a specific compound, even for months of operation including changes of solvent buffers and columns. However, isomers occur frequently in metabolomics, defined by having the same accurate mass and (most often) almost identical MS/MS spectra, but different chromatographic retention times. Such isomers are often closely eluting within 3-5s of each other, meaning that their retention error windows may occasionally overlap. Here, we will add the distance of the experimental to the target retention time within LC-BinBase to yield a Δ rt probability score for confident annotations, especially when both isomers are reported. For initial compound annotation within the MassWiki tool, we will integrate our Retip *retention time prediction* machine learning models¹ that we have now upgraded within Retip 2.0 to superior AutoGluon models. Currently, no database includes retention time predictions for compound confidence scoring. Retip gives plausible ranges for expected retention times for chemicals that are good matches by accurate mass and Δ MS/MS but that do not have measured retention times as authentic compounds in MS/MS libraries.

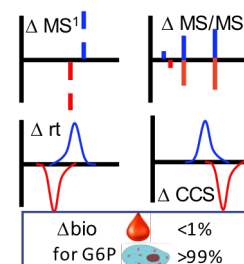


Figure 8. Measures of annotation confidence.

(2) Accurate masses, Δ MS. Similarly, Δ MS in Orbitrap instruments is usually at 1 mDa mass error. Yet, for very low abundant peaks errors may extend 10 mDa, due to lesser ion statistics. Conversely, a highly abundant peak should be detected at 10 mDa mass error except for detector saturation. Errors are also different between +/- ESI modes (due to different levels of noise ions) and between different types of mass spectrometers. Hence, we will adjust the Δ MS error models by precursor ion intensities, ESI mode and instrument types to achieve specific probability distributions. Additionally, we will test the impact of collision energies and merge Bin spectra into consensus spectra over time. We will also study how much inclusion of isotope abundances may assist in compound IDs, for example for brominated and chlorinated exposome chemicals.

(3) Spectral similarities, Δ MS/MS. Next, Δ MS/MS similarity errors get worse for lower abundant precursor ions, as the contribution of noise ions get more pronounced relative to true fragments. Here, we will use the BUDDY formula annotations as a starting point to comprehensively clean from LC-BinBase noise ions that cannot be deduced by formula fragments. LC-BinBase uses entropy similarity that has been shown to be superior to 40 other similarity measures⁶, especially when removing the precursor ion (**Figure 4**) that is used for orthogonal Δ MS assessments. For the cleaned MS/MS spectra, we will compile data distributions from LC-BinBase database to perform statistical assessments of MS/MS similarities when matching experimental spectra to LC-BinBase entries. These distributions may differ between assays (lipidomics or metabolomics, +/- ESI modes), and likely also between instruments (Orbitrap vs. QTOF MS). We will use these assay-specific probability distributions as input into overall probability calculations for annotation confidence levels.

(4) Ion mobility, Δ CCS. Over the past 10 years, ion mobility has advanced to improve the overall separation space for metabolomics and lipidomics. Commercial products that integrate LC-ion mobility-MS/MS are available from different vendors and are in use in the Metz and Fiehn laboratories⁸⁻⁹. We will improve LC-BinBase by adding a module to incorporate ion mobility data. Ion mobility is determined by the size, shape and charge of a molecule in the gas phase, and can therefore be predicted and calculated like an accurate mass. MS-DIAL software can process LC-ion mobility-MS/MS data since version 4.0, and we will implement those modules into the LC-BinBase core. Metz has published AutoCCS and DEIMoS software for extracting and using Δ CCS data^{8,61}, and ISiCLE for predicting collision cross sections based on quantum chemistry⁷. This tool will serve as an additional measure of total confidence in LC-BinBase compound annotations.

(5) Biological probability, Δ bio. As the number of compounds in chemical and biochemical databases grows, the likelihood of false discoveries in LC-BinBase will increase as well. Here, we will explore the use of probability-based ranking of unlikely compound annotations in BinBase. This strategy exploits the idea that experimental detection of a compound that frequently appears in the biomedical literature is more likely than the detection of a completely new structure that has never been confirmed before. For example, glucose-6-phosphate (G6P) is known to be found in all cells, but it should not be found in plasma (**Figure 8**). We will automatically extract the number of literature citations for compounds from PubMed, based on PubChem synonym lists and InChI keys. We will add the number of database links listed from PubChem and ChemSpider databases and add aggregated literature-derived databases such as our BloodExposome DB⁶², MarkerDB⁶³, DrugBank⁶⁴ or FoodDB⁶⁵ as

complementary factor for presence in biological contexts. Importantly, we will not combine the biological likelihood into the total chemistry probability score from Δ rt, Δ M, Δ MS/MS and Δ CCS. Instead, we will only flag (highlight) compound annotations in LC-BinBase that were annotated through MassWiki with a high chemical probability (as calculated above) but a low biological probability score. Such compounds will be subjected to additional administrative review, with high priority to buy chemical standards for confirmation. We will use the chemistry-based annotation probability scores to estimate **chemical False Discovery Rates (FDR)**. As *True Positives*, we will use LC-BinBase compounds that are confirmed on MSI confidence level 1 by authentic chemical standards. As test case, we will use the Broeckling blinded LC-MS/MS datasets for benchmarking our algorithms (see letter of support). *False Positives* (decoy databases) are a harder construct in metabolomics than in proteomics, because even implausible structures (like D-amino acids) have been shown to exist⁶⁶. Instead, we will use the mass spectra of all isobaric and isomeric compounds in the combined NIST20, MassBank and GNPS databases that fall within the maximum accurate mass probability range (as calculated from LC-BinBase) and probe the likelihood that accurate mass and MS/MS spectra of these alternative library compounds could match the probabilities of the *True Positive* identifications, using retention time and ion mobility estimates from Retip 2.0 and ISICLE software. We will limit False Positives to isomeric and isobaric compounds that differ in more than one chemical bond compared to the LC-BinBase annotation target. Finally, we will verify high probability LC-BinBase annotations (low FDR levels) by experimental confirmation using at least 1,000 commercially available standards that we have available.

(SA2c) Chemical classifications for unknown compounds in LC-BinBase

At present, we have enumerated between 6,000-8,000 unique LC-BinBase entries per HILIC- or RPLC-MS/MS assay from the 20,000 samples that we have processed so far. By retention time- and MS/MS matching to libraries, we have annotated more than 1,000 compounds per assay. Hence, more than 70% of all LC-BinBase

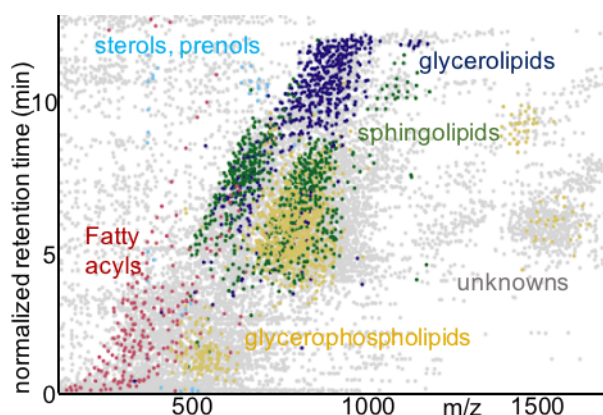


Figure 9. Preliminary data by Evans-Metz-Fiehn³⁸ for standardized data acquisition and processing by MS-DIAL with MassBank. Superclasses labeled by colors.

entries are currently not annotated, despite the vast MS/MS libraries we use. Such compounds are called ‘unknowns’ or the ‘metabolome dark matter’⁴⁰. Over the past two years, the Fiehn-Evans-Metz laboratories have engaged in cross-laboratory tests to screen for both identified and unknown metabolites and lipids in five animal model organs: skeletal muscle, liver, brain, heart and plasma (**Figure 9**)⁵². While we used shared LC methods and internal standards, MS/MS methods were not standardized, inhibiting the use of MS-standardized LC-BinBase algorithms. Nevertheless, using MS-DIAL and MassBank.us assessments manually, we identified 1,572 unique lipids representing 43 lipid classes in positive ESI mode and 952 unique lipids representing 59 lipid classes in negative ESI mode (**Figure 9**)⁵². We assigned compounds names and lipid classes and superclasses according to the LIPIDMAPS system⁶⁷ and the ClassyFire chemotaxonomy schema⁶⁸. Brain tissue contained a higher proportion of sphingolipids, whereas sterol lipids, the least frequently detected lipid class, were more numerous in plasma⁵². Interestingly, most unknown lipids clustered with the major superclasses of identified lipids, showing promise to classify these unknowns in our SA1c proposed here. In addition, unknown lipids were found in separate clusters, e.g. around 6 minutes retention time/1500 Da (**Figure 9**). This finding demonstrates unknown compounds may also comprise new lipid classes. For example, oxidized lipids are not annotated here but are easily characterized by mass shifts of +15.99415 Da (oxygen) and shifts to lower retention times than corresponding non-oxidized parent lipids. To enable the interpretation of unknown compounds in statistical results of biomedical studies, we will **classify such unknowns by adding elemental formulas and chemical class (substructure) information**: (1) We will automatically combine accurate mass MS1 and MS/MS data to obtain elemental formulas for all unknown compounds. We will use the consensus formulas calculated by MS-Finder², BUDDY⁶⁹ and Sirius CSI:Finger ID⁵⁹. While we have previously shown that this approach leads to >95% confident formula annotations⁷⁰, it has never been implemented for large scale databases. Fragment ions that cannot logically originate from the precursor molecules will be automatically removed to improve MS/MS spectral quality and compound annotation scores (see preliminary data). (2) We will classify these unknown compounds into ClassyFire chemical classes⁶⁸ by using CANOPUS software⁷¹. We will complement CANOPUS classifications by extracting maximum common substructures from MassWiki’s ‘open search’, ‘neutral loss search’ and ‘hybrid search’^{21, 54} (see preliminary data **Figure 5**) for all LC-BinBase entries. We will merge these schemes to structurally classify every unknown compound. Substructure-classification of known compounds will

serve to measure accuracy of this tool. Such classifications can be directly used in our chemical set enrichment statistics software ChemRICH⁷², improving the interpretability of biomedical studies.

Pitfalls and Alternative Strategies. We do not foresee difficulties implementing and testing retention time prediction tools in LC-BinBase (Retip 2.0). Retention time matching for our standardized LC-methods to other (similar) methods used by the community will be tested at a later phase. For integrating automatic calculations of elemental compositions, we may eventually restrict ourselves to using only one test algorithm (MS-Finder, BUDDY, Sirius) if differences in performance are minimal or if one algorithm consistently outperforms the others. Classification of unknown compounds remains difficult. If using maximum common substructures and CANOPUS do not perform consistently, we will use MS/MS similarity networks in analogy to GNPS approaches. Here, we will use calculated UMAP network graphs from MS/MS entropy similarity scores of all LC-BinBase spectra and use the ClassyFire ontology⁶⁸ of the top-10 nearest identified neighbors for each unknown database entry. We want to be clear that we use biological background knowledge (Δ bio) only to flag improbable annotations, but not to integrate this information into an overall probability score. Implementing new technologies such as electron activated dissociation MS/MS or UV photodissociation MS/MS will follow in a later phase.

Scientific rigor: To ensure code quality and stability, we will use a combination of unit tests (pytest, scalatest, karma) during the development process, plus integration and stress tests when deploying code to the LC-BinBase production system. Code stability is ensured by our dedicated test, development and staging environment that is isolated from the production environment. Applications are deployed by a combination of docker and serverless configuration files, depending on the specific needs. Deployments are automatically executed by the TeamCity continuous integration and deployment system.

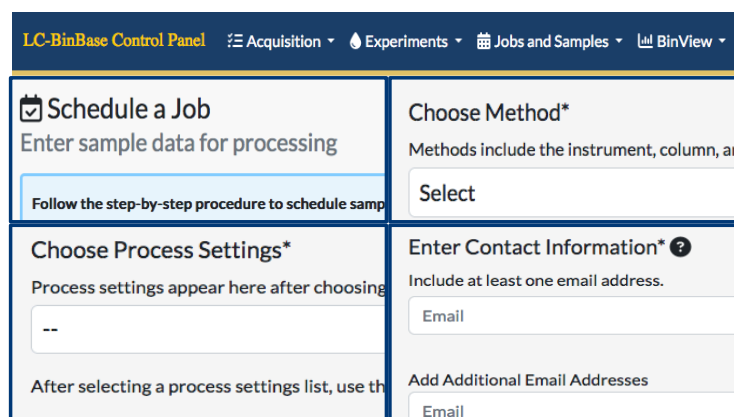
Specific Aim 3: Improve and test the biological usability and scope of LC-BinBase data reports.

Our overarching aim is to ensure that LC-BinBase data reports are useful for biomedical scientists, and that standardized metabolomics and lipidomics data become available across the nation to enhance data confidence, data re-usability and data comparisons across studies. Importantly, metabolite data reports only become meaningful if statistical differences are matched with biological context. Here, LC-BinBase reports need to be improved by adding identifiers for a range of downstream informatics tools and databases, to enable direct import and cross-referencing across software. At the Fiehn, Evans and Metz laboratories, we daily collaborate with NIH-funded investigators. We have collaborated with all colleagues during the past NIH Common Funds Metabolomics Program, and we are deeply engaged in metabolomics associations, including the Metabolomics Society, MANA and mQACC. We will therefore engage with biomedical scientists and analytical chemists to test and benchmark LC-BinBase data processing and LC-BinBase data reports, starting from webinars and direct invitations to community based workshops, both online and at metabolomics conferences.

Approach.

(SA3a) We will extend and validate the usability of the LC-BinBase biomedical user-faced frontends. Specifically, we will improve and implement our recently published MeSH-based frontend for ontology-based description of species and specimens⁷³, SMetaS (Study Metadata Standardizer)⁷⁴. We will first explore the user-friendliness of this **biomedical-focused frontend** internally before inviting external biomedical collaborators to beta-test it. (i) Internally, we will use text and table descriptors that we routinely receive from external collaborators in our three laboratories.

We will first explore if these text and table descriptors would be sufficient in depth and scope to be utilized within SMetaS, and if all partner laboratories can use SMetaS without supervision. (ii) Next, we will invite our biomedical collaborators who have contributed projects to our ongoing metabolomics projects, if these scientists would be able to enter the biological metadata themselves into SMetaS (species, gene descriptors, diets, specimens and related information). We will ensure that these studies include organs of animal models (specifically mice), cell cultures including human cell types, and human biofluids, including plasma, saliva, stool and urine-focused projects. Similarly, we will ensure that study designs show sufficient variation including time-dependent, dose-dependent, genotype-focused and disease-phenotype oriented studies. (iii) Last, we will direct open invites to biomedical and metabolomics scientists through our community engagement networks (specifically MANA and



LC-BinBase Control Panel

Acquisition Experiments Jobs and Samples BinView

☒ Schedule a Job

Enter sample data for processing

Follow the step-by-step procedure to schedule sample

Choose Method*

Methods include the instrument, column, and

Select

Choose Process Settings*

Process settings appear here after choosing

--

After selecting a process settings list, use the

Enter Contact Information* ?

Include at least one email address.

Email

Add Additional Email Addresses

Email

Figure 10. Current LC-BinBase frontend for data acquisition.

mQACC). Here, we will use online surveys to gauge satisfaction and user friendliness, and receive and implement comments for improving the LC-BinBase frontend SMetaS.

(SA3b) Similarly, we will extend and validate the usability of our LC-BinBase frontends that face towards analytical chemists in metabolomics laboratories (**Figure 10**). LC-BinBase relies on standardized naming of LC-MS/MS files and directly links to samples deposited via SMetaS. Again, we will first ensure that all chemists in the Fiehn, Evans, and Metz laboratories can independently and confidently submit study designs and raw data files for cloud-based data processing. We will modify the layout and functionality based on feedback. Collaboratively, the laboratories will work out tutorials and online forms to best instruct future external users of LC-BinBase. The laboratories will also collaboratively evaluate LC-BinBase reports, annotate compounds (using MassWiki), and discuss improvements in LC-BinBase algorithms. Subsequently (beginning Y3), we will direct open invites to biomedical and metabolomics scientists through our community engagement networks (specifically MANA and mQACC). For participants, we will open MassWiki for compound annotations in LC-BinBase (**Figure 11**). Again, we will use online surveys to gauge satisfaction and user friendliness, and receive and implement comments for improving the LC-BinBase frontend panels. In quarterly user meetings we will then discuss benchmarking results for LC-BinBase reports when compared to in-house solutions that metabolomics participants are using.

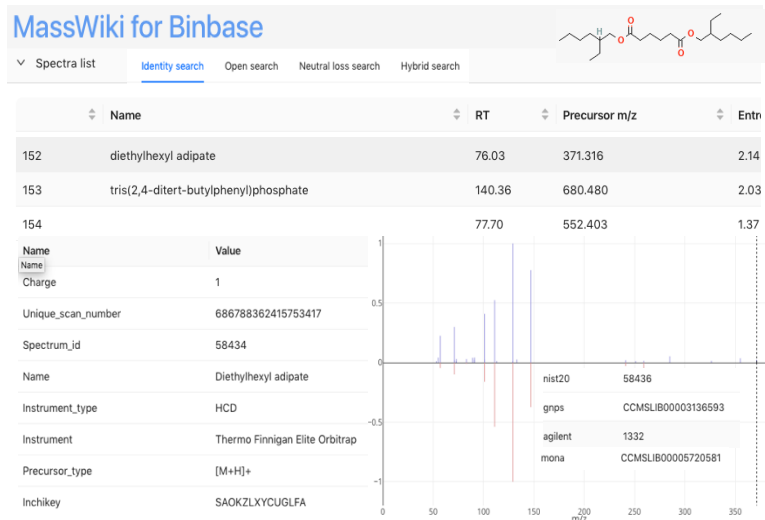


Figure 11. Current MassWiki frontend for compound ID.

(SA3c) Metabolite names are not sufficient for **integrating metabolomic and lipidomic data into downstream bioinformatics software**. Biomedical studies have different aims, from epidemiology analysis of risk factors to molecular mechanisms how lipids and metabolites impact signaling pathways in cells and tissues. We will therefore provide a set of different identifiers to readily utilize metabolites and lipids in external software programs. We will improve usability of LC-BinBase by providing a unified compound metadata list for all annotated metabolites, including PubChem, RefMet, LIPIDMAPS and ChEBI identifiers. The primary identifiers in LC-BinBase reports are the unique spectral hash key identifiers (SPLASH)²⁹ that were developed in coordination with major metabolomics databases such as HMDB and GNPS⁷⁵ to enable interoperability across repositories. To enable large-scale re-use of our data, we will add RefMet identifiers to link to biomedical studies uploaded at the NIH-funded Metabolomics Workbench database¹². We will use International Chemical Identifier keys (InChI keys)⁷⁶ to all identified metabolites, including example InChI keys for multiple possible isomers such as phosphatidylcholine PC (18:1_16:0). We will use classic biochemical identifiers such as Reactome⁷⁷, MetaCyc⁷⁸ and KEGG pathway by updating our online chemical translation service CTS²⁸. IDs will be complemented with computationally readable structure identifiers (SMILES)⁷⁹, chemical database identifiers (PubChem⁸⁰ and ChEBI³⁸), lipid specific pathway identifiers (BioPAN⁸¹ and SwissLipid ids⁸²), HMDB identifiers⁸³, and Rhea DB⁸⁴ systems biology identifiers. Collectively, this suite of identifiers will enable biomedical and bioinformatics scientists to utilize LC-BinBase reports in a wide range of downstream software (**Table 1**).

We had started with RefMet unified clear names but found that about 1/3 of our identified compounds were not yet listed in the corresponding NIH-funded Metabolomics Workbench database. We routinely use PubChem and InChI key identifiers as links to additional databases but understand that mapping tools are not straightforward (even with PubChem's identifier exchange service⁸⁵). We will therefore perform **cross-database mapping with manual curation** and uploading these ID mappers as permanent links to LC-BinBase and LC-BinBase reports. For example, InChI keys are most useful within ClassyFire software⁶⁸ to add a hierarchy of chemical classes to understand the **context of detected chemicals**. Metabolite classifications are directly used in chemical set enrichment statistics like in our ChemRICH software⁷², to complement classic univariate analyses. InChI keys are also mapped in MarkerDB⁶³ that links metabolites, genes and proteins to four biomarker categories (diagnostic, predictive, prognostic and exposure). ChEBI is used in multiple contexts, for example in Reactome software that links metabolites to molecular and cellular pathways⁷⁷. Classic biochemical networks in KEGG must be complemented by better curated BioCyc IDs⁸⁶. Neither KEGG nor BioCyc, unfortunately, use the wealth of lipid-transforming enzymes that are covered in BioPAN⁸¹ and SwissLipids⁸² using LIPIDMAPS identifiers.

BiGG identifiers are necessary for biomedical biologists (and system biologists) who aim to add genome-reconstructed networks and link to diseases⁸⁷. While most repositories and tools use more than one metabolite identifier, no single identifier is used by all tools, and none of the repositories is fully comprehensive. Overall, with this metadata-focused addition of identifiers to LC-BinBase, a wide range of use cases in biomedical metabolomic studies will be assisted by mapping ready-to-use identifiers for each study.

Table 1. Adding metadata identifiers to LC-BinBase data reports for use in systems biology and chemical context analysis.

small molecule ID	use	tool	Content	example use cases in biomedical studies
ChEBI	sys.bio.	Reactome	molecular pathways	signaling pathways, apoptosis
BioCyc ID	sys.bio.	MetaCyc	primary biochemical pathways	pathway tools, genome inferencing, flux, omics
KEGG id	sys.bio.	KEGG	primary biochemical pathways	pathway enrichment (in MetaboAnalyst), networks
ChEBI	sys.bio.	Rhea	chemical and transport reactions	link to UniProt knowledgebase
LIPIDMAPS id	sys.bio.	BioPAN	lipid biochemical pathways	complex lipid interactions
BiGG ids	sys.bio.	BiGG	genome-scale metabolic networks	visualize metabolic pathway maps, export SBML files
BiGG ids	sys.bio.	VMH	virtual metabolic human	linking metabolites to gut microbes and to diseases
ChEBI, LIPIDMAPS	sys.bio.	SwissLipids	lipid biology	links to proteins, literature
InChI key	sys.bio.	MarkerDB	gene, protein, chemical biomarkers	linking metabolites to genetic biomarkers
InChI key	chemistry	ClassyFire	Chemotaxonomy	set enrichment statistics, chemical context
ASCII string	chemistry	SMILES	structure representation	2D structure graphs, structure analysis
CID	chemistry	PubChem	synonyms, chem. Information	DB links, structure data
ChEBI id	chemistry	ChEBI	chemicals of biological interest	for Reactome, rhea software
HMDB id	chemistry	HMDB	Metabolites	reference concentrations
refMet	chemistry	Met.Work.	metabolome result data	unified clear names, lipid classes

Pitfalls and Alternative Strategies.

LC-BinBase is already functional as a prototype. At the end of Y01, Evans and Metz will have evaluated their first LC-BinBase datasets (**Table 2**). We will then hold a zoom-based LC-BinBase workshop for interested scientists through the Metabolomics Association of North America to receive initial feedback by the metabolomics community on the usability of the database, approaches for quantification, and our efforts in standardization and harmonization of data reports. We do not foresee difficulties in cross-mapping identifiers given in **Table 1** to improve usability of LC-BinBase reports. All identifiers will be tested to be accepted by current versions of their respective biological use software, e.g. for Reactome and BioPAN. A data dictionary will be written to inform biomedical researchers and bioinformaticists about use cases. We will incorporate suggestions to further develop LC-BinBase as given in SA1-3 with the aim to be poised for a large-scale nationwide extension of the database through other mechanisms such as RM1 or RC2 funding. For such future phases, but not in the current R01 project, we will then expand our efforts towards standardized quantification reports (estimates of absolute quantifications), and efforts to employ transfer learning to predict retention time shifts from community-based chromatographic methods to the standard LC-BinBase methods⁸⁸⁻⁹⁰. For example, many lipidomic methods use C18 reversed phase columns, but with different buffer conditions or C18 variants).

	Month 01-06	Month 07-12	Month 13-18	Month 19-24	Month 25-30	Month 31-36	Month 37-42	Month 43-48
SA1a data acquisition	Δ rt, Δ MS/MS	Δ LC	robustness	evaluations	matrices	evaluations	community	-
SA1b peak picking	#int.stds.	ADAP	GridMass	others	ISTD.matrix	evaluations	community	-
SA2b Code integration	hyperparam.	review	BUDDY	Binner	mzQuality	review	final version	final review
SA2b Compound ID confidence	data distrib. Δ M, MSMS	Δ rt, retip 2.0 MassWiki	Δ rt, M, MSMS calculations	Δ IM tests	Δ bio tests	improved calculations	improved Δ bio, Δ IM	final review
SA2c Chem. classes	-	-	data retrieval	version 01	validations	final version	final review	-
SA3a Biomed.usability	implement	UCD tests	PNNL, UM	revisions	community	revisions	community	final review
SA3b Chem. usability	PNNL, UM	revisions	tutorials	evaluations	community	revisions	community	final review
SA3c Omics metadata	CID, HMDB	SMILES	refMet, Biocyc	BiGG, KEGG	ChEBI	BioPAN	remaining	final review

Table 2. Timeline and milestones

Multiple PD/PI Leadership Plan

Our proposal is based on a multi-PI collaborative effort between Prof. Oliver Fiehn at the UC Davis Genome Center (Lead Principal Investigator), with Prof. Charles Evans (University of Michigan, UM) and Dr. Thomas Metz (Pacific Northwest National Laboratory, PNNL).

Roles and responsibilities: Principal Investigator Prof. Fiehn will serve as main contact to the NIH program officers, including the submission of project reports. He will lead the team of investigators with monthly meetings, supervision of UC Davis scientists and ensuring that deadlines and milestones are met. He will lead the metabolomics and cheminformatics parts of the project. Prof. Evans will supervise the scientists at UM and ensure that deadlines and milestones are met, same as Dr. Thomas Metz at PNNL who will supervise his scientists. All PIs are committed to collaborate with the NIH and external collaborators, and all PIs will ensure that they and their scientists are participating in the project zoom calls, project in-person meetings and in the planned community engagement workshops. Key person Dr. Shen will serve as PI for administrative purposes, e.g. for project reports, in case Prof. Fiehn is unavailable. Prof. Fiehn, Metz and Evans will collaborate in all research related aspects as well as presenting findings to the scientific community, providing progress reports to NIH, and writing manuscripts.

Management coordination: The three PIs Metz, Evans and Fiehn and their team members, including the UC Davis key scientists Dr. Tong Shen and Gert Wohlgemuth, plus scientists from the UM and PNNL laboratories, will meet monthly to discuss project progress and timelines. They will exchange samples, data, and scientific findings to prepare scientific reports and manuscripts. Authorships will be decided in consensus form, with an understanding that for most anticipated manuscripts, authorships will be shared among all team members. First- and corresponding- authorships will depend on the topic and scientific content of the manuscripts, but we anticipate that first authorships will be shared if several authors contributed equally to a topic. For the submitted manuscripts that we have collaborated already we have shown that this consensus method worked for our teams.

Fiscal coordination: Prof. Fiehn will manage the overall budget for the project, assisted by the UC Davis Genome Center staff. Changes in the fiscal strategy would need to be approved by both UM and PNNL, and the NIH program officer.

Intellectual Property: For sharing data, software, spectra, research protocols and strains we refer to our data sharing plan. NIH specific recommendations such as the Bayh-Dole Act (Patent and Trademark Law Amendments) regarding intellectual properties will be followed. Patent disputes will be resolved through the UC Davis Technology Transfer Office.

Conflict Resolution: The investigators do not foresee points of possible conflicts because access to resources (e.g. mass spectrometers, code, samples, computational infrastructure and budgets) is clearly defined by the respective work areas. In case of unexpected conflicts, the investigators intend to solve issues in good faith. For unforeseen severe cases, we will use the UC Davis resources that support conflict resolution through an ombudsman (for publications), a conflicts resolution center at the UC Davis Extension Center (for management) and conflict management officials at the UC Davis Center for Human Services (for staff and scientist work-related issues).

Change in PI Location: In case the PI transfers to a new academic location within the U.S., the University and the PI would negotiate a change in the management plan with the NIH, transfer of duties to the new location, and subcontracts with the University to ensure that work can progress as planned. Any change in key personnel would be negotiated and approved by the NIH program officer.

Consortium/Contractual Arrangements

Contact PD/PI: Fiehn, Oliver

Consortium arrangements with University of Michigan and Pacific Northwest National Laboratory (PNNL) are requested for all years of the award.

Detailed budgets and justifications are given for both contractual partners.

Note: *PNNL is not required to have a Forward Pricing Rate Agreement that is negotiated or approved by the federal government. The Pacific Northwest National Laboratory (PNNL) is a Department of Energy (DOE) Office of Science (DOE-SC) Federally Funded Research and Development Center (FFRDC). Battelle operates PNNL for DOE under a Management and Operations (M&O) prime contract, DE-AC05-76RL01830 (<http://doeprimecontract.pnnl.gov>).*

Under this M&O contract, in accordance with FAR and DOE requirements (DOE O 522.1A), Battelle is required to annually submit their Billing and Proposal Pricing Rates to DOE and required by federal law and regulations to achieve full cost recovery. Therefore, Battelle can't deviate from these rates. DOE establishes the programmatic controls, oversight functions, customary reporting information, and general categories and procedures for the overhead cost structure for PNNL. Therefore, all the rates and fees are considered allowable cost elements of this proposal.

*Battelle does not share the billing & pricing package outside of the U.S. Federal Government, so it will not be provided to the lead proposer. **However, it can be provided directly to a U.S. Government Contracting Officer upon request.***

NIH policy (NOT-OD-05-004) allows applicants to exclude consortium/contractual F&A costs when determining if an application falls at or beneath any applicable direct cost limit.

Resource Sharing Plan

UC Davis PI Fiehn will (1) assist all project staff researchers to submit peer-reviewed publications to PubMed in a timely manner, (2) collect all data and code from UC Davis, University of Michigan and PNNL staff and to submit data collected in this project to relevant national repositories, and (3) be a liaison with the UC Davis library staff who will help researchers archive their data through the UC Curation Center. Our informatics team, specifically lead IT member Gert Wohlgemuth, provides day-to-day security and bi-weekly back-ups for data saved on Genome Center servers.

(1) Software Resource Sharing Plan

We plan to continue our 20-year experience in making all software and code freely available using the CC-by 4.0 license. The CC-by license explicitly permits and encourages the dissemination and commercialization of enhanced or customized versions of the software, or incorporation of the software or pieces of it into other software packages. Software may be widely disseminated, enhanced by other users, or incorporated in part or in whole into other software packages. Software will be transferable; other individuals or teams may continue software development in the event the investigators of this application are unable to do so. CC-by details are given e.g. for our MassBank.us public repository, see <https://massbank.us/documentation/license>

(A) Code. We will create code and workflows for a variety of tools that collectively are combined in the LC-BinBase data processing and database repository. We use python, and other languages as given in the Research Strategy. All code is made freely available in Github, Bitbucket and as Docker images. Ancillary to LC-BinBase, we will also improve software to better identify metabolites from mass spectra and retention time information, in MassWiki. Again, all code will be publicly available for re-use in Github. Coded software will be accessible by web interface and public download at GitHub and linked from project websites and webpages. We will use forks to allow any contributor to make own copies for further development. Our lab has successfully contributed software tools (SPLASH software, CTS, BinBase, Seven Golden Rules software, MS2Analyzer) and databases (FiehnLib, MoNA, LipidBlast) for many years. All of the databases and software tools are accompanied by publications and in many cases extensive documentations and tutorials were created. Actively used software requires code updates, bug fixings and improvements. Commonly such improvements are made by collaborative developments on GitHub and Bitbucket and other code sharing platforms. All software tools have issue trackers. Members can actively follow the development and improve the software. Additional improvements or novel plug-ins can be easily managed on Github. For example many different software developers have added many language ports to the spectral hash (SPLASH) including JAVA, C++, Python, R, JavaScript, Scala as well as REST services. In conclusion, we have all methods for data and software sharing in place and we plan to maintain and improve those actions.

As evidence of our current efforts, here are links to our public code disseminations: (I) Github. We implicitly satisfy guidelines attributable to the use of GitHub in that (a) all software is transferable. Other individuals or teams simply “fork” the public repository to continue development, or ownership of the official (i.e. originating) repository may be transferred. (b) modifications can be made and shared. Researchers outside the immediate project team can fork the repository and make modifications to the source code, which can then be shared with other colleagues. (c) improvements/customizations can be disseminated or integrated. Changes made by others to forks of the repository can be disseminated as such (noting proper license terms and attribution of work), or a pull request may be submitted, which would, pending approval, apply suggested changes to the official core software. If multiple customizations become relevant, we can host and support separate branches, effectively maintaining multiple functional versions of the source code.

- UC Davis (Fiehn) has generated public 51 repositories in Github (<https://github.com/berlinguyinca>) , including the official West Coast Metabolomics Center Github repository (<https://github.com/metabolomics-us>), which has 30 public repositories, with lead DB developer Wohlgemuth as an Arctic Code Vault Contributor. The six most popular projects are SPLASH (SpectralHash keys) in C++ with 11 forks (the international hash key to exchange mass spectra across databases) and four python-coded projects including pycuda, backtesting.py, and osra-docker (providing the osra structural recognition software).

- PNNL (Metz) also uses GitHub as trackable, downloadable, and modifiable open revision control and source code management system. GitHub increases transparency of the development process and provides computational tools to the community for immediate use, feedback, and bug reporting. Software components will be easily installable via the Python Package Installer (PyPI) and/or Conda build recipes, which will automatically download and build from source, though this feature will not be developed until a functional version of the software is complete. PNNL software and source code resulting will be freely available for non-

profit use, e.g. universities, research institutions, government laboratories, under an Apache 2.0 license. However, specific, third-party dependencies within the PNNL code may require the user to obtain a license from the appropriate vendor.

- Finally, University of Michigan (Evans) also uses GitHub repositories to enable to fork and download software source code. Any improvements which are made to the software will be uploaded to these websites and to github as they are completed. University of Michigan will write full documentation for any software they release. Examples of such manuals and GitHub code can be found on the corresponding websites for the Michigan tools MetabCombiner: <https://github.com/hhabra/metabCombiner>, Binner: <http://binner.med.umich.edu>, MetIDTracker: <https://github.com/araskind/MRC2Toolbox>, and MetaboDrift: <https://github.com/evanscr/MetaboDrift>

(II) Bitbucket Deposited in <https://bitbucket.org/fiehnlab/workspace/repositories>, UC Davis has overseen the generation of 50 active repositories within 7 projects from 2015-2022 including Python REST client for OSSA OpenCL Similarity Search, Stasis result AGregAtoR, MS-FLO (to improve the quality of feature lists/peak tables after initial processing), Chemical Translation Service (as RESTful architecture, spring-boot based, with javascript plugin frontend), MGF file parser for AngularJS, a Peak Annotation Software Benchmarking Utility, and MassBank.us (MoNA; as metadata-centric, auto-curating public MS repository), and others.

(III) Published Docker Images Located at <https://gallery.ecr.aws/?searchTerm=fiehnlab>, we manage all public and private docker images, for different applications, developed for the UC Davis West Coast Metabolomics Center over the last years. These docker images provide the community and contributors an easy access to pre-build ready to run images of our applications.

(B) Maintenance of free web-services: We have developed several web services; however, updates and server improvements can be critical. In many cases the software and database content can be distributed to other groups so multiple instances of the same service (Europe, Asia, North America) can be created. For MassBank.us (that is directly linked to MassWiki and LC-BinBase), such server instances have been installed as MassBank.jp and MassBank.eu in Japan and Europe, respectively.

(C) Timeline. Code is uploaded to Github for immediate use in the Bioinformatics community (and beyond) at the very moment when it is successfully deployed from test environments into the production environment. Unit tests are described in the Research Strategy that outline the requirements for production deployment. Production environment code for the current prototype LC-BinBase is already available to the public. We foresee updates of the production DB code every six months after project start. For ancillary software that may not be included in the LC-BinBase production environment, all resources will be disseminated to public repositories and the scientific community no later than one year of the completion of the funded project period or upon acceptance of the data for publication, whichever is earlier.

(2) Plans for intellectual property protection. We will follow all rules as outlined by local UC Davis and collaborating laboratories at PNNL and University of Michigan with regards to regulations of protecting and commercializing intellectual property. Should any intellectual property arise which requires a patent, we would ensure that the technology remains widely available to the research community in accordance with the NIH Principles and Guidelines document. Ownership of sole or joint inventions developed under the project will be owned by the institution(s) employing the inventor(s). Inventors shall be determined by U.S. Patent law, Title 35 SC. University and Participating investigators/institutions will disclose any inventions developed under the project and such inventions will be reported and managed as provided by NIH policies. Sole inventions will be administered by the institution employing the inventor. Joint inventions shall be administered based on mutual consultation between the parties. We will abide to the principles and guidelines for recipients of NIH research grants and contracts on obtaining and disseminating biomedical research resources as outlined in the federal register. Specifically, the principles of academic freedom and publication, the Bayh-Dole Act (Patent and Trademark Law Amendments) and the dissemination of research with resources developed with NIH funds. In accordance with these rules we plan to widely disseminate and share all materials, data, results and software with the metabolomics community. Similar procedures will be followed for copyrights. Materials generated under the project will be disseminated in accordance with University/Participating institutional and NIH policies. Depending on such policies, materials may be transferred to others under the terms of a material transfer agreement.

Data Management, Data Sharing and Resource Sharing Plan

UC Davis PI Fiehn will (1) assist all project staff researchers to submit peer-reviewed publications to PubMed in a timely manner, (2) collect all data and code from UC Davis, University of Michigan and PNNL staff and to submit data collected in this project to relevant national repositories, and (3) be a liaison with the UC Davis library staff who will help researchers archive their data through the UC Curation Center. Our informatics team, specifically lead IT member Gert Wohlgemuth, provides day-to-day security and bi-weekly back-ups for data saved on Genome Center servers.

Research data outputs, types of data to be available. UC Davis (PI Fiehn), University of Michigan (co-PI Evans) and PNNL (co-PI Metz) will generate metabolomics data and software code. The following sections detail our data sharing and software dissemination plan for this project. Generally, we will broadly and freely share data, mass spectra, developed software and methods with the metabolomics and biomedical community and follow the outlined rules of data deposition, quality control metrics, standardization and metadata requirements. All our software tools support standard input raw file formats whenever possible (mzXML/mzML), and will support standard spectral library formats (.MSP or NIST). Open dissemination of data, software tools, and standard operating procedures (SOPs) to easily-accessible computational and analytical platforms is an important aspect of publicly-funded research. Scientific progress relies in part on the effective sharing of software, methods, data, and results as they become available for easy dissemination among the Investigators, our partners, and the broader scientific community. To this end, we will abide by the resource and data sharing standards developed and agreed upon by the NIH, and will employ those standards to ensure ease of access, collaboration, and use of all data, software, and SOPs that result from this project and related work. These open-science practices enable increased productivity, quicker and deeper impact on the community, and low-friction collaborations among groups.

We are committed to disseminating all project-related data, tools, algorithms, SOPs, and results to the community at large. All data and resources will be disseminated to the greater metabolomics and biomedical research communities as soon as possible to facilitate progress in these fields, as well as early adoption of methods and tools. Any software modifications or developments will employ modern development and design practices, including transparent version control use, platform-independent installation, scalable performance across arbitrary HPC resources, and use of feature-rich APIs to provide provenance, fault tolerance, and checkpointing. To ensure free access and the long-term impact of the data generated and any computational tools modified or developed during the program, we will deposit or facilitate the deposition of these resources in appropriate public repositories, per the Specific Plans below.

(1) Publication in scientific journals and scientific meetings. The team intends to publish all substantive research findings in high impact, PubMed-indexed journals. We have an exceptionally strong track record of accomplishing this aim as outlined in our biosketches. The PIs of this application and their research teams of students, postdocs and staff scientists regularly attend and present research findings at national and international conferences in the fields of metabolomics, mass spectrometry, and biomedical research. We will continue these practices for the data generated in the context of this proposal and are enthusiastic to share our results with the broader research community.

(2) Metabolomic Data Sharing Plan

We will abide by all data deposition, quality control metrics, standardization, metadata requirements, data and software release, and public copyright requirements set forth by the NIH. As participants in multiple NIH consortia, PIs Fiehn, Metz and Evans are familiar with and active supporters of efforts to ensure all relevant data are submitted to public repositories in a Findable, Accessible, Interoperable, and Reusable (FAIR) format.

(I) Metabolomic results data and methods. All experimental data and data reports from the new LC-BinBase DB will be compliant to the Metabolomics Standards Initiative levels with full disclosure of accurate masses, relative retention times, MS/MS data, quantification masses, and matches to library spectra and experimental retention times (for authentic chemicals) or deviation to machine-learning predicted retention times (for annotated metabolites that are not available as authentic chemicals). We intend to upload all metabolomics data generated, in both raw and processed format, to the Metabolomics Workbench (www.metabolomicsworkbench.org), and/or to other publicly accessible archives selected by our team or recommended by NIH program officers. Protocols and methods are public in the MetabolomicsWorkbench.org data repository and are regularly updated. All data acquisition protocols or computational metabolomics protocols will be outlined in detailed step-by-step instructions including the

sources of publicly or commercially available software and databases. The analytical protocols will be structured in a way that a scientist would be able to recreate our results from our experimental dataset using our methods. We plan to publish our protocols and results in the peer-reviewed literature but we will independently make the methods publicly available at the MetabolomicsWorkbench Protocol Exchange website as soon as we have assembled all results or by the end of the granting cycle. Data for quality controls, including pools samples, blank samples and instrument suitability samples will be included in the data result files. We will use other well-structured, community-accepted repositories such as MetaboLights and MassIVE if the MetabolomicsWorkbench becomes unavailable or if other repositories outperform MetabolomicsWorkbench.

(II) Mass spectra. All our mass spectra will be freely downloadable as JSON or MSP files so that they can be easily mirrored by many groups. All mass spectra and metadata (such as InChI keys, protocols and retention times, mol files (SDF), database IDs and identifiers such as spectral hashes (SPLASH)) will be available via MassBank.us that is maintained by the Fiehn laboratory. Software and spectra will be freely available to academic researchers. We use the least restrictive license that allows for wide commercial and non-commercial use. To obtain academic acknowledgements, we commonly utilize the CC-BY license. This license lets anybody freely use and modify the software if proper credits or citations are given. That means anybody outside this collaborative project will be able to use, share and further develop software and mass spectra. We do not foresee any problems with data retention, because the data is publicly available and backed up in many different places. Funding for these tools and databases is usually covered by many sources and not only a single grant or granting organization. At current, MassBank.us is entirely funded through metabolomics services at the West Coast Metabolomics Center.

(III) Timing of data sharing. All data will be shared per plans and rules set by MetabolomicsWorkbench, latest after publication in peer-reviewed journals.

(IV) Safeguarding human subjects by HIPAA

We will not perform human subjects research. Neither UC Davis, nor University of Michigan or PNNL will receive any identifiable subject information for any biosamples. In case human tissues, cells or biofluids are used, samples or data will remain be completely anonymized per HIPAA compliance rules. All HIPAA-relevant data will follow the rules of the NIH and the Clinical Center's institutional rules. The Federal Health Insurance Privacy and Portability Act (HIPAA) Privacy Rule requires data will be free of identifiers that would permit linkages to individual research participants, and will exclude variables that could lead to deductive disclosure of the identity of individual subjects.

(3) Plans for intellectual property protection. We will follow all rules as outlined by local UC Davis and collaborating laboratories at PNNL and University of Michigan with regards to regulations of protecting and commercializing intellectual property. Should any intellectual property arise which requires a patent, we would ensure that the technology remains widely available to the research community in accordance with the NIH Principles and Guidelines document. Ownership of sole or joint inventions developed under the project will be owned by the institution(s) employing the inventor(s). Inventors shall be determined by U.S. Patent law, Title 35 SC. University and Participating investigators/institutions will disclose any inventions developed under the project and such inventions will be reported and managed as provided by NIH policies. Sole inventions will be administered by the institution employing the inventor. Joint inventions shall be administered based on mutual consultation between the parties.

We will abide to the principles and guidelines for recipients of NIH research grants and contracts on obtaining and disseminating biomedical research resources as outlined in the federal register. Specifically, the principles of academic freedom and publication, the Bayh-Dole Act (Patent and Trademark Law Amendments) and the dissemination of research with resources developed with NIH funds. In accordance with these rules we plan to widely disseminate and share all materials, data, results and software with the metabolomics community. Similar procedures will be followed for copyrights. Materials generated under the project will be disseminated in accordance with University/Participating institutional and NIH policies. Depending on such policies, materials may be transferred to others under the terms of a material transfer agreement.

PHS Assignment Request Form

Funding Opportunity Number: PAR-22-127

Funding Opportunity Title: Focused Technology Research and Development (R01 Clinical Trial Not Allowed)

Awarding Component Assignment Request (*optional*)

If you have a suggestion for an awarding component (e.g., NIH Institute/Center) assignment, use the link below to identify the appropriate short abbreviation (e.g. "NCI" for National Cancer Institute") and enter it below in the boxes for "Suggested Awarding Components". All suggestions will be considered; however, not all assignment suggestions can be honored.

Awarding Components: https://grants.nih.gov/grants/phs_assignment_information.htm#AwardingComponents

Suggested Awarding Component:

Study Section Assignment Request (*optional*)

If you have a suggestion for a study section assignment, use the link below to identify a study section(s). Enter the short abbreviation for that study section in the boxes for "Suggested Study Sections." Remove all hyphens, parentheses, and spaces. All suggestions will be considered; however, not all assignment suggestions can be honored.

For example, enter "CAMP" if you wish to suggest assignment to the NIH Cancer Molecular Pathobiology study section, or "ZRG1HDMR" if you wish to suggest assignment to the NIH Healthcare Delivery and Methodologies SBIR/STTR panel for informatics.

Study Sections: https://grants.nih.gov/grants/phs_assignment_information.htm#StudySection

Assign to Study Section:

Each entry is limited to 20 characters

PHS Assignment Request Form

Rationale for assignment suggestions (optional)

*Entry is limited to
1000 characters*

List individuals who should not review your application and why (optional)

*Entry is limited to
1000 characters*

Identify scientific areas of expertise needed to review your applications (optional)

Note: Do not provide names of individuals

Expertise:

*Each entry is
limited to 40
characters*

PHS Human Subjects and Clinical Trials Information

Use of Human Specimens and/or Data

* Does any of the proposed research involve human specimens and/or data?

Yes

No

Are Human Subjects Involved

Yes

No

Is the Project Exempt from Federal regulations?

Yes

No

Exemption Number

1

2

3

4

5

6

7

8

Other Requested information

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

Proposal Summary

Proposal Number: Proposal Status:
Sponsor Deadline: 10/05/2023 Submission Method:
Submission Type: Application

INVESTIGATOR DATA

PROJECT DIRECTOR / PRINCIPAL INVESTIGATOR CONTACT INFORMATION

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
<u>Dr.</u>	<u>Oliver</u>		<u>Fiehn</u>	<u>Ph.D</u>
Position/Title:	<u>Professor</u>	Organization:	<u>The Regents of the University of California (Davis)</u>	
Department:	<u>UC Davis Genome Center</u>	Division:	<u>College of Biological Sciences</u>	
Street1:	<u>1850 Research Park Drive</u>	Street2:	<u>Suite 300</u>	
City:	<u>Davis</u>	County:	<u>Yolo</u>	
State:	<u>CA</u>	Zip Code:	<u>95618-6513</u>	
Country:	<u>USA</u>	Employee ID:	<u>10211261</u>	
Phone:	<u>530-754-8258</u>	Fax:		
Email:	<u>ofiehn@ucdavis.edu</u>			

First Budget Period Effort: Calendar: 2.40 Academic: Summer:

Status of PI:
Status Waiver Required?
Signed Intellectual Property Waiver Attached?
Signed Conflict of Interest Disclosure Attached?
Agency Certification Documentation Attached?
Cost Sharing Authorization Form Attached?

SPONSOR DATA

Agency: National Institutes of Health
Proposal Type
Sponsor Mechanism: Focused Technology Research and Development (R01 Clinical Trial Not Allowed)
Sponsor Type:
Sponsor Code:
Sponsor Name:
SubDivision 1:
SubDivision 2:

PROJECT DATA

Title of Project: A reproducible database for untargeted metabolomics data processing across laboratories
Is This a Subcontract? No
If Yes, who is prime?
Type of Proposal:
Type of Agency: Federal
Kind of Application: New
Previous Grant # or Federal Identifier:
Change in grantee institution?
Type of Project:

PROJECT ADMINISTRATION

Who is responsible for this research?		
Departmental Identification Number:	Primary:	Secondary:
Departmental Name:	Primary:	Secondary:
Primary Dept. Contact Info:		
Account Classification:	Primary:	Secondary:
Other Institutional Code:		
NAICS Code:		

COMPLIANCE DATA

Proposal Summary (cont'd)

Are animal subjects used? No
Is IACUC review pending?
IACUC Protocol #
IACUC Approval Date:
Are human subjects used? No
Is IRB review pending?
IRB Protocol #
IRB Approval Date:

Does this project involve use of any of the following? Radioactive Material(s), Radiation Producing Devices(s), Recombinant DNA, Biohazardous Chemical(s), Class IIIb or IV Lasers, Other certifications of health, safety and/or environmental compliance.

BUDGET DATA

Performance Dates	Begin Date	End Date
First Budget Period:	<u>07/01/2024</u>	<u>06/30/2025</u>
Cumulative Budget Period:	<u>07/01/2024</u>	<u>06/30/2028</u>

Cost Sharing Information	Mandatory	Voluntary
Committed:		
Amount:		
Source:		

Budget Period	Direct Cost	Indirect Cost	Total Cost
Period 1:	<u>604,120</u>	<u>213,499</u>	<u>817,619</u>
Period 2:	<u>606,773</u>	<u>182,999</u>	<u>789,772</u>
Period 3:	<u>606,708</u>	<u>182,999</u>	<u>789,707</u>
Period 4:	<u>606,187</u>	<u>182,999</u>	<u>789,186</u>
Total:	<u>2,423,788</u>	<u>762,496</u>	<u>3,186,284</u>

AWARD DATA

Award #: Contract #: Date:

Budget Period	Direct Cost	Indirect Cost	Total Cost
Period 1:			
Period 2:			
Period 3:			
Period 4:			
Total:			

EXPORT CONTROL

1. Will the project involve participation, collaboration or access to information by foreign nationals, defined as: individuals with foreign citizenship, foreign governments, foreign associations and corporations, or foreign political parties? Note: Foreign nationals granted US citizenship, or permanent residence "green card" or granted status as a "protected individual", e.g., political refugees and political asylum holders are "EXEMPT" from deemed export rule.
2. Will the project involve the shipment of equipment, technology, software, materials data or other information?
3. Will the project involve a foreign subcontract or other foreign contractual agreement?

COMMENTS AND EXPLANATIONS

PLEASE INDICATE ANY SPECIAL INSTRUCTIONS BELOW:

BUDGET JUSTIFICATION

Institute: University of California, Davis

PI: Oliver Fiehn, Ph.D

KEY PERSONNEL

Prof. Oliver Fiehn, Ph.D. (Principal Investigator), will commit 2.4 calendar months (20.0% efforts) to coordinate the work for this project, including study designs, curation and validation of MS/MS data, integration of genomic and biochemical metadata, and engagement with the metabolomics and cancer communities. Prof. Fiehn will be responsible for overall project oversight, reporting, and data dissemination of all findings throughout the project. Specifically, Prof. Fiehn will interact with lead database engineer Gert Wohlgemuth for all aspects of software integration for compound annotations and metadata import, as well as usability of export results. Prof. Fiehn will personally merge, curate and integrate external database metabolite identifiers for updating LC-BinBase compounds for usability in biomedical and systems biology projects. Prof. Fiehn will interact with Dr. Yuanyue Li and Fanzhou Kong on all aspects of FDR calculations of metabolite and lipid annotations, and integration of retention time modeling and use of QCs and internal standards for quantifications. Prof. Fiehn will engage with his team in weekly project meetings to ensure that timelines and expectations are met and to discuss optimal data validation and code deployment. Prof. Fiehn will coordinate the work with external partner laboratories Metz and Evans in monthly team meetings. He will oversee the work by Dr. Tong Shen on results of specimen testing between the three laboratories and comparison between manual and automatic data processing. Prof. Fiehn will initiate and oversee manuscript writing and data and code sharing.

Senior software- and database developer **Gert Wohlgemuth** will commit 50% efforts in year 01-04 to the overall program. Gert Wohlgemuth joined the Fiehn laboratory in 2002 and has been responsible for all major database and software developments in the West Coast Metabolomics Center for the past 21 years, ranging from the original GC-TOF MS focused BinBase database and miniX study design database to the CTS chemical translation service and the public MassBank.us database with its more than 2 million mass spectra of authentic metabolites. Over the past three years he has programmed the prototype Amazon Web Service-based LC-BinBase software that automatically aligns chromatograms and generates reports for untargeted lipidomic and metabolomic assays. For this project, Mr. Wohlgemuth will develop code to integrate a suite of software tools developed by Dr. Yuanyue Li and Dr. Huaxu Yu into a common pipeline, from library MS/MS reports (in MassBank.us) to retention time predictions (using Retip.app) and adding genomics metadata (from Reactome to BioPAN). Mr. Wohlgemuth will use his excellent expertise in Python, JavaScript, Node.js, MongoDB, Java Springboot, Express.js, and Oracle SQL programming environments and coding languages for tool integration, unit testing, code reviews and overall functionality tests. He will also be responsible for code sharing with external communities and implementing QA/QC reports, consistent data standards and data security protocols. Gert Wohlgemuth will work with informatics team members in the Metz and Evans laboratories for integrating code on ion mobility and adduct formations into the LC-BinBase routines.

Dr. Tong Shen, Project Scientist, is the lead analytical chemist in the Fiehn laboratory. She will commit 25% efforts in years 01-02 and 20% efforts in years 03-04 to this project to oversee the experimental analyses in specific aim 01 and the implementation of BioPAN lipid pathway identifiers and increasing usability of lipidomics data for specific aim 03. She will also oversee the pilot tests of LC-BinBase with external lipidomics and metabolomics laboratories and perform quality controls of metabolite and lipid annotations in LC-BinBase. Dr. Shen has joined the UC Davis West Coast Metabolomics Center in 2016 after obtaining her PhD in Singapore in 2014 in the lipidomics group of Prof. Markus Wenk. Since 2018, Dr. Shen leads the analytical chemistry unit of the UC Davis West Coast Metabolomics Center in all aspects, including throughput, quantification, compound identifications and quality control. She will acquire mass spectrometry data for validation specimen sent by the Evans laboratory and compare manually processed data to LC-BinBase

reports. She will advise the data processing informatics team on issues in result reports and parameters of data processing such as m/z accuracy windows, retention index tolerances, compound identification MSI-annotations and upper and lower QC warning and intervention limits. She will engage with the Metz and Evans laboratories on protocol transfers, analytical chemistry optimizations, and compound annotation confidence as well as on comparison of result data between the laboratories for the different cells, organs and biofluids.

OTHER PERSONNEL

Dr. Yuanyue Li, Project Scientist, is the lead cheminformatics expert in the Fiehn laboratory. He will commit 25% efforts in years 01-02 and 20% efforts in years 03-04 to this project. He joined the West Coast Metabolomics Center in December 2019 and has since developed the Entropy Similarity MS/MS algorithm that was published in *Nature Methods* in 2021 and FlashEntropy published in *Nature Methods* in 2023. He was promoted to Project Scientist in 2022. Dr. Li was trained in multi-omics analyses, including proteomics and genomics, at the European Molecular Biology Laboratory (EMBL) at Peer Bork's group in Heidelberg, Germany. He had obtained his PhD in 2015 at Xianmen University (China) on analyzing high-throughput sequencing data for RNA-seq and for analyzing interactome changes across cellular phenotypes. He will now use his expertise to analyze LC-BinBase high resolution MS/MS spectra using his Entropy Similarity algorithm and integrate this tool with the results of retention time predictions, exact mass distributions and biological likelihood estimates to form a metabolomic confidence score. He will oversee the addition and use of cancer-relevant genomic and biochemical database identifiers to improve the usability of LC-BinBase data reports. Together with Prof. Fiehn, he will advise cheminformatics postdoctoral scholar Huaxu Yu on algorithm development, testing and validation. Dr. Li will report to Prof. Fiehn and write first drafts of the outcome manuscripts. He will engage with the informatics team members in the Evans and Metz laboratories with respect to confidence measures in metabolite annotations, and with respect to LC-BinBase metadata and identifiers for systems biology use.

Dr. Huaxu Yu, postdoctoral scholar in cheminformatics, will commit 100% allowable time to the project in years 01-04. Before joining UC Davis in Spring 2023, he worked at University of British Columbia, Vancouver/Canada, where he was lead author on publications on metabolomics and cheminformatics projects including use of maximal density fold changes, feature specific data transformation, nonlinear electrospray ionization responses in mass spectrometry, patterned signal ratio biases, and computational variation in automated data processing in untargeted metabolomics. Hence, he is well versed in all aspects of metabolomics data curation, normalization, and data transformation for hypothesis-based and discovery-driven informatics data analyses. For specific aim 1, Dr. Huaxu Yu oversee data acquisitions and analysis of exported results from LC-BinBase. For specific aim 2, he will implement and apply his MS/MS autocuration code on automated mass spectral library building for LC-BinBase by cleaning mass spectra from noise ions using fragmentation trees. Together with Dr. Li, he will work on building probability models to integrate retention index, accurate mass and MS/MS similarities into a unified compound ID confidence score. He will also work with Gert Wohlgemuth to test and implement retip2.0 code into the LC-BinBase environment and assist Dr. Yuanyue Li to assess metabolomics and lipidomics data from the Metz and Evans laboratories for integrating code on ion mobility and adduct formations into the LC-BinBase routines.

FRINGE BENEFITS

Fringe benefits are calculated per the Colleges and Universities Rate Agreement approved on 6/28/2021.

OTHER COSTS

Consumables.

For improving, testing and validating the *LC-BinBase* database, using recurrent analyses of LC-MS/MS data, we will use Amazon Web Services, for example for testing automatic parameter optimization in raw data processing. Based on the experience over the past three years, costs of \$12,000 per year are requested. Metabolomics and lipidomics analyses will be performed in the Fiehn laboratory for raw data acquisition of biological specimens (organs, body fluids and cells) purchased from BioIVT. Metabolomics costs for consumables are requested for \$13,129 in year 01, \$8,852 in year 02, \$15,808 in year 03 and \$11,341 in year 04 for LC-MS/MS raw data acquisitions. These costs include BioIVT sample costs, fractional costs for instrument use and maintenance that include consumables such as vials, plates, liners, nuts, ferrules, septa, valves, columns, guard columns, syringes, fractional costs for computer soft- and hardware, chemicals such as internal standards, reference compounds, buffers solvents, gases, 96-well plates and plastic ware, tubing, paper, and printer necessities.

Travel

For travel to the annual ASMS or MANA conferences, we request \$3,000 per year to cover economy transport, lodging and registrations for two scientists (total) per year.

Publication costs

To defray publication costs, including costs for online access, we request \$2000 per year.

Data Management and Sharing Justification

For data curation and the development of related documentation, the project is requesting \$ 1,000 per year to offset hourly staff costs for data transfer. These funds will allow us to prepare data for sharing including deidentification of data, the incorporation of metadata to ensure discoverability and the data transfer process to the MetabolomicsWorkbench.org repository for preservation and access.

Subaward Costs

To enable external metabolomics partners to develop and integrate algorithms for metabolite annotation confidence testing, and to beta-test and validate the *LC-BinBase* database for biomedical projects, costs for subcontracts are requested for the Metz laboratory (PNNL) and for the Evans laboratory (U Michigan) for \$100,000 per year (from years 01-04) direct costs each, plus corresponding indirect costs applicable to their institutions. These costs will cover activities for both informatics efforts and mass spectrometry work at these labs. Specifically, the Evans and Metz laboratories will perform raw data acquisitions for experimental robustness tests in Specific Aim 1. Both laboratories will acquire data for subsequent validation studies, process the data using their in-house routines, and upload and compare data processed through *LC-BinBase*. Both the Evans and Metz laboratories will also actively support overall informatics efforts to optimize *LC-BinBase* workflows, including the integration of adduct formation software from the U Michigan lab (Binner), and integration of ion mobility calculation software from the PNNL lab (ISiCle and autoCCS). Both laboratories will engage in common data evaluation, manuscript writing, code testing and dissemination of results.

FACILITIES AND ADMINISTRATION

F&A Costs were estimated in accordance with UC Davis's approved indirect cost rate agreement with the Department of Health and Human Services (DHHS), dated June 28, 2021, the Federal Cognizant Agency for UC Davis. Indirect costs are approved with a Modified Total Direct Costs (MTDC) base. The costs of any proposed equipment, capital expenditures, graduate fees and tuition, rental costs of off-site facilities, scholarships, fellowships, participant support costs, and the portion of each sub-award in excess of \$25K are excluded from the MTDC base.